
Applied Econometrics

A Textbook for ECON 621

University of Northern British Columbia

School of Business and Economics

Based on ECON 712: Applied Econometrics (Winter 2026)

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*Drawing on: Angrist & Pischke (2009) · Davidson & MacKinnon (2004) · Cunningham
(2021) · Wooldridge (2019)*

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These lecture notes are prepared for course use only.
They draw extensively on the following foundational texts:

- Angrist, J. D., and Pischke, J. S. (2009). *Mostly Harmless Econometrics*. Princeton University Press.
- Davidson, R., and MacKinnon, J. G. (2004). *Econometric Theory and Methods*. Oxford University Press.
- Cunningham, S. (2021). *Causal Inference: The Mixtape*. Yale University Press.
- Wooldridge, J. M. (2019). *Introductory Econometrics*, 7th ed. Cengage.
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Preface

This textbook is written for ECON 621: Applied Econometrics at the University of Northern British Columbia. The course is cross-listed as a senior undergraduate and graduate course, and these notes are designed to serve both audiences: they develop the material rigorously enough for graduate students while maintaining the accessibility and applied focus appropriate for advanced undergraduates.

The notes are organized in three parts. **Part I: Foundations** (Chapters 1–5) covers the regression framework rigorously, from the potential outcomes model through OLS algebra, finite-sample theory, and asymptotic inference with robust standard errors. **Part II: The Regression Toolkit** (Chapters 6–7) extends the framework with the Frisch-Waugh-Lovell theorem, omitted variable bias, and flexible specifications. **Part III: Identification Strategies** (Chapters 8–13) covers instrumental variables, the Local Average Treatment Effect, panel data, differences-in-differences, matching, and regression discontinuity — the core of modern applied econometrics.

Every chapter includes formal definitions and proofs, economic intuition, canonical empirical examples from the literature, exercises at multiple levels, and a complete **R programming lab** with working code that students can run directly.

The approach follows the *Credibility Revolution* tradition: we are as concerned with *research design* as with estimation. Understanding what variation identifies an effect, and whether that identification strategy is credible, is the central skill of applied econometrics.

Note to students: Do not just read these notes — work through the derivations with pencil and paper, run the R code, and modify it to test your understanding. Applied econometrics is learned by doing.

Part I

Foundations

Chapter 1

Chapter 1: The Quest for Causality — An Introduction to Applied Econometrics

1.1 Chapter Overview

This chapter introduces the central challenge of applied econometrics: distinguishing **correlation** from **causation** in economic data. We begin by asking what kinds of questions economists care about, explain why answering them with observational data is hard, and introduce the conceptual framework — the **potential outcomes model** — that modern applied economists use to think clearly about causality.

By the end of this chapter you will understand why simply comparing group averages can be deeply misleading, what it means to "identify" a causal effect, and what tools the rest of this book will give you to do so.

1.2 Learning Objectives

After completing this chapter, students will be able to:

1. Define applied econometrics and explain its goals.
2. Formulate a well-posed empirical question in terms of a causal relationship.
3. Write down the **potential outcomes** (counterfactual) model for a binary treatment.
4. Define the **Average Treatment Effect (ATE)**, the **Average Treatment Effect on the Treated (ATT)**, and relate them to the observable difference in means.
5. Identify and explain **selection bias** and why it contaminates simple comparisons.
6. Explain how **random assignment** solves the selection bias problem.

7. Describe, at a high level, the identification strategies studied in this book (regression, IV, DiD, matching, RDD).
-

1.3 1.1 What Is Applied Econometrics?

Economists study a wide variety of questions: Does access to health insurance improve health outcomes? Do minimum wage laws cause unemployment? Does attending a smaller class improve student test scores? Does going to university raise lifetime earnings?

What unites all of these questions is that they are **causal**. We do not merely want to know whether smaller class sizes and higher test scores tend to go together in data — we want to know whether *reducing* class size *causes* test scores to rise, and by how much. The distinction matters enormously for policy. If better-resourced schools happen to have both smaller classes and better teachers, then we might find a strong correlation between class size and test scores even if class size has no causal effect whatsoever.

Applied econometrics is the branch of economics that uses **data and statistical methods** to estimate causal relationships and test economic theories. It sits at the intersection of economic theory, statistics, and data science, but its animating concern — separating cause from correlation — makes it distinctive.

1.3.1 1.1.1 A Map of the Terrain

Applied econometrics divides naturally into two broad activities:

1. **Estimating causal effects.** How large is the return to an additional year of schooling? Does a job-training program raise employment? By how much did a particular policy change crime rates?
1. **Testing economic theories.** Do consumers respond to prices as standard theory predicts? Is the labour supply curve upward sloping? Are financial markets informationally efficient?

This course focuses primarily on (1), but the tools we develop are equally useful for (2). Throughout, we will be careful to distinguish between **descriptive** statements (what is the world like?) and **causal** statements (what would happen if we intervened?).

1.3.2 1.1.2 Data in Applied Econometrics

Economists work with many types of data:

- **Cross-sectional data** — observations on many units (individuals, firms, countries) at a single point in time.

- **Time-series data** — observations on a single unit over many time periods.
- **Panel (longitudinal) data** — observations on many units over multiple time periods. This is the richest structure and features prominently in this book.
- **Experimental data** — data collected through randomized controlled trials (RCTs).

Each data type raises its own challenges. Observational data — data collected without deliberate experimental design — is by far the most common, and learning to draw valid causal inferences from it is the central skill of applied econometrics.

1.4 1.2 Empirical Questions and the Causal Ideal

The starting point of any empirical project is a **well-defined question**. Precision matters. Consider the following progression:

Vague Question	Precise Causal Question
Do hospitals make people healthier?	Does hospitalization <i>cause</i> better health outcomes
Does education matter?	Does an additional year of schooling <i>cause</i> higher l
Is the minimum wage bad for employment?	Does a \$1 increase in the minimum wage <i>cause</i> a r
The precise version specifies: (a) the cause (the treatment or intervention), (b) the effect (the outcome), and (c) the population of interest.	

1.4.1 1.2.1 Fundamental Concepts

Definition 1.1 (Treatment): A *treatment* is a variable D_i that takes the value 1 if unit i receives the intervention (is "treated") and 0 otherwise.

Definition 1.2 (Outcome): An *outcome* Y_i is the variable we care about — earnings, health, test scores, employment.

Definition 1.3 (Causal Effect): The causal effect of the treatment on unit i is the difference between the outcome they would experience *with* treatment and the outcome they would experience *without* treatment.

This definition immediately reveals the **fundamental problem of causal inference**: we can never observe the same unit simultaneously with and without the treatment. Whatever we observe is only one side of a counterfactual comparison.

1.5 1.3 The Potential Outcomes Framework

The most productive way to think about causal questions is the **potential outcomes framework**, originally developed by Neyman (1923) and formalized by Rubin (1974). This framework is sometimes called the **Rubin Causal Model** or the **Neyman-Rubin framework**.

1.5.1 1.3.1 Potential Outcomes Defined

For each individual i in the population, define two **potential outcomes**:

Y_{0i} = the outcome for unit i if $D_i = 0$ (not treated)

Y_{1i} = the outcome for unit i if $D_i = 1$ (treated)

Key Insight: Both Y_{0i} and Y_{1i} are defined for *every* individual, but we can only ever *observe* one of them — the one corresponding to the treatment the individual actually received.**

The **individual causal effect** for unit i is:

$$\tau_i = Y_{1i} - Y_{0i}$$

This is the difference that would result from "switching" the individual's treatment status, everything else held constant.

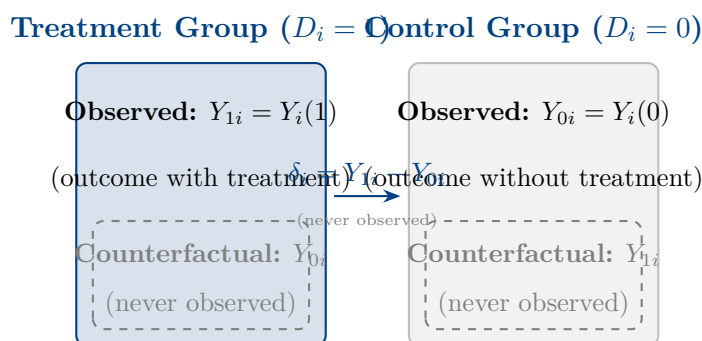


Figure 1.1: The fundamental problem of causal inference. For each unit, only one potential outcome is observed. The causal effect $\delta_i = Y_{1i} - Y_{0i}$ can never be computed directly.

1.5.2 1.3.2 Population-Level Causal Estimands

Because τ_i varies across individuals and we cannot observe it for any individual, economists focus on **average** causal effects:

Definition 1.4 (Average Treatment Effect):

$$\text{ATE} = E[Y_{1i} - Y_{0i}] = E[\tau_i]$$

The ATE is the average causal effect for a randomly chosen unit from the population.

Definition 1.5 (Average Treatment Effect on the Treated):

$$\text{ATT} = E[Y_{1i} - Y_{0i} \mid D_i = 1]$$

The ATT is the average causal effect for individuals who actually received the treatment. This is often the more policy-relevant quantity: when a government wants to know if a training program helped the participants, it cares about the ATT.

Definition 1.6 (Average Treatment Effect on the Untreated):

$$\text{ATU} = E[Y_{1i} - Y_{0i} \mid D_i = 0]$$

The ATU describes what the average effect would be if the untreated were brought into the program.

Note on ATE vs ATT: When treatment effects are heterogeneous (vary by individual) and treatment is not randomly assigned, ATE and ATT need not be equal. Understanding the distinction is important for interpreting estimates and for policy.**

1.5.3 1.3.3 What We Observe

In any dataset, we observe the *realized* outcome:

$$Y_i = Y_{0i} + (Y_{1i} - Y_{0i})D_i = \begin{cases} Y_{1i} & \text{if } D_i = 1 \\ Y_{0i} & \text{if } D_i = 0 \end{cases}$$

This is the **switching equation**. It links the observable data to the underlying potential outcomes.

1.6 1.4 Selection Bias: Why Simple Comparisons Mislead

The most naive approach to estimating a causal effect is to compare the average outcomes of treated and untreated units. Let us see precisely why this fails.

1.6.1 1.4.1 The Decomposition of Observed Differences

Consider the difference in mean outcomes between treated and untreated groups:

$$\underbrace{E[Y_i | D_i = 1] - E[Y_i | D_i = 0]}_{\text{Observed Difference in Means}}$$

Substituting the switching equation:

$$= E[Y_{1i} | D_i = 1] - E[Y_{0i} | D_i = 0]$$

Now add and subtract $E[Y_{0i} | D_i = 1]$:

$$= \underbrace{E[Y_{1i} - Y_{0i} | D_i = 1]}_{\text{ATT}} + \underbrace{E[Y_{0i} | D_i = 1] - E[Y_{0i} | D_i = 0]}_{\text{Selection Bias}}$$

This is the fundamental decomposition of observed differences into causal effects and bias.

The first term is the ATT — what we actually want to know.

The second term, **selection bias**, is the difference in average untreated outcomes between the treated and untreated groups. It captures the fact that people who select into treatment may be systematically different from those who do not.

Equation 1.1: Bias Decomposition

$$\underbrace{E[Y_i | D_i = 1] - E[Y_i | D_i = 0]}_{\text{Observed difference}} = \underbrace{\text{ATT}}_{\text{Causal effect}} + \underbrace{E[Y_{0i} | D_i = 1] - E[Y_{0i} | D_i = 0]}_{\text{Selection bias}}$$

1.6.2 1.4.2 A Canonical Example: Hospitals and Health

Consider the question: do hospitals make people healthier? The National Health Interview Survey (NHIS) data reveal a troubling fact — individuals who were hospitalized in the past year report *worse* health status on average than those who were not. Does this mean hospitals make you sick?

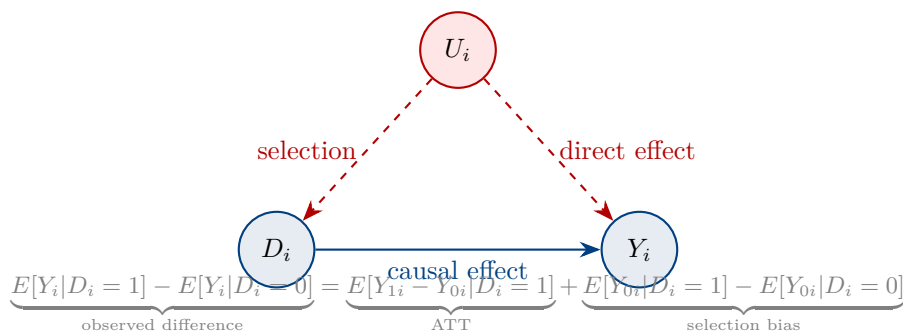


Figure 1.2: A directed acyclic graph (DAG) illustrating selection bias. The unobserved confounder U_i (red, dashed) simultaneously affects treatment assignment D_i and the outcome Y_i , creating a spurious association between treatment and outcome beyond the true causal effect.

Of course not. The selection bias here runs in a clear direction: people who end up in hospitals are already, on average, much sicker than the general population. They have worse Y_{0i} — they would have had worse health *even without* hospitalization. So:

$$E[Y_{0i} | D_i = 1] < E[Y_{0i} | D_i = 0]$$

The selection bias is *negative*, masking what may well be a large positive causal effect of hospitalization on health. The observable comparison between hospitalized and non-hospitalized individuals is dominated by the fact that hospitalized people were sicker to begin with.

This example, from Angrist and Pischke (2009), illustrates how selection bias can flip the apparent sign of a causal relationship.

1.6.3 1.4.3 Selection Bias in Labour Economics

A similar issue arises in estimating returns to education. People who choose to obtain more education tend to be more able, more motivated, and from higher-income families. Even if education had *zero* causal effect on earnings, we would observe a positive correlation between education and earnings because of these underlying differences.

Formally:

$$E[Y_{0i} | D_i = 1] > E[Y_{0i} | D_i = 0]$$

where $D_i = 1$ denotes more education. The selection bias is positive, and simple OLS regressions of wages on education will overstate the causal return to schooling. (We

return to this in Chapters 6 and 8.)

1.7 1.5 The Gold Standard: Randomized Experiments

The cleanest solution to the selection bias problem is **random assignment**. In a randomized controlled trial (RCT), whether an individual receives treatment is determined by a coin flip (or random number generator), not by any characteristic of the individual.

1.7.1 1.5.1 Why Randomization Works

If D_i is assigned randomly, then it is independent of potential outcomes:

$$\{Y_{0i}, Y_{1i}\} \perp\!\!\!\perp D_i$$

This independence condition ensures that:

$$E[Y_{0i} | D_i = 1] = E[Y_{0i} | D_i = 0] = E[Y_{0i}]$$

That is, the untreated outcomes of treated and untreated individuals are the same in expectation. **Selection bias is zero.**

As a result:

$$E[Y_i | D_i = 1] - E[Y_i | D_i = 0] = E[Y_{1i} - Y_{0i}] = \text{ATE}$$

The simple difference in group means identifies the Average Treatment Effect.

1.7.2 1.5.2 Linking Randomization to Regression

To connect experiments to the regression framework, write:

$$Y_i = \alpha + \rho D_i + \epsilon_i$$

where $\alpha = E[Y_{0i}]$, $\rho = Y_{1i} - Y_{0i}$ (assumed constant here), and $\epsilon_i = Y_{0i} - E[Y_{0i}]$.

Under random assignment, D_i is independent of Y_{0i} , so D_i is uncorrelated with ϵ_i . The OLS estimator of ρ from this regression recovers the causal effect:

$$\hat{\rho}^{OLS} \xrightarrow{p} \rho$$

This is the bridge between the potential outcomes framework and regression. Regression with a randomly assigned treatment variable identifies causal effects. The

problem — and most of this course — is about what to do when treatment is *not* randomly assigned.

1.7.3 1.5.3 Example: Project STAR

One of the most celebrated experiments in economics is Project STAR (Student Teacher Achievement Ratio), conducted in Tennessee from 1985 to 1989. Over 11,000 students and their teachers were randomly assigned to small classes (13–17 students) or regular classes (22–26 students) from kindergarten through grade 3.

Because assignment was random, differences in test scores between students in small and large classes can be attributed to class size, not to preexisting differences in student ability or family background. Krueger (1999) found that students assigned to smaller classes scored roughly 4–5 percentile points higher on standardized tests — and that these effects were larger for minority and disadvantaged students.

This example illustrates two things: the power of randomization to answer causal questions cleanly, and the policy relevance of the answers.

Box 1.1 — Fundamental Questions (FUQs)

Not every empirical question can be answered even in principle. A question is a *Fundamental Unidentified Question* (FUQ) if it requires observing a counterfactual that can never be defined. For example: "What would Canada's GDP have been if it had never adopted a central bank?" The counterfactual here is so distant from any observed data that no statistical method can recover it. The potential outcomes framework helps identify which questions are FUQs and which, with clever design, can be answered.

1.8 1.6 When Experiments Are Impossible: Identification Strategies

In most economic settings, running experiments is not feasible — for ethical, practical, or financial reasons. We cannot randomly assign people to different levels of schooling, different countries to different policies, or firms to different market structures.

Applied econometrics has developed a set of **identification strategies** — methods for estimating causal effects from non-experimental (observational) data. The key insight behind each strategy is to find a source of variation in treatment that is "as good as random," even if treatment itself is not randomly assigned.

The major identification strategies covered in this book are:

Strategy	Core Idea
Multiple Regression / OLS	Control for observed confounders
Instrumental Variables (IV)	Use exogenous variation in a "third variable"
Panel Data / Fixed Effects	Control for time-invariant unobservables
Differences-in-Differences	Compare changes across treated and untreated groups
Matching	Match treated and untreated units on observable characteristics
Regression Discontinuity	Exploit arbitrary thresholds in treatment assignment

Each strategy rests on a **key identifying assumption** that must be plausible given the research question and context. A major theme of this book is that we must be transparent about these assumptions and subject them to scrutiny.

1.8.1 1.6.1 The Credibility Revolution

Beginning in the 1990s, empirical economics underwent a methodological transformation sometimes called the **Credibility Revolution**. Economists increasingly insisted that empirical claims rest on transparent identification assumptions that could be argued on substantive grounds, rather than on parametric assumptions that are difficult to verify.

This shift was associated with economists like Joshua Angrist, David Card, Guido Imbens, and Alan Krueger (and recognized by the 2021 Nobel Prize in Economic Sciences, awarded to Card, Angrist, and Imbens). Their work emphasized the use of natural experiments — situations where policy changes, institutional rules, or other features of the world generate quasi-random variation in treatment — to identify causal effects.

This textbook teaches econometrics in that tradition. We will be as concerned with *research design* (which variation are we using?) as with *estimation* (how are we fitting curves to data?).

1.9 1.7 Roadmap of This Textbook

This book is organized into three parts.

Part I: Foundations (Chapters 2–5) develops the regression framework rigorously. We cover the Conditional Expectation Function, OLS algebra and geometry, finite-sample and asymptotic statistical properties, and modern robust inference (heteroskedasticity- and cluster-robust standard errors). This is the toolkit you will use throughout.

Part II: The Regression Toolkit (Chapters 6–7) extends the basic framework with essential tools: the Frisch-Waugh-Lovell theorem, omitted variable bias, dummy variables, interaction terms, and non-linearities.

Part III: Identification Strategies (Chapters 8–13) covers each major strategy in turn: instrumental variables, LATE, panel data, differences-in-differences, matching, and regression discontinuity. For each, we develop the theory, discuss the key assumptions, and study canonical empirical applications.

Appendices provide reviews of probability, matrix algebra, and R programming.

1.10 Chapter Summary

- Applied econometrics aims to estimate **causal relationships** from data.
 - The **potential outcomes framework** defines the causal effect of a treatment for each individual as the difference between two hypothetical outcomes — one with treatment, one without.
 - Because we cannot observe both potential outcomes for the same individual, causal inference requires comparing different groups.
 - **Selection bias** arises when treated and untreated units differ in their untreated potential outcomes.
 - The simple observed difference in means equals the ATT plus selection bias.
 - **Random assignment** eliminates selection bias by ensuring treatment is independent of potential outcomes.
 - When randomization is infeasible, applied econometricians use **identification strategies** — research designs that generate quasi-random variation in treatment.
 - The **Credibility Revolution** has reoriented empirical economics around transparent, design-based identification.
-

1.11 Key Terms

Applied econometrics — the use of statistical methods and data to estimate economic relationships and test economic theories.

Potential outcomes — the hypothetical outcomes Y_{1i} and Y_{0i} that a unit would experience under treatment and control, respectively.

Causal effect — the individual-level difference $\tau_i = Y_{1i} - Y_{0i}$.

ATE (Average Treatment Effect) — $E[Y_{1i} - Y_{0i}]$; the average causal effect across all individuals.

ATT (Average Treatment Effect on the Treated) — $E[Y_{1i} - Y_{0i} \mid D_i = 1]$.

Selection bias — $E[Y_{0i} \mid D_i = 1] - E[Y_{0i} \mid D_i = 0]$; the difference in counterfactual outcomes between treated and untreated groups.

Switching equation — $Y_i = Y_{0i} + (Y_{1i} - Y_{0i})D_i$.

Randomized controlled trial (RCT) — a study in which treatment is assigned randomly.

Identification strategy — a research design that exploits quasi-random variation in treatment to estimate causal effects.

Fundamental problem of causal inference — the impossibility of observing both Y_{1i} and Y_{0i} for the same unit.

1.12 Exercises

1.12.1 Conceptual Questions

1.1 Distinguish between correlation and causation. Give an example (not in this chapter) where a strong correlation does not reflect a causal relationship.

1.2 Suppose we observe that people who wear helmets while cycling have a higher rate of cycling injuries than those who don't. Does this mean helmets cause injuries? Use the potential outcomes framework to explain the selection bias present in this comparison.

1.3 Write down the switching equation for the following scenario: $D_i = 1$ if individual i participates in a job-training program; Y_i is earnings one year after the program.

1.4 In your own words, explain the difference between the ATE and the ATT. In what situations would these two quantities be very different? In what situations would they be approximately equal?

1.5 Explain why, in a randomized experiment, the simple difference in mean outcomes between treated and control groups identifies the ATE.

1.6 Consider Project STAR. Explain why the fact that some students switched class-size assignments after randomization (non-compliance) complicates the interpretation of the results.

1.12.2 Analytical Questions

1.7 Suppose:

- $E[Y_{1i}] = 10$
- $E[Y_{0i}] = 6$

- $E[Y_{0i} | D_i = 1] = 8$
 - $E[Y_{0i} | D_i = 0] = 5$
- (a) Compute the ATE. (b) Compute the selection bias. (c) Compute the observed difference in means $E[Y_i | D_i = 1] - E[Y_i | D_i = 0]$, assuming constant treatment effects. (d) Does the observed difference in means overstate or understate the ATE? Why?

1.8 Show algebraically that when $D_i \perp \{Y_{0i}, Y_{1i}\}$ (random assignment), the selection bias term is zero and the observed difference in means equals the ATE.

1.9 Consider a government program that provides subsidized childcare to low-income families. Describe the selection bias you would expect if you estimated the effect of the program on children's school readiness by comparing children who received subsidized care to those who did not.

1.12.3 Applied Questions

1.10 The following table shows average earnings (in \$000/year) for workers who attended university and those who did not, in a hypothetical survey:

	Attended University	Did Not Attend	Difference
Earnings	\$65	\$40	\$25

(a) What assumption would you need to make to interpret this \$25 difference as the causal return to university education? (b) Is this assumption plausible? Discuss the likely direction of selection bias. (c) Propose a research design that would allow you to estimate the causal effect more credibly.

1.11 Project STAR involved four conditions: small class, regular class, regular class with a teacher's aide. Briefly describe how you would estimate the effect of small classes relative to regular classes, assuming random assignment was perfect.

1.13 R Lab 1: Introduction to R and Causal Thinking

1.13.1 Objectives

- Learn basic R syntax and data manipulation
- Simulate potential outcomes data
- Illustrate the selection bias problem with simulated data
- Perform a simple comparison of means (as a naive estimator)

1.13.2 Setup

```
1 # Install and load necessary packages
2 # install.packages(c("tidyverse", "haven", "stargazer"))
3 library(tidyverse)
4 set.seed(42)
```

1.13.3 Exercise 1.1: Simulating Potential Outcomes

```
1 # Simulate a population of 1000 individuals
2 n <- 1000
3
4 # Each individual has two potential outcomes
5 # Y0: outcome without treatment (e.g., earnings without training)
6 # Y1: outcome with treatment (e.g., earnings with training)
7
8 # True individual-level treatment effect is 5 (in $000)
9 # But individuals with higher ability (theta) also have higher Y0
10 theta <- rnorm(n, mean = 10, sd = 3) # unobserved ability
11
12 Y0 <- 30 + 2 * theta + rnorm(n, sd = 5) # untreated earnings
13 Y1 <- Y0 + 5                             # treated earnings
14     (constant +5)
15
16 # True ATE
17 true_ATE <- mean(Y1 - Y0)
18 cat("True ATE:", true_ATE, "\n") # Should be approximately 5
```

1.13.4 Exercise 1.2: Simulating Selection Bias

```
1 # Now simulate NON-random treatment:
2 # High-ability individuals are more likely to select into treatment
3 prob_treat <- plogis(-5 + 0.5 * theta) # probability of selecting
4                                         treatment
5
6 D <- rbinom(n, 1, prob_treat)           # treatment indicator
7
8 # Observed outcome
9 Y <- D * Y1 + (1 - D) * Y0
10
11 # Naive comparison of means
12 mean_treated <- mean(Y[D == 1])
13 mean_untreated <- mean(Y[D == 0])
14 naive_estimate <- mean_treated - mean_untreated
15
16 cat("Mean outcome, treated:", round(mean_treated, 2), "\n")
17 cat("Mean outcome, untreated:", round(mean_untreated, 2), "\n")
18 cat("Naive difference in means:", round(naive_estimate, 2), "\n")
19 cat("True ATE:", round(true_ATE, 2), "\n")
```

```
18 cat("Selection bias:", round(naive_estimate - true_ATE, 2), "\n")
```

Questions:

- How large is the selection bias in your simulation?
- In what direction does it go? Why?
- What would happen if you ran an OLS regression of Y on D? Would you recover the true ATE?

1.13.5 Exercise 1.3: Simulating a Randomized Experiment

```
1 # Now randomly assign treatment
2 D_random <- rbinom(n, 1, 0.5)
3 Y_random <- D_random * Y1 + (1 - D_random) * Y0
4
5 # Compare means
6 rct_estimate <- mean(Y_random[D_random == 1]) -
7   mean(Y_random[D_random == 0])
8 cat("RCT estimate:", round(rct_estimate, 2), "\n")
9 cat("True ATE:", round(true_ATE, 2), "\n")
10 cat("Bias under RCT:", round(rct_estimate - true_ATE, 2), "\n")
11
12 # OLS regression under random assignment
13 lm_rct <- lm(Y_random ~ D_random)
14 summary(lm_rct)
```

Questions:

- How does the RCT estimate compare to the true ATE?
- What is the role of the constant in the regression?
- Run the simulation 1000 times (use `replicate()`) and show that the RCT estimate is unbiased on average.

1.13.6 Exercise 1.4: Visualizing Selection Bias

```
1 # Create data frame
2 df <- data.frame(Y0 = Y0, Y1 = Y1, D = factor(D, labels =
3   c("Untreated", "Treated")))
4
5 # Plot distributions of Y0 by treatment status
6 ggplot(df, aes(x = Y0, fill = D)) +
7   geom_density(alpha = 0.5) +
8   labs(
9     title = "Distribution of Untreated Potential Outcome (Y0) by
10     Treatment Status",
```

```
9     subtitle = "Selection bias is visible: treated group has higher
10           Y0 on average",
11     x = "Untreated Potential Outcome (Y0)",
12     y = "Density",
13     fill = "Treatment Status"
14 ) +
15 theme_minimal()
```

Interpretation: If treatment were randomly assigned, the two distributions would overlap almost perfectly. The fact that they are separated visually confirms the presence of selection bias.

End of Chapter 1

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Chapter 2

Chapter 2: The Conditional Expectation Function and Linear Regression

2.1 Chapter Overview

Before we can understand why regression works — and when it doesn't — we need a clear statistical framework for describing relationships between variables. This chapter develops the **Conditional Expectation Function (CEF)**, the population object that regression is trying to estimate. We show that the CEF has deep and useful mathematical properties, that OLS regression is the best linear approximation to the CEF, and that the two coincide when the CEF happens to be linear.

This chapter also connects the regression framework to the causal questions introduced in Chapter 1. Understanding the relationship between regression and the CEF is essential for interpreting regression output correctly and avoiding common mistakes.

2.2 Learning Objectives

After completing this chapter, students will be able to:

1. Define the **Conditional Expectation Function (CEF)** and explain what it represents.
2. State and apply the **Law of Iterated Expectations (LIE)**.
3. Explain the **CEF Decomposition Property** and its implications.
4. Prove that the CEF is the **best predictor** of Y given X in the mean-squared-error sense.
5. Explain the **ANOVA Theorem** (variance decomposition).
6. State the **OLS linear regression problem** and derive the population parameter β^{OLS} .

7. Explain why OLS is the **best linear approximation** to the CEF even when the CEF is non-linear.
 8. Interpret regression coefficients as statements about the CEF.
 9. Apply these concepts to real economic data using R.
-

2.3 2.1 Describing Relationships: From Scatter Plots to Functions

Consider the question: how do wages vary with education? Figure 2.1 (described below) shows a scatter plot of log weekly earnings against years of schooling, constructed from Current Population Survey (CPS) data. There is a clear positive relationship, but with enormous variation: two people with identical schooling can have very different earnings. What summary of this cloud of points do we want?

The obvious answer is: we want to know the **average wage for each level of education**. If we could calculate the mean of Y_i (log earnings) for every value of X_i (schooling), we would trace out a curve through the scatter plot. This curve is the **Conditional Expectation Function**.

Figure 2.1 Description:** Scatter plot of log weekly earnings (Y) against years of schooling (X) for a random sample of workers. The cloud of points slopes upward on average. Within each level of schooling, wages are spread over a wide range. The line running through the scatter plot represents $E[Y_i | X_i = x]$ for each value of x .

2.4 2.2 The Conditional Expectation Function (CEF)

Definition 2.1 (CEF): Let Y_i be a random variable (the outcome) and $\mathbf{X}_i$ be a vector of conditioning variables (regressors). The **Conditional Expectation Function** is:

$$E[Y_i | \mathbf{X}_i = \mathbf{x}]$$

This is a function of $\mathbf{x}$ that gives the mean of Y_i in the subpopulation where $\mathbf{X}_i$ equals $\mathbf{x}$.

2.4.1 2.2.1 The CEF with a Single Binary Variable

The simplest case has $X_i \in \{0, 1\}$:

$$E[Y_i | X_i = 1] = \text{mean outcome among treated units}$$

$$E[Y_i | X_i = 0] = \text{mean outcome among untreated units}$$

These are the two conditional means we discussed in Chapter 1. The CEF here takes only two values.

2.4.2 2.2.2 The CEF with a Continuous Variable

When X_i is continuous, $E[Y_i | X_i = x]$ traces a potentially complex curve. For the education-wage example, it might look roughly linear but could be non-linear (e.g., the return to the 12th year of schooling may differ from the return to the 16th).

2.4.3 2.2.3 The Multivariate CEF

In practice, we condition on multiple variables $\mathbf{X}_i = (X_{1i}, X_{2i}, \dots, X_{ki})^T$:

$$E[Y_i | \mathbf{X}_i = \mathbf{x}] = E[Y_i | X_{1i} = x_1, X_{2i} = x_2, \dots, X_{ki} = x_k]$$

This is a function from $\mathbb{R}^k$ to $\mathbb{R}$. It answers the question: among all individuals with *exactly* these values of all k covariates, what is the average value of Y ?

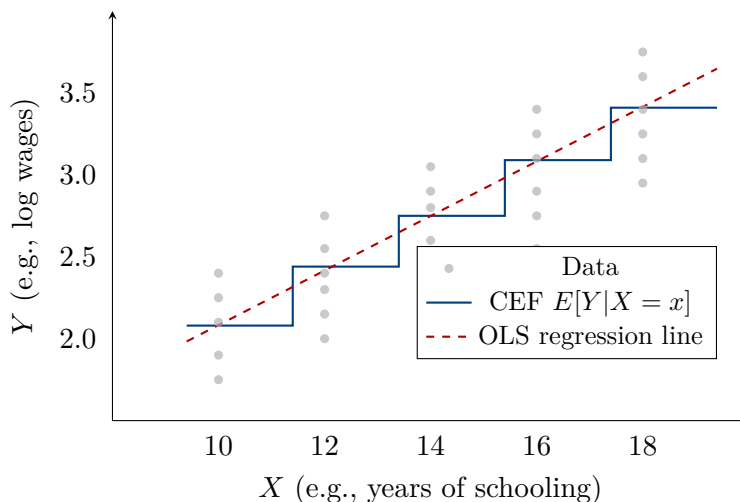


Figure 2.1: The Conditional Expectation Function (CEF) and the OLS regression line. The CEF $E[Y_i|X_i = x]$ (solid blue step function) gives the population mean of Y at each value of X . The OLS regression line (dashed red) is the best linear approximation to the CEF in the minimum mean-squared-error sense.

2.5 2.3 The Law of Iterated Expectations

The Law of Iterated Expectations (LIE) is one of the most useful identities in probability and statistics. We will use it repeatedly throughout this course.

Theorem 2.1 (Law of Iterated Expectations):

$$E[Y_i] = E[E[Y_i | \mathbf{X}_i]]$$

Interpretation: You can compute the unconditional expectation of Y_i in two steps: first compute $E[Y_i | \mathbf{X}_i = \mathbf{x}]$ for each value $\mathbf{x}$, then average this over the distribution of $\mathbf{X}_i$.

Proof (continuous case): Let $g(\mathbf{x}) = E[Y_i | \mathbf{X}_i = \mathbf{x}]$ and let $f_X(\mathbf{x})$ be the marginal density of $\mathbf{X}_i$. Then:

$$E[E[Y_i | \mathbf{X}_i]] = \int g(\mathbf{x})f_X(\mathbf{x}) d\mathbf{x} = \int E[Y_i | \mathbf{X}_i = \mathbf{x}]f_X(\mathbf{x}) d\mathbf{x} = E[Y_i] \quad \square$$

2.5.1 2.3.1 Practical Application of the LIE

Example 2.1: Suppose we want to compute average wages in an economy where workers are either male (M) or female (F). The LIE tells us:

$$E[\text{Wage}] = E[\text{Wage} | \text{Male}] \cdot P(\text{Male}) + E[\text{Wage} | \text{Female}] \cdot P(\text{Female})$$

This is intuitive: the average wage is a probability-weighted average of average wages within each group.

Corollary 2.1: $E[h(\mathbf{X}_i)\epsilon_i] = 0$ for any function h , where $\epsilon_i = Y_i - E[Y_i | \mathbf{X}_i]$.

Proof: By the LIE, $E[h(\mathbf{X}_i)\epsilon_i] = E[E[h(\mathbf{X}_i)\epsilon_i | \mathbf{X}_i]] = E[h(\mathbf{X}_i)E[\epsilon_i | \mathbf{X}_i]] = 0$, since $E[\epsilon_i | \mathbf{X}_i] = 0$ by construction. $\square$

2.6 2.4 The CEF Decomposition and Its Properties

The CEF defines a natural **decomposition** of any random variable into a systematic part and a residual part.

2.6.1 2.4.1 The CEF Decomposition

Proposition 2.1 (CEF Decomposition): Any random variable Y_i can be written as:

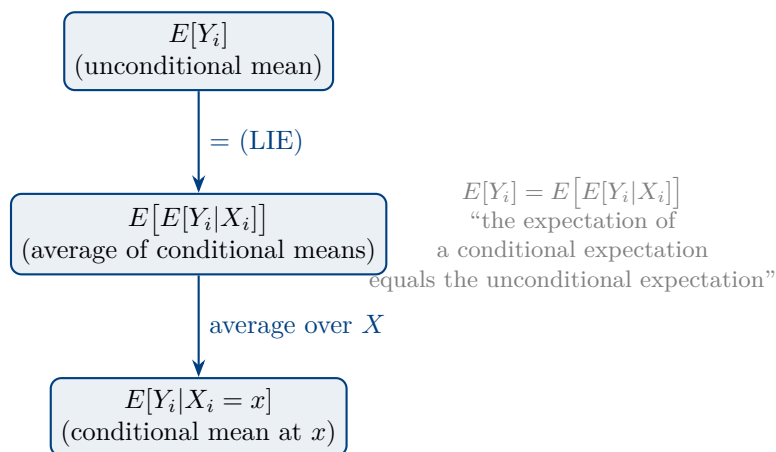


Figure 2.2: The Law of Iterated Expectations (LIE). The unconditional mean equals the expectation of the conditional mean function taken over the distribution of X .

$$Y_i = \underbrace{E[Y_i | \mathbf{X}_i]}_{\text{Systematic part}} + \underbrace{\epsilon_i}_{\text{Residual part}}$$

where the residual $\epsilon_i = Y_i - E[Y_i | \mathbf{X}_i]$ satisfies:

1. **Mean Independence:** $E[\epsilon_i | \mathbf{X}_i] = 0$ for all values of $\mathbf{X}_i$.
2. **Zero Unconditional Mean:** $E[\epsilon_i] = 0$ (by the LIE).
3. **Uncorrelated with Any Function of $\mathbf{X}_i$:** $E[h(\mathbf{X}_i)\epsilon_i] = 0$ for any function h .

Proof: Property (1) follows directly from the definition: $E[\epsilon_i | \mathbf{X}_i] = E[Y_i | \mathbf{X}_i] - E[Y_i | \mathbf{X}_i] = 0$. Properties (2) and (3) follow from (1) and the LIE. $\square$

This decomposition is fundamental. It says that any variable can be split into a part predictable from $\mathbf{X}_i$ (the CEF) and a part that is, in the strongest possible sense, unpredictable given $\mathbf{X}_i$ (the residual ϵ_i).**

2.6.2 2.4.2 The CEF is the Best Predictor

Theorem 2.2 (CEF as Best Predictor): Among *all* functions $m(\mathbf{X}_i)$, the CEF $E[Y_i | \mathbf{X}_i]$ minimizes the **Mean Squared Prediction Error (MSPE)**:

$$E[Y_i | \mathbf{X}_i] = \operatorname{argmin}_{m(\cdot)} E[(Y_i - m(\mathbf{X}_i))^2]$$

Proof: For any function m , write:

$$\begin{aligned} E[(Y_i - m(\mathbf{X}_i))^2] &= E[(Y_i - E[Y_i | \mathbf{X}_i] + E[Y_i | \mathbf{X}_i] - m(\mathbf{X}_i))^2] \\ &= E[\epsilon_i^2] + E[(E[Y_i | \mathbf{X}_i] - m(\mathbf{X}_i))^2] + 2E[\epsilon_i(E[Y_i | \mathbf{X}_i] - m(\mathbf{X}_i))] \end{aligned}$$

The cross term is zero by Corollary 2.1. The second term is non-negative and equals zero only when $m(\mathbf{X}_i) = E[Y_i | \mathbf{X}_i]$. $\square$

Intuition: $E[Y_i]$ is the best constant predictor of Y_i . The CEF does better by conditioning on available information: by the LIE, $E[E[Y_i | \mathbf{X}_i]] = E[Y_i]$, so the CEF is as good on average and better in specific strata.

2.6.3 2.4.3 The ANOVA Theorem

Theorem 2.3 (Variance Decomposition / ANOVA):

$$\underbrace{V(Y_i)}_{\text{Total variance}} = \underbrace{V(E[Y_i | \mathbf{X}_i])}_{\text{Explained variance}} + \underbrace{E[V(Y_i | \mathbf{X}_i)]}_{\text{Unexplained variance}}$$

Interpretation: The total variation in Y_i decomposes into:

- The variation in the conditional mean (what $\mathbf{X}$ can "explain")
- The average within-group variation (what remains after conditioning on $\mathbf{X}$)

This is the population analogue of the sample R^2 decomposition $\text{TSS} = \text{ESS} + \text{SSR}$.

Example 2.2 (ANOVA in Wage Data): In the CPS data, approximately 25–30% of the variation in log wages is explained by education, age, gender, and occupation. The remaining 70–75% is within-group variation: workers with the same observable characteristics still have very different wages.

2.7 2.5 Why Linear Regression?

The CEF can take any shape — it need not be linear. So why do economists almost always estimate *linear* regressions? This section provides two justifications.

2.7.1 2.5.1 Justification 1: The CEF is Linear

If the true population CEF is linear:

$$E[Y_i | \mathbf{X}_i] = \mathbf{X}_i^T \boldsymbol{\beta}^*$$

then by the CEF decomposition theorem:

$$E[\mathbf{X}_i(Y_i - \mathbf{X}_i^T \beta^*)] = \mathbf{0}$$

Solving for β^* :

$$\beta^* = E[\mathbf{X}_i \mathbf{X}_i^T]^{-1} E[\mathbf{X}_i Y_i] \equiv \beta^{OLS}$$

So when the CEF is linear, the OLS population parameter β^{OLS} equals the true parameter β^* of the CEF. The linear regression model is correctly specified.

When is the CEF exactly linear? The CEF is linear in X when (Y, X) follow a jointly normal distribution (because $E[Y | X] = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X}(X - \mu_X)$ for the bivariate normal). But this is a very strong assumption.

2.7.2 2.5.2 Justification 2: OLS Provides the Best Linear Approximation

What if the CEF is *not* linear? Does OLS still make sense? The answer is yes.

Theorem 2.4 (OLS as Best Linear Approximation): Even when $E[Y_i | \mathbf{X}_i]$ is not linear, β^{OLS} minimizes the mean squared error of linear prediction:

$$\beta^{OLS} = \underset{\mathbf{b}}{\operatorname{argmin}} E[(Y_i - \mathbf{X}_i^T \mathbf{b})^2]$$

Moreover, the linear projection $\mathbf{X}_i^T \beta^{OLS}$ is the **best linear approximation to the CEF**:

$$\beta^{OLS} = \underset{\mathbf{b}}{\operatorname{argmin}} E[(E[Y_i | \mathbf{X}_i] - \mathbf{X}_i^T \mathbf{b})^2]$$

Proof of the second result: Taking the first-order condition:

$$E[\mathbf{X}_i(E[Y_i | \mathbf{X}_i] - \mathbf{X}_i^T \beta)] = \mathbf{0}$$

$$\Rightarrow \beta = E[\mathbf{X}_i \mathbf{X}_i^T]^{-1} E[\mathbf{X}_i E[Y_i | \mathbf{X}_i]]$$

By the CEF decomposition, $E[\mathbf{X}_i E[Y_i | \mathbf{X}_i]] = E[\mathbf{X}_i Y_i] - E[\mathbf{X}_i \epsilon_i] = E[\mathbf{X}_i Y_i]$.

Therefore: $\beta = E[\mathbf{X}_i \mathbf{X}_i^T]^{-1} E[\mathbf{X}_i Y_i] = \beta^{OLS}$. $\square$

Key Takeaway: OLS provides the best linear fit to the data, regardless of whether the CEF is linear or not. When the CEF is not**

linear, the OLS regression is the best *linear* approximation to the CEF — it fits a straight line through a potentially curved cloud as well as any straight line can.

2.7.3 2.5.3 Visualizing the Linear Approximation

Consider the education-wage CEF. The true CEF may curve (with higher returns to college years than to early secondary years). But the OLS regression line through the scatter plot is the straight line that minimizes the sum of squared vertical distances to all data points. This line is the best linear summary of the relationship.

For many purposes — especially for computing conditional means at typical values of the covariates — the linear approximation is excellent. For prediction at extreme values or for studying non-linearities, we might prefer more flexible methods.

2.8 2.6 OLS as the Best Linear Approximation to the CEF

Let us be more explicit about the **population linear projection model**.

Definition 2.2 (Linear Projection / Population OLS): The linear projection of Y_i on $\mathbf{X}_i$ is:

$$L(Y_i | \mathbf{X}_i) = \mathbf{X}_i^T \boldsymbol{\beta}^{OLS}$$

where $\boldsymbol{\beta}^{OLS}$ solves:

$$\boldsymbol{\beta}^{OLS} = \left(E[\mathbf{X}_i \mathbf{X}_i^T] \right)^{-1} E[\mathbf{X}_i Y_i]$$

This is the population parameter vector that OLS estimates from a sample. The sample OLS estimator (Chapter 3) converges to this population quantity as sample size grows.

Interpreting the Components: Consider the simple bivariate case with $\mathbf{X}_i = (1, X_i)^T$ and $\boldsymbol{\beta} = (\alpha, \beta)^T$. Then:

$$\beta^{OLS} = \frac{\text{Cov}(X_i, Y_i)}{V(X_i)}$$

This is the **population regression slope**: the ratio of the covariance of X and Y to the variance of X . It equals the change in the conditional mean of Y for a one-unit change in X , under the linear approximation.

2.8.1 2.6.1 The Linear Regression Equation

Given β^{OLS} , we can always write:

$$Y_i = \mathbf{X}_i^T \beta^{OLS} + \underbrace{e_i}_{\text{Population OLS residual}}$$

where $e_i = Y_i - \mathbf{X}_i^T \beta^{OLS}$ is the **population OLS residual** (not the CEF residual ϵ_i , though they coincide when the CEF is linear). By construction:

$$E[\mathbf{X}_i e_i] = \mathbf{0}$$

This **zero covariance condition** (or **orthogonality condition**) is the key property of the OLS estimator — it is weaker than strict mean independence $E[e_i | \mathbf{X}_i] = 0$.

2.8.2 2.6.2 When Does Regression Identify Causal Effects?

Regression identifies the CEF (up to a linear approximation). But the CEF is not necessarily a causal relationship. Whether the regression coefficient β has a causal interpretation depends on *why* $E[Y_i | X_i = x]$ varies with x .

Three distinct reasons for $E[Y_i | X_i = x]$ to vary with x :

1. **Causality:** changing X for a given individual changes their Y .
2. **Selection:** people with higher X would have had higher Y even without the higher X (like the wage-education example with ability selection).
3. **Reverse causality:** Y causes X (e.g., healthier people exercise more).

OLS recovers the CEF regardless of the reason. The **CEF is causal** only if X is randomly assigned or if we can credibly argue that, conditional on $\mathbf{X}$, the treatment variable of interest is "as good as randomly assigned." This is the identification problem we discussed in Chapter 1.

2.9 2.7 Empirical Illustration: Education and Wages

Let us apply these concepts to a classic economic question: how does schooling affect wages?

2.9.1 2.7.1 Data and Setting

We use data from the Current Population Survey (CPS), a representative survey of the US labour market. The outcome variable Y_i is the log of weekly earnings; the regressor X_i is years of completed schooling.

2.9.2 2.7.2 The Wage-Schooling CEF

When we calculate $E[\log(\text{wage}) \mid \text{schooling} = s]$ for each value of $s = 0, 1, \dots, 20$, we trace out the wage-schooling CEF. Key features:

- The CEF is approximately **linear** over most of the schooling range.
- There is a notable **kink** at 12 years (high school completion) and at 16 years (college completion) — the "sheepskin" effects.
- The overall slope is approximately **0.08–0.11** in OLS regressions, implying each additional year of schooling is associated with an 8–11% increase in wages.

2.9.3 2.7.3 The Linear Regression Approximation

The OLS regression line:

$$\widehat{\log(\text{wage})}_i = \hat{\alpha} + \hat{\beta} \times \text{schooling}_i$$

provides the best linear fit to the wage-schooling scatter. The slope $\hat{\beta} \approx 0.10$ says: on average, each additional year of schooling is associated with a 10% increase in weekly wages.

Is this causal? Not necessarily. People who stay in school longer differ in many ways — cognitive ability, family background, patience — from those who leave school earlier. The OLS regression does not distinguish between the causal effect of schooling and these selection differences. We return to this in Chapters 6 (omitted variable bias) and 8 (instrumental variables).

2.10 Chapter Summary

- The **Conditional Expectation Function (CEF)** $E[Y_i \mid \mathbf{X}_i]$ is the best predictor of Y_i given $\mathbf{X}_i$ in the mean-squared-error sense.
- The **CEF Decomposition** writes $Y_i = E[Y_i \mid \mathbf{X}_i] + \epsilon_i$, where ϵ_i has zero mean conditional on $\mathbf{X}_i$.
- The **Law of Iterated Expectations** relates the unconditional mean to an average of conditional means.
- The **ANOVA Theorem** decomposes total variance into explained and unexplained components.
- The OLS population coefficient vector β^{OLS} equals the CEF parameter when the CEF is linear.

- When the CEF is non-linear, OLS still provides the **best linear approximation** to the CEF.
 - Whether the OLS regression has a **causal interpretation** depends on the research design, not the statistical method.
-

2.11 Key Terms

Conditional Expectation Function (CEF) — the function $E[Y_i | \mathbf{X}_i = \mathbf{x}]$ giving the expected value of Y for each value of $\mathbf{X}$.

Law of Iterated Expectations (LIE) — $E[Y_i] = E[E[Y_i | \mathbf{X}_i]]$.

CEF Decomposition — $Y_i = E[Y_i | \mathbf{X}_i] + \epsilon_i$ where $E[\epsilon_i | \mathbf{X}_i] = 0$.

Mean Independence — the condition $E[\epsilon_i | \mathbf{X}_i] = 0$.

ANOVA Theorem — $V(Y_i) = V(E[Y_i | \mathbf{X}_i]) + E[V(Y_i | \mathbf{X}_i)]$.

Best linear approximation — the linear function of $\mathbf{X}_i$ that is closest to the CEF in the mean-squared-error sense.

Population OLS parameter — $\beta^{OLS} = (E[\mathbf{X}_i \mathbf{X}_i^T])^{-1} E[\mathbf{X}_i Y_i]$.

Linear projection — $L(Y_i | \mathbf{X}_i) = \mathbf{X}_i^T \beta^{OLS}$.

Orthogonality condition — $E[\mathbf{X}_i e_i] = \mathbf{0}$, where e_i is the population OLS residual.

2.12 Exercises

2.12.1 Conceptual Questions

2.1 Explain in plain language what the CEF $E[Y_i | X_i = x]$ means. Give an example from economics where the CEF is likely non-linear.

2.2 State the Law of Iterated Expectations. Give an example where you would use it.

2.3 Is it possible for $E[Y_i | X_i = x]$ to be decreasing in x even though the population regression coefficient β^{OLS} is positive? Give an example or prove impossibility.

2.4 Explain the difference between the CEF residual $\epsilon_i = Y_i - E[Y_i | \mathbf{X}_i]$ and the population OLS residual $e_i = Y_i - \mathbf{X}_i^T \beta^{OLS}$. When are they the same?

2.5 "OLS regression always identifies the causal effect of X on Y ." Is this statement true or false? Explain.

2.12.2 Analytical Questions

2.6 Let $Y_i \in \{0, 1\}$ (binary outcome) and $X_i \in \{0, 1\}$ (binary treatment), with joint distribution:

	$X_i = 0$	$X_i = 1$
$Y_i = 0$	0.40	0.10
$Y_i = 1$	0.20	0.30

(a) Compute $E[Y_i]$, $E[Y_i | X_i = 0]$, and $E[Y_i | X_i = 1]$. (b) Compute β^{OLS} from the bivariate regression of Y_i on X_i using $\beta^{OLS} = \text{Cov}(X, Y)/V(X)$. (c) Interpret the slope coefficient. Does it have a causal interpretation here?

2.7 Using the ANOVA theorem, show that the population $R^2 = V(E[Y_i | \mathbf{X}_i])/V(Y_i)$ lies between 0 and 1.

2.8 Suppose $E[Y_i | X_i] = \alpha + \beta X_i + \gamma X_i^2$ (a quadratic CEF). You run the OLS regression of Y_i on X_i only (without the quadratic term). Show that the OLS estimate of the slope will generally not equal β , and derive the formula for the bias in terms of γ and the moments of X_i .

2.9 Prove Corollary 2.1: if $\epsilon_i = Y_i - E[Y_i | \mathbf{X}_i]$, then $E[h(\mathbf{X}_i)\epsilon_i] = 0$ for any function $h(\cdot)$.

2.10 Consider the regression of Y_i on $\mathbf{X}_i$. Show that adding a variable Z_i to the regression does not change the population OLS coefficient on X_{1i} if and only if Z_i is uncorrelated with X_{1i} after partialling out the other regressors.

2.12.3 Applied Questions

2.11 Describe how you would construct the wage-schooling CEF from survey data. What would you do for each value of schooling from 0 to 20?

2.12 The CEF of health status on income slopes upward: people with higher income have better health on average. List at least three reasons this positive association might hold, and classify each as: (a) causal (income causes health), (b) reverse causality (health causes income), (c) confounding.

2.13 In your own words, explain why the fact that "OLS provides the best linear approximation to the CEF" is a useful result even when the CEF is non-linear.

2.13 R Lab 2: Estimating the CEF and OLS Regression

2.13.1 Objectives

- Compute the empirical CEF from data
- Compare the CEF to OLS regression
- Visualize the relationship between a continuous variable and an outcome

2.13.2 Setup

```
1 library(tidyverse)
2 # install.packages("quantreg")
3 # install.packages("AER")
4 library(AER) # contains CPS1988 data
5
6 # Load CPS data on wages and education
7 data("CPS1988")
8 head(CPS1988)
```

2.13.3 Exercise 2.1: Exploring the Data

```
1 # Summary statistics
2 summary(CPS1988[, c("wage", "education", "experience", "ethnicity")])
3
4 # Log wages
5 CPS1988 <- CPS1988 %>%
6   mutate(log_wage = log(wage))
7
8 # Distribution of education
9 ggplot(CPS1988, aes(x = education)) +
10   geom_histogram(bins = 16, fill = "steelblue", color = "white") +
11   labs(title = "Distribution of Years of Schooling", x = "Years of
12     Education", y = "Count") +
13   theme_minimal()
```

2.13.4 Exercise 2.2: Estimating the Empirical CEF

```
1 # Compute the empirical CEF: mean log wage for each education level
2 cef_data <- CPS1988 %>%
3   group_by(education) %>%
4   summarise(
5     mean_log_wage = mean(log_wage),
6     median_log_wage = median(log_wage),
7     n = n(),
8     se_log_wage = sd(log_wage) / sqrt(n)
9   )
10
11 print(cef_data)
12
13 # Plot the empirical CEF with confidence intervals
14 ggplot(cef_data, aes(x = education, y = mean_log_wage)) +
15   geom_point(aes(size = n), color = "steelblue") +
16   geom_line(color = "steelblue") +
17   geom_ribbon(aes(ymin = mean_log_wage - 1.96 * se_log_wage,
18     ymax = mean_log_wage + 1.96 * se_log_wage),
19     alpha = 0.2, fill = "steelblue") +
```

```
20 labs(  
21   title = "Empirical CEF: Mean Log Wage by Years of Education",  
22   subtitle = "The CEF shows a generally increasing, approximately  
      linear relationship",  
23   x = "Years of Education",  
24   y = "Mean Log Wage",  
25   size = "Sample Size"  
26 ) +  
27 theme_minimal()
```

2.13.5 Exercise 2.3: OLS Regression vs. the CEF

```
1 # Run OLS regression  
2 ols_model <- lm(log_wage ~ education, data = CPS1988)  
3 summary(ols_model)  
4  
5 # Add OLS fitted line to the CEF plot  
6 ggplot(cef_data, aes(x = education, y = mean_log_wage)) +  
7   geom_point(color = "steelblue", size = 3) +  
8   geom_line(color = "steelblue", linetype = "dashed") +  
9   geom_smooth(  
10     data = CPS1988,  
11     aes(x = education, y = log_wage),  
12     method = "lm",  
13     se = TRUE,  
14     color = "red"  
15 ) +  
16 labs(  
17   title = "Empirical CEF vs. OLS Regression Line",  
18   subtitle = "Blue: Empirical CEF; Red: OLS best linear fit",  
19   x = "Years of Education",  
20   y = "Log Wage"  
21 ) +  
22 theme_minimal()
```

Questions:

1. Is the empirical CEF approximately linear?
2. Where does it deviate most from linearity? (Hint: look at 12 and 16 years of schooling.)
3. How does the slope of the OLS regression compare to the slope of the CEF in the linear range?

2.13.6 Exercise 2.4: Residuals and the ANOVA Decomposition

```
1 # Compute OLS residuals  
2 CPS1988 <- CPS1988 %>%
```

```

3   mutate(
4     fitted_ols = predict(ols_model),
5     resid_ols  = residuals(ols_model)
6   )
7
8   # ANOVA decomposition
9   TSS <- var(CPS1988$log_wage)
10  ESS <- var(CPS1988$fitted_ols)
11  SSR_mean <- mean(CPS1988$resid_ols^2)
12
13  cat("Total Variance (TSS equivalent):", round(TSS, 4), "\n")
14  cat("Explained Variance (ESS):", round(ESS, 4), "\n")
15  cat("Average Residual Variance:", round(SSR_mean, 4), "\n")
16  cat("R-squared:", round(summary(ols_model)$r.squared, 4), "\n")
17  cat("Check (ESS/TSS):", round(ESS/TSS, 4), "\n")
18
19  # Does  $E[X * resid] = 0$ ?
20  cat("Cov(education, residuals):", round(cov(CPS1988$education,
      CPS1988$resid_ols), 6), "\n")

```

2.13.7 Exercise 2.5: Multiple Regression CEF

```

1   # Multiple regression: education + experience
2   ols_multi <- lm(log_wage ~ education + experience + I(experience^2),
3     data = CPS1988)
4   summary(ols_multi)
5
6   # Partial effect of education (controlling for experience)
7   # Using the FWL theorem (preview):
8   # Regress education on experience, get residuals
9   educ_resid <- residuals(lm(education ~ experience + I(experience^2),
10     data = CPS1988))
11   wage_resid <- residuals(lm(log_wage ~ experience +
12     I(experience^2), data = CPS1988))
13   fwl_model <- lm(wage_resid ~ educ_resid)
14   cat("FWL estimate of education coefficient:",
15     round(coef(fwl_model)[2], 4), "\n")
16   cat("Multiple regression estimate:",
17     round(coef(ols_multi)["education"], 4), "\n")

```

Interpretation: The two estimates should be identical, illustrating the Frisch-Waugh-Lovell Theorem (covered in Chapter 6).

End of Chapter 2

References

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Chapter 3

Chapter 3: Ordinary Least Squares — Algebra and Geometry

3.1 Chapter Overview

This chapter develops the OLS estimator from the ground up. We begin with the multiple regression model written in matrix notation, derive the OLS estimator through two routes (method-of-moments and minimization), and then provide a geometric interpretation using linear algebra. This geometric perspective is not merely elegant — it reveals deep structural properties of OLS that are impossible to see from the algebraic derivation alone. We end with the Frisch-Waugh-Lovell insight, goodness-of-fit measures, and invariance properties of OLS.

Throughout this chapter, we work at the **sample** level: we have n observations and we are fitting an estimator to data, not analyzing a population parameter.

3.2 Learning Objectives

After completing this chapter, students will be able to:

1. Write the multiple regression model in **matrix notation**.
2. Derive the **OLS estimator** using the method-of-moments and the least-squares criterion.
3. State the conditions under which $(\mathbf{X}^T \mathbf{X})$ is invertible (full rank condition).
4. Interpret OLS geometrically as an **orthogonal projection** onto the column space of $\mathbf{X}$.
5. Define the **projection matrices** $\mathbf{P}_\mathbf{X}$ and $\mathbf{M}_\mathbf{X}$ and state their key properties.
6. State and apply the **Pythagorean decomposition** of sum of squares.
7. Compute and interpret R^2 (centered and uncentered).
8. State and apply the **Frisch-Waugh-Lovell (FWL) Theorem**.

9. Discuss how OLS coefficients change with linear transformations of regressors.

3.3 3.1 The Multiple Regression Model

We observe n units (individuals, firms, countries), each described by:

- an **outcome variable** $y_i \in \mathbb{R}$
- a $k \times 1$ vector of **regressors** (including a constant in the first position): $\mathbf{x}_i = (1, x_{i2}, \dots, x_{ik})^T$

The multiple regression equation for unit i is:

$$y_i = \beta_1 + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + u_i = \mathbf{x}_i^T \boldsymbol{\beta} + u_i$$

where $\boldsymbol{\beta} = (\beta_1, \beta_2, \dots, \beta_k)^T$ is the $k \times 1$ vector of **coefficients** and u_i is the **error term**.

3.3.1 3.1.1 Matrix Notation

Stack all n observations:

$$\underset{n \times 1}{\mathbf{y}} = \underset{n \times k}{\mathbf{X}} \underset{k \times 1}{\boldsymbol{\beta}} + \underset{n \times 1}{\mathbf{u}}$$

where:

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_n^T \end{pmatrix} = \begin{pmatrix} 1 & x_{12} & \cdots & x_{1k} \\ 1 & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n2} & \cdots & x_{nk} \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \vdots \\ \beta_k \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}$$

The matrix $\mathbf{X}$ is called the **design matrix**. Each row is an observation; each column is a regressor.

Convention:** We use bold lowercase letters for vectors ($\mathbf{y}$, $\mathbf{u}$) and bold uppercase letters for matrices ($\mathbf{X}$, $\mathbf{P}$, $\mathbf{M}$). Scalars are unbolded.

Example 3.1 (Wage Regression): In a regression of log wage on years of education and years of experience:

$$\log(\text{wage}_i) = \beta_1 + \beta_2 \cdot \text{educ}_i + \beta_3 \cdot \text{exp}_i + u_i$$

Here n is the number of workers, $k = 3$, and the design matrix $\mathbf{X}$ has three columns: a column of ones, a column of education values, and a column of experience values.

3.4 3.2 The OLS Estimator: Derivation via Method of Moments

The **Method of Moments (MM)** estimator is derived by finding $\hat{\beta}$ such that the **sample moment conditions** are satisfied.

From the population orthogonality condition $E[\mathbf{x}_i u_i] = \mathbf{0}$, the sample analogue is:

$$\frac{1}{n} \sum_{i=1}^n \mathbf{x}_i (y_i - \mathbf{x}_i^T \hat{\beta}) = \mathbf{0}$$

In matrix notation:

$$\mathbf{X}^T (\mathbf{y} - \mathbf{X} \hat{\beta}) = \mathbf{0}$$

These are the k **Normal Equations** (one for each regressor):

$$\mathbf{X}^T \mathbf{X} \hat{\beta} = \mathbf{X}^T \mathbf{y}$$

If $\mathbf{X}^T \mathbf{X}$ is **invertible** (i.e., $\mathbf{X}$ has full column rank k), we can solve:

$$\boxed{\hat{\beta}^{OLS} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}}$$

This is the **OLS estimator** — the central formula of classical econometrics.

3.4.1 3.2.1 The Full Rank Condition

The matrix $\mathbf{X}$ has **full column rank** (rank k) if and only if no column of $\mathbf{X}$ can be written as a linear combination of the other columns. Violations include:

1. **Perfect multicollinearity:** two regressors are proportional or one is an exact linear combination of others.
2. **Dummy variable trap:** including a dummy for every category while also including an intercept.
3. **Too many variables relative to observations:** $k > n$.

Example 3.2: If we include both a male dummy and a female dummy (along with an intercept), then (male dummy) + (female dummy) = (column of ones). The three columns are linearly dependent; $\mathbf{X}^T \mathbf{X}$ is singular and the OLS estimator does not exist.

3.5 3.3 OLS Derivation via Minimization

Alternatively, define the **sum of squared residuals (SSR)**:

$$SSR(\hat{\beta}) = (\mathbf{y} - \mathbf{X}\hat{\beta})^T (\mathbf{y} - \mathbf{X}\hat{\beta}) = \sum_{i=1}^n (y_i - \mathbf{x}_i^T \hat{\beta})^2 = \|\mathbf{y} - \mathbf{X}\hat{\beta}\|^2$$

The **Ordinary Least Squares** estimator minimizes the SSR:

$$\hat{\beta}^{OLS} = \underset{\hat{\beta}}{\operatorname{argmin}} \|\mathbf{y} - \mathbf{X}\hat{\beta}\|^2$$

Derivation: Expand the objective function:

$$SSR = \mathbf{y}^T \mathbf{y} - 2\hat{\beta}^T \mathbf{X}^T \mathbf{y} + \hat{\beta}^T \mathbf{X}^T \mathbf{X} \hat{\beta}$$

Taking the derivative with respect to $\hat{\beta}$ and setting to zero:

$$\frac{\partial SSR}{\partial \hat{\beta}} = -2\mathbf{X}^T \mathbf{y} + 2\mathbf{X}^T \mathbf{X} \hat{\beta} = \mathbf{0}$$

This gives exactly the normal equations. The second-order condition requires $\mathbf{X}^T \mathbf{X}$ to be positive definite, which holds when $\mathbf{X}$ has full rank. $\square$

The equivalence of MM and OLS follows because the moment conditions and the first-order conditions of the least-squares problem are identical. This is a special property of linear models.**

3.6 3.4 Linear Regression as Orthogonal Projection

The geometric interpretation of OLS is the most insightful way to understand what the estimator is doing. This section requires thinking of vectors and matrices in $\mathbb{R}^n$.

3.6.1 3.4.1 Vectors in $\mathbb{R}^n$

Think of $\mathbf{y} = (y_1, \dots, y_n)^T$ as a single point (or arrow from the origin) in n -dimensional Euclidean space E^n . Similarly, each column of $\mathbf{X}$ is a vector in E^n .

The **scalar product** (inner product) of two vectors $\mathbf{a}, \mathbf{b} \in E^n$ is:

$$\langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a}^T \mathbf{b} = \sum_{i=1}^n a_i b_i$$

The **length** (norm) of a vector is:

$$\|\mathbf{a}\| = \sqrt{\langle \mathbf{a}, \mathbf{a} \rangle} = \sqrt{\sum_{i=1}^n a_i^2}$$

Two vectors are **orthogonal** if $\langle \mathbf{a}, \mathbf{b} \rangle = 0$. The **Cauchy-Schwarz inequality** states: $|\langle \mathbf{a}, \mathbf{b} \rangle| \leq \|\mathbf{a}\| \|\mathbf{b}\|$.

3.6.2 3.4.2 The Column Space of $\mathbf{X}$

The **column space** of $\mathbf{X}$, denoted $\mathcal{S}(\mathbf{X})$, is the set of all vectors that can be written as a linear combination of the columns of $\mathbf{X}$:

$$\mathcal{S}(\mathbf{X}) = \left\{ \mathbf{z} \in E^n : \mathbf{z} = \mathbf{X}\mathbf{b} \text{ for some } \mathbf{b} \in \mathbb{R}^k \right\}$$

When $\mathbf{X}$ has rank k , $\mathcal{S}(\mathbf{X})$ is a k -dimensional **subspace** of E^n .

Key insight: The fitted values $\mathbf{X}\hat{\boldsymbol{\beta}}$ lie in $\mathcal{S}(\mathbf{X})$ for *any* choice of $\hat{\boldsymbol{\beta}}$.

3.6.3 3.4.3 OLS as Closest Point in the Column Space

The OLS problem asks: **which point in $\mathcal{S}(\mathbf{X})$ is closest to $\mathbf{y}$?**

Formally: find $\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} \in \mathcal{S}(\mathbf{X})$ that minimizes $\|\mathbf{y} - \hat{\mathbf{y}}\|^2$.

The answer is the orthogonal projection of $\mathbf{y}$ onto $\mathcal{S}(\mathbf{X})$.

This is the point in $\mathcal{S}(\mathbf{X})$ such that the residual vector $\hat{\mathbf{u}} = \mathbf{y} - \hat{\mathbf{y}}$ is **orthogonal** to every vector in $\mathcal{S}(\mathbf{X})$:

$$\mathbf{X}^T \hat{\mathbf{u}} = \mathbf{0}$$

This is exactly the normal equation, confirming the geometric interpretation.

Figure 3.1 (Geometric Interpretation): Imagine $n = 3$ observations and $k = 2$ regressors (including the constant). The column space $\mathcal{S}(\mathbf{X})$ is a 2-dimensional plane in 3-dimensional space. The vector $\mathbf{y}$ is a point in 3D space, generally not lying in the plane. The OLS fitted value $\hat{\mathbf{y}}$ is the point in the plane closest to $\mathbf{y}$, obtained by dropping a perpendicular from $\mathbf{y}$ onto the plane. The residual vector $\hat{\mathbf{u}} = \mathbf{y} - \hat{\mathbf{y}}$ is the perpendicular.

3.7 3.5 Projection Matrices: $\mathbf{P}_\mathbf{X}$ and $\mathbf{M}_\mathbf{X}$

3.7.1 3.5.1 The Hat Matrix $\mathbf{P}_\mathbf{X}$

The **orthogonal projection matrix** onto $\mathcal{S}(\mathbf{X})$ is:

$$\mathbf{P}_\mathbf{X} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$$

This is also called the **hat matrix** (because it "puts a hat on" $\mathbf{y}$):

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \mathbf{P}_\mathbf{X} \mathbf{y}$$

3.7.2 3.5.2 The Annihilator Matrix $\mathbf{M}_\mathbf{X}$

The **orthogonal projection onto the orthogonal complement** of $\mathcal{S}(\mathbf{X})$ is:

$$\mathbf{M}_\mathbf{X} = \mathbf{I}_n - \mathbf{P}_\mathbf{X}$$

This matrix produces the OLS residuals:

$$\hat{\mathbf{u}} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{P}_\mathbf{X} \mathbf{y} = (\mathbf{I} - \mathbf{P}_\mathbf{X}) \mathbf{y} = \mathbf{M}_\mathbf{X} \mathbf{y}$$

$\mathbf{M}_\mathbf{X}$ is called the **annihilator** because it destroys (annihilates) any vector in $\mathcal{S}(\mathbf{X})$: $\mathbf{M}_\mathbf{X} \mathbf{X} = \mathbf{0}$.

3.7.3 3.5.3 Key Properties of $\mathbf{P}_\mathbf{X}$ and $\mathbf{M}_\mathbf{X}$

Property 3.1 (Symmetry): $\mathbf{P}_\mathbf{X}^T = \mathbf{P}_\mathbf{X}$ and $\mathbf{M}_\mathbf{X}^T = \mathbf{M}_\mathbf{X}$.

Property 3.2 (Idempotency): $\mathbf{P}_\mathbf{X}^2 = \mathbf{P}_\mathbf{X}$ and $\mathbf{M}_\mathbf{X}^2 = \mathbf{M}_\mathbf{X}$.

Proof: $\mathbf{P}_\mathbf{X}^2 = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \cdot \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} (\mathbf{X}^T \mathbf{X}) (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T = \mathbf{P}_\mathbf{X}$. $\square$

Interpretation: Projecting a vector twice onto the same subspace returns it to where it was after the first projection. Idempotency is the algebraic expression of this geometric fact.

Property 3.3 (Annihilation): $\mathbf{P}_X \mathbf{M}_X = \mathbf{M}_X \mathbf{P}_X = \mathbf{0}$.

Proof: $\mathbf{P}_X \mathbf{M}_X = \mathbf{P}_X (\mathbf{I} - \mathbf{P}_X) = \mathbf{P}_X - \mathbf{P}_X^2 = \mathbf{P}_X - \mathbf{P}_X = \mathbf{0}$. $\square$

Interpretation: The subspace $\mathcal{S}(\mathbf{X})$ and its orthogonal complement are perpendicular; projecting onto one annihilates any component in the other.

Property 3.4 (Trace): $\text{tr}(\mathbf{P}_X) = k$ and $\text{tr}(\mathbf{M}_X) = n - k$.

Proof: $\text{tr}(\mathbf{P}_X) = \text{tr}(\mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T) = \text{tr}((\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X}) = \text{tr}(\mathbf{I}_k) = k$, using the cyclic property of trace. $\square$

Interpretation: The trace of a projection matrix equals the dimension of the subspace it projects onto. The hat matrix projects onto a k -dimensional space; the annihilator projects onto an $(n - k)$ -dimensional space.

Property 3.5 (Invariance of projection): $\mathbf{P}_X \mathbf{X} = \mathbf{X}$.

Proof: Any vector in $\mathcal{S}(\mathbf{X})$ maps to itself under the projection. $\mathbf{X}$ is in $\mathcal{S}(\mathbf{X})$ (each column is). $\square$

Property 3.6 (OLS residuals and regressors are orthogonal): $\mathbf{X}^T \hat{\mathbf{u}} = \mathbf{0}$.

Proof: $\mathbf{X}^T \hat{\mathbf{u}} = \mathbf{X}^T \mathbf{M}_X \mathbf{y} = (\mathbf{M}_X \mathbf{X})^T \mathbf{y} = \mathbf{0}^T \mathbf{y} = \mathbf{0}$. $\square$

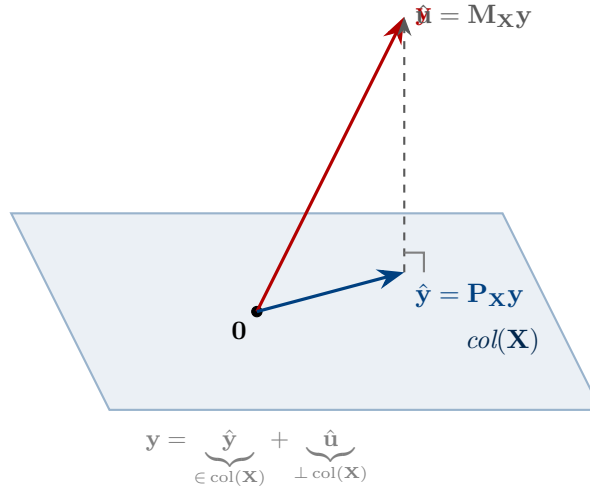


Figure 3.1: The geometry of OLS. The OLS estimator projects $\mathbf{y}$ orthogonally onto the column space of $\mathbf{X}$. The fitted values $\hat{\mathbf{y}} = \mathbf{P}_X \mathbf{y}$ lie in $\text{col}(\mathbf{X})$, and the residual vector $\hat{\mathbf{u}} = \mathbf{M}_X \mathbf{y}$ is perpendicular to $\text{col}(\mathbf{X})$ (i.e., $\mathbf{X}^T \hat{\mathbf{u}} = \mathbf{0}$).

3.8 3.6 The Pythagorean Theorem and Goodness of Fit (R^2)

3.8.1 3.6.1 Sum of Squares Decomposition

By the geometry of projections (Pythagorean theorem in E^n):

$$\|\mathbf{y}\|^2 = \|\mathbf{P}_{\mathbf{X}}\mathbf{y}\|^2 + \|\mathbf{M}_{\mathbf{X}}\mathbf{y}\|^2$$

In more familiar notation (when an intercept is included in $\mathbf{X}$):

$$\underbrace{\|\mathbf{y} - \bar{y}\mathbf{1}\|^2}_{TSS} = \underbrace{\|\hat{\mathbf{y}} - \bar{y}\mathbf{1}\|^2}_{ESS} + \underbrace{\|\hat{\mathbf{u}}\|^2}_{SSR}$$

where:

- $TSS = \sum_{i=1}^n (y_i - \bar{y})^2$ is the **Total Sum of Squares**
- $ESS = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$ is the **Explained Sum of Squares**
- $SSR = \sum_{i=1}^n \hat{u}_i^2$ is the **Sum of Squared Residuals**

Proof that $TSS = ESS + SSR$: Write $\mathbf{y} - \bar{y}\mathbf{1} = (\hat{\mathbf{y}} - \bar{y}\mathbf{1}) + \hat{\mathbf{u}}$. Squaring: $\|\mathbf{y} - \bar{y}\mathbf{1}\|^2 = \|\hat{\mathbf{y}} - \bar{y}\mathbf{1}\|^2 + \|\hat{\mathbf{u}}\|^2 + 2(\hat{\mathbf{y}} - \bar{y}\mathbf{1})^T \hat{\mathbf{u}}$. The cross term is zero because $\hat{\mathbf{u}} \perp \mathcal{S}(\mathbf{X})$ and $(\hat{\mathbf{y}} - \bar{y}\mathbf{1}) \in \mathcal{S}(\mathbf{X})$. $\square$

3.8.2 3.6.2 The Coefficient of Determination (R^2)

Definition 3.1 (Centered R^2):

$$R_c^2 = \frac{ESS}{TSS} = 1 - \frac{SSR}{TSS} = \frac{\|\mathbf{P}_{\mathbf{X}}\mathbf{M}_t\mathbf{y}\|^2}{\|\mathbf{M}_t\mathbf{y}\|^2}$$

where $\mathbf{M}_t = \mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^T$ is the demeaning matrix.

Definition 3.2 (Uncentered R^2):

$$R_u^2 = \frac{\|\mathbf{P}_{\mathbf{X}}\mathbf{y}\|^2}{\|\mathbf{y}\|^2}$$

Properties of R^2 :

1. $0 \leq R^2 \leq 1$ (when an intercept is included).
2. R^2 is the square of the sample correlation between $\hat{y}_i$ and y_i .
3. Adding more regressors never decreases R^2 (but may be pure noise fitting).
4. R^2 measures **in-sample fit**, not predictive accuracy or causal validity.

Warning:** A high R^2 does not mean the regression is correctly specified or that the coefficients are causally interpretable. A low R^2 does not mean the regression is useless — in many empirical settings, including program evaluation, R^2 values of 0.05–0.20 are common even when the estimates are valid.

Definition 3.3 (Adjusted R^2): To penalize for the number of regressors:

$$\bar{R}^2 = 1 - \frac{SSR/(n-k)}{TSS/(n-1)} = 1 - (1 - R^2) \frac{n-1}{n-k}$$

The adjusted R^2 can decrease when an irrelevant variable is added.

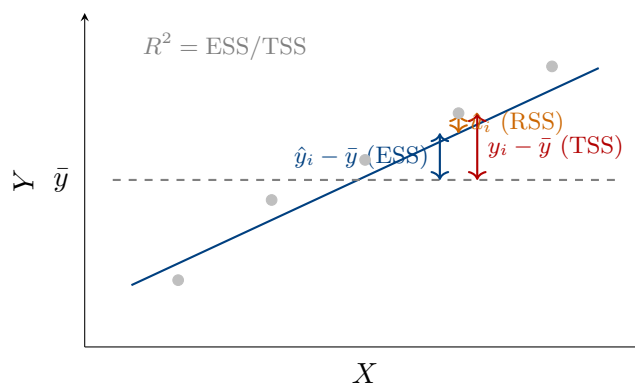


Figure 3.2: Decomposition of variation in the OLS model. For each observation, the total deviation from the mean $y_i - \bar{y}$ (TSS component) is split into the explained part $\hat{y}_i - \bar{y}$ (ESS) and the residual $\hat{u}_i$ (RSS). $R^2 = ESS/TSS$ measures the share of variance explained.

3.9 3.7 Linear Transformations and Invariance Results

3.9.1 3.7.1 Rescaling Regressors

If we replace $\mathbf{X}$ with $\mathbf{Z} = \mathbf{X}\mathbf{A}$ where $\mathbf{A}$ is a $k \times k$ full-rank matrix, then the OLS fitted values are unchanged:

$$\mathbf{P}_Z = \mathbf{P}_X$$

and the coefficient estimates satisfy:

$$\mathbf{A}\hat{\alpha} = \hat{\beta}$$

where $\hat{\alpha}$ is the OLS estimate from regressing $\mathbf{y}$ on $\mathbf{Z}$.

Implication: OLS coefficients change predictably when units change. If we measure income in \$000 instead of dollars, the coefficient on income is multiplied by 1000. If we take the log of income, the coefficient changes entirely. But the fitted values, residuals, and R^2 are invariant to linear transformations.

3.9.2 3.7.2 Demeaning as a Linear Transformation

An important special case: subtracting the mean $\bar{x}$ from each regressor (demeaning).

$$Z_i = X_i - \bar{X}$$

Under this transformation:

- The intercept changes but all slope coefficients are unchanged.
- The R^2 and residuals are unchanged.
- The demeaned regression $\tilde{y}_i = \mathbf{M}_l \mathbf{y}$ on $\tilde{X}_i = \mathbf{M}_l \mathbf{X}$ gives the same slopes as the original regression.

This is a special case of the **Frisch-Waugh-Lovell (FWL) Theorem**.

3.9.3 3.7.3 The Frisch-Waugh-Lovell Theorem

This theorem is one of the most useful results in regression analysis. It shows that the coefficients on a subset of regressors can be obtained by "partialling out" the other regressors.

Theorem 3.1 (Frisch-Waugh-Lovell): Partition the design matrix as $\mathbf{X} = [\mathbf{X}_1 \mathbf{X}_2]$, where $\mathbf{X}_1$ is $n \times k_1$ and $\mathbf{X}_2$ is $n \times k_2$, with $k_1 + k_2 = k$. The OLS estimator of β_2 from the full regression $\mathbf{y} = \mathbf{X}_1 \beta_1 + \mathbf{X}_2 \beta_2 + \mathbf{u}$ equals the OLS estimator from the "residualized" regression:

$$\hat{\beta}_2^{FWL} = (\tilde{\mathbf{X}}_2^T \tilde{\mathbf{X}}_2)^{-1} \tilde{\mathbf{X}}_2^T \tilde{\mathbf{y}}$$

where $\tilde{\mathbf{y}} = \mathbf{M}_{\mathbf{X}_1} \mathbf{y}$ and $\tilde{\mathbf{X}}_2 = \mathbf{M}_{\mathbf{X}_1} \mathbf{X}_2$ are the residuals from regressing $\mathbf{y}$ and $\mathbf{X}_2$, respectively, on $\mathbf{X}_1$.

Moreover, $\hat{\beta}_2^{FWL} = \hat{\beta}_2^{OLS}$ and the residuals from both regressions are identical.

Proof sketch: Let $\tilde{\mathbf{y}} = \mathbf{M}_{\mathbf{X}_1} \mathbf{y}$ and $\tilde{\mathbf{X}}_2 = \mathbf{M}_{\mathbf{X}_1} \mathbf{X}_2$. Then $\tilde{\mathbf{X}}_2^T \tilde{\mathbf{y}} = \mathbf{X}_2^T \mathbf{M}_{\mathbf{X}_1} \mathbf{y}$ (since $\mathbf{M}_{\mathbf{X}_1}$ is symmetric). One can show by substitution that the normal equations of the FWL regression are equivalent to the normal equations of the full regression projected

onto the β_2 block. The residuals are the same because $\mathbf{M}_\mathbf{X} = \mathbf{M}_{\mathbf{X}_1} \mathbf{M}_{[\mathbf{M}_{\mathbf{X}_1} \mathbf{X}_2]}$. (See Davidson and MacKinnon (2004), Chapter 2 for a complete proof.) $\square$

Why the FWL Theorem Matters

1. **Interpreting multiple regression.** The coefficient on X_{2i} in a multiple regression is the effect of X_{2i} on y_i after "controlling for" X_{1i} — specifically, after removing the part of both y_i and X_{2i} that is linearly predictable from X_{1i} .
2. **Partial effects.** The FWL theorem shows that OLS coefficients measure *partial* effects: the marginal association between y and one regressor, holding the other regressors fixed.
3. **Omitted variable bias.** If we omit $\mathbf{X}_1$ from the regression, $\hat{\beta}_2$ will be biased if $\mathbf{X}_1$ is correlated with $\mathbf{X}_2$ (since we won't residualize $\mathbf{X}_2$ on $\mathbf{X}_1$).

Example 3.3 (FWL in the Wage Regression): In the wage regression with education and experience, the FWL theorem says:

1. Regress log wages on experience; save residuals $\tilde{y}_i$.
2. Regress education on experience; save residuals $\tilde{\text{educ}}_i$.
3. Regress $\tilde{y}_i$ on $\tilde{\text{educ}}_i$ (no intercept needed); the slope is the education coefficient from the full regression.

The slope in step 3 measures the return to education *after* removing the variation in both wages and education that is due to experience.

3.10 Chapter Summary

- The multiple regression model in matrix notation is $\mathbf{y} = \mathbf{X}\beta + \mathbf{u}$.
- The OLS estimator $\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ is derived by minimizing the SSR or equating sample moments to zero.
- OLS is geometrically an **orthogonal projection** of $\mathbf{y}$ onto the column space of $\mathbf{X}$.
- The **projection matrices** $\mathbf{P}_\mathbf{X} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$ and $\mathbf{M}_\mathbf{X} = \mathbf{I} - \mathbf{P}_\mathbf{X}$ are symmetric, idempotent, and annihilate each other.
- The **Pythagorean decomposition** $TSS = ESS + SSR$ follows from the geometry of projections.
- The R^2 measures in-sample fit as the fraction of variance explained by the regressors.

- The **FWL Theorem** shows that OLS coefficients measure partial effects after residualizing on the other regressors.
-

3.11 Key Terms

Design matrix ($\mathbf{X}$) — the $n \times k$ matrix of observations on the regressors.

Normal equations — $\mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^T \mathbf{y}$; the first-order conditions for the OLS estimator.

Full rank — $\mathbf{X}$ has full column rank k (no perfect multicollinearity).

OLS estimator — $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$.

Column space ($\mathcal{S}(\mathbf{X})$) — the set of all vectors of the form $\mathbf{X}\mathbf{b}$ for $\mathbf{b} \in \mathbb{R}^k$.

Projection matrix ($\mathbf{P}_\mathbf{X}$) — $\mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$; projects onto $\mathcal{S}(\mathbf{X})$.

Annihilator matrix ($\mathbf{M}_\mathbf{X}$) — $\mathbf{I} - \mathbf{P}_\mathbf{X}$; projects onto the orthogonal complement of $\mathcal{S}(\mathbf{X})$.

Idempotent — a matrix $\mathbf{A}$ such that $\mathbf{A}^2 = \mathbf{A}$.

Frisch-Waugh-Lovell (FWL) Theorem — the coefficient on $\mathbf{X}_2$ in the full regression equals the coefficient from regressing residualized $\mathbf{y}$ on residualized $\mathbf{X}_2$.

R^2 — fraction of total variation in y explained by $\mathbf{X}$; $R^2 = ESS/TSS$.

3.12 Exercises

3.12.1 Conceptual Questions

3.1 Explain in your own words why the OLS estimator can be interpreted as an orthogonal projection.

3.2 What is the geometric interpretation of the OLS residuals $\hat{\mathbf{u}}$? What does it mean for $\hat{\mathbf{u}}$ to be orthogonal to the columns of $\mathbf{X}$?

3.3 "If we add a variable that is completely uncorrelated with all existing regressors and the outcome variable, R^2 does not change." Is this true? Explain.

3.4 You run a regression of test scores on class size and find $R^2 = 0.02$. A colleague says the regression is worthless because R^2 is so low. How would you respond?

3.5 State the FWL Theorem. Give an intuitive explanation of why the "residualized regression" gives the same coefficient as the full regression.

3.12.2 Analytical Questions

3.6 Prove that the hat matrix $\mathbf{P}_\mathbf{X}$ is symmetric and idempotent.

3.7 Show that $\hat{\beta}$ from a regression of $\mathbf{y}$ on $\mathbf{X}$ and $\hat{\beta}$ from a regression of $\mathbf{M}_\iota \mathbf{y}$ on $\mathbf{M}_\iota \mathbf{X}$ (i.e., demeaned regression) give the same slope coefficients.

3.8 Consider the simple bivariate regression $y_i = \alpha + \beta x_i + u_i$.

(a) Show that $\hat{\beta} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$.

(b) Show that $\hat{\alpha} = \bar{y} - \hat{\beta}\bar{x}$.

(c) Show that the OLS regression line passes through the point $(\bar{x}, \bar{y})$.

3.9 Consider the regression of y_i on x_i and z_i . If x_i and z_i are orthogonal in the sample ($\sum x_i z_i = 0$ after demeaning), show that the OLS coefficient on x_i from the multiple regression equals the OLS coefficient on x_i from the simple regression of y_i on x_i alone.

3.10 Show that $\text{tr}(\mathbf{M}_\mathbf{X}) = n - k$.

3.11 Let $\mathbf{X}_1 = \iota$ (column of ones). Show that $\mathbf{M}_{\mathbf{X}_1} = \mathbf{M}_\iota$ is the demeaning matrix: $\mathbf{M}_\iota \mathbf{y} = \mathbf{y} - \bar{y}\iota$.

3.12.3 Applied Questions

3.12 You have data on $n = 500$ workers with: log wages (y), years of education (x_1), and a dummy for urban residence (x_2).

(a) Write out the full OLS estimator in matrix form. (b) Describe what the FWL theorem says about the coefficient on education in this regression. (c) If education is positively correlated with urban residence, and urban workers earn higher wages, what happens to the education coefficient if you omit the urban dummy?

3.13 Using the data from R Lab 2, verify the FWL Theorem numerically: (a) run the full regression of log wages on education and experience; (b) residualize log wages and education on experience; (c) regress the residualized wages on residualized education; (d) confirm the slopes in (a) and (c) match.

3.13 R Lab 3: OLS in Matrix Form and the Geometry of Regression

3.13.1 Objectives

- Compute OLS directly using matrix algebra
- Construct and verify projection matrices
- Verify the FWL theorem numerically
- Visualize the sum-of-squares decomposition

3.13.2 Setup

```
1 library(tidyverse)
2 library(AER)
3 data("CPS1988")
4 CPS1988 <- CPS1988 %>% mutate(log_wage = log(wage))
```

3.13.3 Exercise 3.1: OLS via Matrix Algebra

```
1 # Create the design matrix
2 y <- as.matrix(CPS1988$log_wage)
3 X <- cbind(1, CPS1988$education, CPS1988$experience,
4           CPS1988$experience^2)
5 colnames(X) <- c("intercept", "education", "experience",
6                 "experience_sq")
7
8 n <- nrow(X)
9 k <- ncol(X)
10
11 # OLS estimator:  $\hat{\beta} = (X'X)^{-1} X'y$ 
12 XtX <- t(X) %*% X
13 Xty <- t(X) %*% y
14 beta_hat <- solve(XtX) %*% Xty
15 print(round(beta_hat, 4))
16
17 # Compare with lm()
18 lm_model <- lm(log_wage ~ education + experience + I(experience^2),
19               data = CPS1988)
20 print(round(coef(lm_model), 4))
```

3.13.4 Exercise 3.2: Projection Matrices

```
1 # Hat matrix  $P_X$ 
2 P_X <- X %*% solve(XtX) %*% t(X)
3
4 # Annihilator  $M_X$ 
5 M_X <- diag(n) - P_X
6
7 # Verify properties
8 cat("Is P_X idempotent?", all.equal(P_X %*% P_X, P_X,
9   check.attributes = FALSE), "\n")
10 cat("Is M_X idempotent?", all.equal(M_X %*% M_X, M_X,
11   check.attributes = FALSE), "\n")
12
13 # Fitted values and residuals
14 y_hat <- P_X %*% y # fitted values
15 u_hat <- M_X %*% y # residuals
```

```

15 # Check:  $X'u_{\hat{}} = 0$  (should be numerically zero)
16 cat("Max |X'u_hat|:", max(abs(t(X) %*% u_hat)), "\n")
17
18 # Trace of projection matrices
19 cat("tr(P_X) =", round(sum(diag(P_X)), 2), "(should be", k, ")\n")
20 cat("tr(M_X) =", round(sum(diag(M_X)), 2), "(should be", n - k,
    "\n")

```

3.13.5 Exercise 3.3: Sum of Squares Decomposition

```

1 # Demeaning matrix
2 iota <- matrix(1, n, 1)
3 M_iota <- diag(n) - (1/n) * (iota %*% t(iota))
4
5 TSS <- as.numeric(t(y) %*% M_iota %*% y) # Total SS (demeaned)
6 ESS <- as.numeric(t(y_hat) %*% M_iota %*% y_hat) # Explained SS
7 SSR <- as.numeric(t(u_hat) %*% u_hat) # Residual SS
8
9 cat("TSS:", round(TSS, 4), "\n")
10 cat("ESS:", round(ESS, 4), "\n")
11 cat("SSR:", round(SSR, 4), "\n")
12 cat("ESS + SSR:", round(ESS + SSR, 4), "(should equal TSS)\n")
13
14 R_squared <- ESS / TSS
15 cat("R-squared:", round(R_squared, 4), "\n")
16 cat("R-squared from lm():", round(summary(lm_model)$r.squared, 4),
    "\n")

```

3.13.6 Exercise 3.4: Verifying the FWL Theorem

```

1 # Full regression: log_wage ~ education + experience + experience^2
2 # FWL for the education coefficient
3
4 # Step 1: Partition X1 (intercept + experience) and X2 (education)
5 X1 <- cbind(1, CPS1988$experience, CPS1988$experience^2)
6 X2 <- as.matrix(CPS1988$education)
7
8 # Step 2: Residualize y and X2 on X1
9 M_X1 <- diag(n) - X1 %*% solve(t(X1) %*% X1) %*% t(X1)
10 y_tilde <- M_X1 %*% y
11 X2_tilde <- M_X1 %*% X2
12
13 # Step 3: FWL regression
14 beta_FWL <- solve(t(X2_tilde) %*% X2_tilde) %*% t(X2_tilde) %*%
    y_tilde
15 cat("FWL estimate of education coefficient:", round(beta_FWL, 6),
    "\n")

```

```
16 cat("Full OLS estimate of education coefficient:",
      round(beta_hat["education"], 6), "\n")
17
18 # Confirm residuals are the same
19 u_FWL <- y_tilde - X2_tilde %*% beta_FWL
20 cat("Max difference in residuals:", max(abs(u_hat - u_FWL)), "\n")
```

3.13.7 Exercise 3.5: Visualizing OLS Geometry (2D Approximation)

```
1 # Simplified example: 3 observations, 2 regressors (for
  # visualization)
2 set.seed(123)
3 n_small <- 3
4 x_small <- c(1, 2, 4)
5 y_small <- c(2, 3, 5)
6 X_small <- cbind(1, x_small)
7 beta_small <- solve(t(X_small) %*% X_small) %*% t(X_small) %*%
  y_small
8 y_hat_small <- X_small %*% beta_small
9 u_small <- y_small - y_hat_small
10
11 cat("Observations:", y_small, "\n")
12 cat("Fitted values:", round(y_hat_small, 3), "\n")
13 cat("Residuals:", round(u_small, 3), "\n")
14 cat("OLS coefficients:", round(beta_small, 3), "\n")
15 cat("SSR:", round(sum(u_small^2), 4), "\n")
16
17 # Verify orthogonality
18 cat("X'u_hat:", round(t(X_small) %*% u_small, 6), "\n")
```

End of Chapter 3

References

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Chapter 4

Chapter 4: Statistical Properties of OLS — Finite Sample Theory

4.1 Chapter Overview

Chapter 3 established the *algebraic* properties of OLS — results that hold in any sample, without any distributional assumptions. This chapter turns to the *statistical* properties of OLS: is $\hat{\beta}$ close to the true parameter β ? How precise is it? How do we conduct valid inference (hypothesis tests and confidence intervals)?

We answer these questions first in the **finite-sample** framework, which requires stronger assumptions but gives exact distributional results. We state the classical **Gauss-Markov assumptions**, prove **unbiasedness** and the **Gauss-Markov Theorem** (OLS is BLUE), and derive inference procedures under normality. Chapter 5 relaxes the normality and homoskedasticity assumptions using large-sample (asymptotic) theory.

4.2 Learning Objectives

After completing this chapter, students will be able to:

1. State the **classical (Gauss-Markov) assumptions** for the linear regression model.
2. Prove that OLS is **unbiased** under strict exogeneity.
3. Derive the **variance-covariance matrix** of the OLS estimator.
4. State and prove the **Gauss-Markov Theorem**.
5. Derive the ***t*-statistic** and construct **confidence intervals** for individual coefficients.
6. Construct and interpret the ***F*-statistic** for joint hypothesis tests.
7. Explain the **estimated variance** of OLS and why $\hat{\sigma}^2 = SSR/(n - k)$.

8. Discuss the sources of estimator variance (sample size, multicollinearity, error variance).
-

4.3 4.1 Setting Up the Framework: Classical Assumptions

To derive statistical properties of OLS, we need assumptions about the data-generating process (DGP). The **Classical Linear Regression Model (CLRM)** rests on five assumptions.

Assumption 4.1 (Linearity): The population model is correctly specified as:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$$

where $\boldsymbol{\beta}$ is the true (unknown) parameter vector.

Assumption 4.2 (Full Rank): The design matrix $\mathbf{X}$ has full column rank k (no perfect multicollinearity).

Assumption 4.3 (Strict Exogeneity):

$$E[\mathbf{u} \mid \mathbf{X}] = \mathbf{0}$$

This says that the conditional expectation of the error vector, given the entire design matrix, is zero. This implies $E[u_i \mid \mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n] = 0$: the error for observation i is mean-zero conditional on *all* observations' covariates.

Assumption 4.4 (Spherical Errors / Homoskedasticity and No Serial Correlation):

$$E[\mathbf{u}\mathbf{u}^T \mid \mathbf{X}] = \sigma^2 \mathbf{I}_n$$

This combines two conditions:

- *Homoskedasticity*: $E[u_i^2 \mid \mathbf{X}] = \sigma^2$ for all i (constant variance).
- *No serial correlation*: $E[u_i u_j \mid \mathbf{X}] = 0$ for all $i \neq j$ (uncorrelated errors).

Assumption 4.5 (Normality): The errors are jointly normally distributed:

$$\mathbf{u} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$$

Note on Assumptions: Assumptions 4.1–4.4 are the Gauss-Markov assumptions**. Assumption 4.5 (normality) is additional and is needed only for *exact* finite-sample inference. As we show in Chapter 5, Assumption 4.5 can be dropped in large samples.

4.3.1 4.1.1 Discussion of Strict Exogeneity

Assumption 4.3 is the critical identifying assumption. It requires that the error u_i is uncorrelated with *all* regressors in *all* observations. This is stronger than merely requiring $E[u_i | \mathbf{x}_i] = 0$ (conditional mean zero for observation i 's own regressors).

When strict exogeneity fails:

- **Omitted variable bias:** An important determinant of y that is correlated with $\mathbf{x}_i$ is left out of the model.
- **Reverse causality:** y causes $\mathbf{x}$ as well as vice versa.
- **Measurement error:** $\mathbf{x}_i$ is measured with error that is correlated with the true regressor.
- **Panel data with lagged dependent variables:** $E[u_{it} | x_{i,t-1}]$ may be non-zero even if $E[u_{it} | x_{it}] = 0$.

When Assumption 4.3 fails, OLS is **biased** and potentially **inconsistent**. Addressing this is the focus of Chapters 6–13.

4.4 4.2 Unbiasedness of OLS

Theorem 4.1 (Unbiasedness of OLS): Under Assumptions 4.1–4.3, the OLS estimator is **unbiased**: $E[\hat{\beta} | \mathbf{X}] = \beta$.

Proof:

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T (\mathbf{X} \beta + \mathbf{u}) = \beta + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{u}$$

Taking the conditional expectation given $\mathbf{X}$:

$$E[\hat{\beta} | \mathbf{X}] = \beta + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \underbrace{E[\mathbf{u} | \mathbf{X}]}_{=0} = \beta$$

By the **Law of Iterated Expectations**: $E[\hat{\beta}] = E[E[\hat{\beta} | \mathbf{X}]] = E[\beta] = \beta$. $\square$

Interpretation: On average (over repeated samples), the OLS estimator hits the true parameter. In any given sample, $\hat{\beta} \neq \beta$ due to sampling variation, but there is no systematic error.

4.4.1 4.2.1 Importance of Strict Exogeneity

The proof relies critically on Assumption 4.3. If $E[\mathbf{u} | \mathbf{X}] \neq \mathbf{0}$, then:

$$E[\hat{\beta} \mid \mathbf{X}] = \beta + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T E[\mathbf{u} \mid \mathbf{X}] \neq \beta$$

The OLS estimator is biased. The bias is:

$$\text{Bias}(\hat{\beta}) = E[\hat{\beta}] - \beta = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T E[\mathbf{u}]$$

4.5 4.3 The Variance-Covariance Matrix of OLS

Theorem 4.2 (Variance of OLS under Spherical Errors): Under Assumptions 4.1–4.4:

$$\text{Var}(\hat{\beta} \mid \mathbf{X}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

Proof: From the proof of Theorem 4.1: $\hat{\beta} - \beta = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{u}$.

$$\begin{aligned} \text{Var}(\hat{\beta} \mid \mathbf{X}) &= E \left[(\hat{\beta} - \beta)(\hat{\beta} - \beta)^T \mid \mathbf{X} \right] \\ &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \underbrace{E[\mathbf{u}\mathbf{u}^T \mid \mathbf{X}]}_{=\sigma^2 \mathbf{I}_n} \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \\ &= \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1} \quad \square \end{aligned}$$

4.5.1 4.3.1 The Variance of Individual Coefficients

The variance of the j -th coefficient $\hat{\beta}_j$ is the j -th diagonal element of $\sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$:

$$\text{Var}(\hat{\beta}_j \mid \mathbf{X}) = \sigma^2 [(\mathbf{X}^T \mathbf{X})^{-1}]_{jj}$$

Using the FWL theorem, one can show:

$$\text{Var}(\hat{\beta}_j \mid \mathbf{X}) = \frac{\sigma^2}{\text{SST}_j (1 - R_j^2)}$$

where $\text{SST}_j = \sum_{i=1}^n (x_{ij} - \bar{x}_j)^2$ is the total variation in x_{ij} , and R_j^2 is the R^2 from regressing x_{ij} on all other regressors.

Three sources of estimator variance:

Factor	Effect on $\text{Var}(\hat{\beta}_j)$	Intuition
Larger σ^2 (more noise)	Increases	More noise $\rightarrow$ harder to estimate sign
Larger SST_j (more variation in x_j)	Decreases	More variation $\rightarrow$ more information
Larger R_j^2 (more multicollinearity)	Increases	Collinearity $\rightarrow x_j$ explains little net i

4.5.2 4.3.2 Estimating σ^2

The population error variance $\sigma^2 = E[u_i^2]$ is unknown. We estimate it using the OLS residuals:

$$\hat{\sigma}^2 = \frac{SSR}{n - k} = \frac{\hat{\mathbf{u}}^T \hat{\mathbf{u}}}{n - k}$$

This is an **unbiased estimator** of σ^2 under Assumptions 4.1–4.4.

Why divide by $n - k$ instead of n ?

$$E[SSR | \mathbf{X}] = E[\hat{\mathbf{u}}^T \hat{\mathbf{u}} | \mathbf{X}] = E[\mathbf{u}^T \mathbf{M}_X \mathbf{u} | \mathbf{X}]$$

$$= \text{tr}(\mathbf{M}_X E[\mathbf{u} \mathbf{u}^T | \mathbf{X}]) = \sigma^2 \text{tr}(\mathbf{M}_X) = \sigma^2(n - k)$$

Therefore $E[\hat{\sigma}^2] = E[SSR/(n - k)] = \sigma^2$. $\square$

The degrees of freedom correction $n - k$ accounts for the fact that OLS "uses up" k degrees of freedom to estimate k parameters. With k parameters perfectly fitting k observations, there are no degrees of freedom left for estimation.

4.6 4.4 The Gauss-Markov Theorem: OLS is BLUE

Theorem 4.3 (Gauss-Markov Theorem): Under Assumptions 4.1–4.4, the OLS estimator $\hat{\beta}$ is the **Best Linear Unbiased Estimator (BLUE)** of β . That is, among all linear unbiased estimators $\tilde{\beta} = \mathbf{C}\mathbf{y}$ (where $\mathbf{C}$ is any non-random matrix with $E[\tilde{\beta}] = \beta$), OLS has the smallest variance matrix in the sense that:

$$\text{Var}(\tilde{\beta} | \mathbf{X}) - \text{Var}(\hat{\beta} | \mathbf{X}) = \text{Var}(\tilde{\beta} | \mathbf{X}) - \sigma^2(\mathbf{X}^T \mathbf{X})^{-1} \geq 0 \quad (\text{positive semi-definite})$$

Proof: Let $\tilde{\beta} = \mathbf{C}\mathbf{y}$ be any linear unbiased estimator. For unbiasedness: $E[\tilde{\beta} | \mathbf{X}] = \mathbf{C}\mathbf{X}\beta = \beta$ for all β , which requires $\mathbf{C}\mathbf{X} = \mathbf{I}_k$.

Write $\mathbf{C} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T + \mathbf{D}$ where $\mathbf{D} = \mathbf{C} - (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$. Unbiasedness requires $\mathbf{D}\mathbf{X} = \mathbf{0}$.

Then:

$$\begin{aligned}\text{Var}(\tilde{\beta} \mid \mathbf{X}) &= \mathbf{C} \cdot \sigma^2 \mathbf{I}_n \cdot \mathbf{C}^T = \sigma^2 \mathbf{C} \mathbf{C}^T \\ &= \sigma^2 \left[(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} + \mathbf{D} \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{D}^T + \mathbf{D} \mathbf{D}^T \right]\end{aligned}$$

Since $\mathbf{D}\mathbf{X} = \mathbf{0}$, the cross terms vanish:

$$\text{Var}(\tilde{\beta} \mid \mathbf{X}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1} + \sigma^2 \mathbf{D} \mathbf{D}^T = \text{Var}(\hat{\beta} \mid \mathbf{X}) + \sigma^2 \mathbf{D} \mathbf{D}^T$$

Since $\sigma^2 \mathbf{D} \mathbf{D}^T \geq 0$ (positive semi-definite), we have $\text{Var}(\tilde{\beta}) \geq \text{Var}(\hat{\beta})$. $\square$

Interpreting Gauss-Markov: The theorem says OLS is the most efficient estimator among all linear unbiased estimators. No other linear combination of the data produces unbiased coefficient estimates with smaller standard errors. This is a strong optimality result within the class of linear estimators.**

Important caveats:

1. Gauss-Markov only compares OLS to *linear* estimators. Non-linear estimators (e.g., maximum likelihood under non-normal errors) may be more efficient.
 2. The theorem requires spherical errors (Assumption 4.4). Under heteroskedasticity, **Feasible Generalized Least Squares (FGLS)** beats OLS in efficiency.
 3. The theorem says nothing about whether OLS is *consistent* (consistent requires Assumption 4.3 to be weakened, as we show in Chapter 5).
-

4.7 4.5 Inference Under Normality

Under all five Gauss-Markov assumptions including normality (Assumption 4.5):

$$\hat{\beta} \mid \mathbf{X} \sim N\left(\beta, \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}\right)$$

This follows immediately from the fact that $\hat{\beta} = \beta + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{u}$ is a linear transformation of $\mathbf{u}$, which is normal.

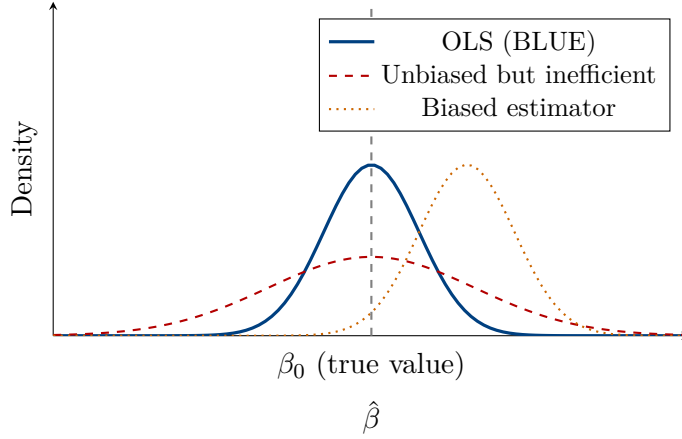


Figure 4.1: Sampling distributions of three estimators of β_0 (true value marked by dashed vertical line). The OLS estimator (solid blue) is both centered on the truth (unbiased) and has the smallest variance among linear unbiased estimators (BLUE). An alternative unbiased estimator (dashed red) has higher variance. A biased estimator (dotted orange) is centered away from β_0 .

For any individual coefficient $\hat{\beta}_j$:

$$\hat{\beta}_j \mid \mathbf{X} \sim N\left(\beta_j, \sigma^2[(\mathbf{X}^T \mathbf{X})^{-1}]_{jj}\right)$$

4.7.1 4.5.1 The t -Statistic

Since σ^2 is unknown, we replace it with $\hat{\sigma}^2$. The resulting statistic follows a t -distribution:

$$t_j = \frac{\hat{\beta}_j - \beta_j^0}{\widehat{SE}(\hat{\beta}_j)} \sim t_{n-k}$$

where β_j^0 is the value under the null hypothesis, and $\widehat{SE}(\hat{\beta}_j) = \hat{\sigma} \sqrt{[(\mathbf{X}^T \mathbf{X})^{-1}]_{jj}}$ is the estimated standard error.

Proof: We have $(\hat{\beta}_j - \beta_j)/(\sigma \sqrt{[(\mathbf{X}^T \mathbf{X})^{-1}]_{jj}}) \sim N(0, 1)$ and $(n - k)\hat{\sigma}^2/\sigma^2 \sim \chi_{n-k}^2$ (under normality), and these two are independent. The ratio of a standard normal to the square root of an independent chi-squared divided by its degrees of freedom has a t -distribution. $\square$

Hypothesis testing: To test $H_0 : \beta_j = 0$ against $H_1 : \beta_j \neq 0$:

1. Compute $t_j = \hat{\beta}_j / \widehat{SE}(\hat{\beta}_j)$.

2. Compare to $t_{n-k, \alpha/2}$ (critical value from t -distribution with $n - k$ degrees of freedom).
3. Reject H_0 if $|t_j| > t_{n-k, \alpha/2}$.
4. The p -value is $P(|T| > |t_j|)$ where $T \sim t_{n-k}$.

For large n , t_{n-k} is well approximated by $N(0, 1)$, and the "rule of thumb" for significance at the 5% level is $|t| > 1.96$.

4.7.2 4.5.2 Confidence Intervals

A $(1 - \alpha)$ confidence interval for β_j is:

$$\hat{\beta}_j \pm t_{n-k, \alpha/2} \cdot \widehat{SE}(\hat{\beta}_j)$$

Interpretation: In repeated sampling, $(1 - \alpha)\%$ of such intervals will contain the true β_j . **It is not true** that the probability the true β_j falls in a specific realized interval is $(1 - \alpha)\%$ — the true β_j is fixed; the interval is random.

4.7.3 4.5.3 Practical Significance vs. Statistical Significance

A statistically significant coefficient is not necessarily economically significant. A study with $n = 100,000$ observations can detect very small effects with high precision. Conversely, a study with $n = 50$ may fail to reject the null even for economically large effects.

Good practice: Report both the coefficient estimate (magnitude of economic effect) and the standard error (precision), not just the t -statistic.

4.8 4.6 Joint Hypothesis Tests: The F -Statistic

Often we want to test joint hypotheses about multiple coefficients. For example:

- $H_0 : \beta_2 = \beta_3 = 0$ (neither experience nor experience² matters)
- $H_0 : \beta_2 = \beta_3$ (two effects are equal)

General setup: Consider the null hypothesis $H_0 : \mathbf{R}\boldsymbol{\beta} = \mathbf{r}$, where $\mathbf{R}$ is a $q \times k$ matrix of linear restrictions (rank q) and $\mathbf{r}$ is a $q \times 1$ vector.

Definition 4.1 (F-Statistic):

$$F = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})^T [\mathbf{R}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{R}^T]^{-1} (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})}{q\hat{\sigma}^2}$$

Under H_0 and Assumptions 4.1–4.5: $F \sim F_{q,n-k}$.

Equivalent Formulation via Restricted and Unrestricted Regressions:

$$F = \frac{(SSR_R - SSR_U)/q}{SSR_U/(n-k)}$$

where SSR_R is from the **restricted regression** (imposing H_0) and SSR_U is from the **unrestricted regression**.

Intuition: If the restrictions are true, imposing them should not increase the SSR by much. If the F -statistic is large, the restrictions are inconsistent with the data.

Special case: When $q = 1$ and $H_0 : \beta_j = 0$, $F = t_j^2$ (the F -statistic is the square of the t -statistic).

The overall F -test: With $\mathbf{R} = [\mathbf{0} \mid \mathbf{I}_{k-1}]$ and $\mathbf{r} = \mathbf{0}_{k-1}$ (testing all slope coefficients jointly), the F -statistic tests whether *any* regressor matters:

$$F = \frac{R^2/(k-1)}{(1-R^2)/(n-k)}$$

4.9 Chapter Summary

- Under Assumptions 4.1–4.3 (including strict exogeneity), OLS is **unbiased**.
 - Under Assumptions 4.1–4.4 (adding spherical errors), the variance of OLS is $\sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$, and OLS is the **Best Linear Unbiased Estimator (BLUE)** — the **Gauss-Markov Theorem**.
 - The variance of $\hat{\beta}_j$ depends inversely on the variation in x_j and positively on the error variance and multicollinearity (R_j^2).
 - $\hat{\sigma}^2 = SSR/(n-k)$ is an unbiased estimator of σ^2 ; the degrees of freedom are $n-k$.
 - Under normality (Assumption 4.5), $\hat{\beta}$ is normally distributed, enabling exact t -tests and F -tests.
 - The t -statistic tests single hypotheses; the F -statistic tests joint hypotheses.
-

4.10 Key Terms

Strict exogeneity — $E[\mathbf{u} \mid \mathbf{X}] = \mathbf{0}$.

Homoskedasticity — $E[u_i^2 \mid \mathbf{X}] = \sigma^2$ (constant error variance).

Spherical errors — $E[\mathbf{u}\mathbf{u}^T | \mathbf{X}] = \sigma^2 \mathbf{I}_n$.

Unbiasedness — $E[\hat{\beta}] = \beta$.

Variance-covariance matrix of OLS — $\text{Var}(\hat{\beta} | \mathbf{X}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$.

Gauss-Markov Theorem — OLS is BLUE (Best Linear Unbiased Estimator) under Assumptions 4.1–4.4.

BLUE — Best (minimum variance) Linear Unbiased Estimator.

Degrees of freedom — $n - k$; the number of observations minus the number of estimated parameters.

***t*-statistic** — $t_j = \hat{\beta}_j / \widehat{SE}(\hat{\beta}_j) \sim t_{n-k}$ under $H_0 : \beta_j = 0$.

***F*-statistic** — statistic for joint hypothesis tests; $F \sim F_{q, n-k}$ under H_0 .

4.11 Exercises

4.11.1 Conceptual Questions

4.1 Explain the difference between **unbiasedness** and **consistency**. Can an estimator be unbiased but inconsistent? Consistent but biased?

4.2 Assumption 4.3 (strict exogeneity) requires $E[u_i | \mathbf{x}_1, \dots, \mathbf{x}_n] = 0$, not just $E[u_i | \mathbf{x}_i] = 0$. Give an example where the weaker condition holds but strict exogeneity fails.

4.3 Suppose we run OLS on data where $V(u_i) = \sigma^2 x_i^2$ (variance increases with x_i). The Gauss-Markov assumptions are violated. What specific assumption fails? Is OLS still unbiased? Is it still BLUE?

4.4 Explain the degrees of freedom correction in $\hat{\sigma}^2 = SSR/(n - k)$. Why is dividing by n instead of $n - k$ problematic?

4.5 A researcher has a sample of $n = 1000$ and runs a regression with 3 regressors. She obtains $\hat{\beta}_2 = 0.5$ with $\widehat{SE}(\hat{\beta}_2) = 0.4$ (t -stat = 1.25). She concludes "the effect of x_2 is economically large but statistically insignificant." Comment on this interpretation.

4.11.2 Analytical Questions

4.6 Under Assumptions 4.1–4.5, derive the distribution of $\hat{\beta} | \mathbf{X}$.

4.7 Prove that $E[SSR | \mathbf{X}] = (n - k)\sigma^2$, showing that $\hat{\sigma}^2 = SSR/(n - k)$ is unbiased.

4.8 Let $\mathbf{c}$ be a known $k \times 1$ vector. Find the variance of the linear combination $\mathbf{c}^T \hat{\beta}$ under Assumptions 4.1–4.4. (This is useful for constructing confidence intervals for marginal effects or predictions.)

4.9 Derive the F -statistic for the overall test of significance (H_0 : all slope coefficients = 0) and show that it equals $\frac{R^2/(k-1)}{(1-R^2)/(n-k)}$.

4.10 Suppose you add an irrelevant variable z (with $\text{Cov}(z, u) = 0$ in the population) to a correctly specified regression. What happens to: (a) unbiasedness of existing coefficient estimates? (b) the variances of existing coefficient estimates? (c) R^2 ? (d) adjusted R^2 ?

4.11.3 Applied Questions

4.11 Using the CPS data from previous labs, run a regression of log wages on education, experience, and experience².

(a) Test $H_0 : \beta_{\text{edu}} = 0$ at the 5% level. Report the t -statistic, p -value, and conclusion. (b) Construct a 95% confidence interval for β_{edu} . (c) Test the joint hypothesis $H_0 : \beta_{\text{exp}} = \beta_{\text{exp}^2} = 0$. (d) What is the estimated peak of the experience-earnings profile?

4.12 Conduct a Monte Carlo study to verify unbiasedness of OLS:

- Generate data: $y_i = 2 + 3x_i + u_i$, $x_i \sim N(0, 1)$, $u_i \sim N(0, 1)$, $n = 50$.
 - Repeat 1000 times; in each repetition, compute $\hat{\beta}_1$ and $\hat{\beta}_2$.
 - Report the mean of $\hat{\beta}_1$ and $\hat{\beta}_2$ across replications.
 - Show that the sampling distributions are centred on the true values.
-

4.12 R Lab 4: Statistical Properties and Inference

4.12.1 Objectives

- Verify unbiasedness via Monte Carlo simulation
- Compute standard errors and construct confidence intervals
- Test hypotheses using t and F statistics
- Explore the effects of multicollinearity

4.12.2 Exercise 4.1: Verifying Unbiasedness via Monte Carlo

```

1 set.seed(42)
2 n_sims <- 2000
3 n <- 100
4 beta_true <- c(2, 3, -1) # intercept, beta_1, beta_2
5
6 # Storage
7 beta_hat_store <- matrix(NA, nrow = n_sims, ncol = 3)

```

```

8
9 for (s in 1:n_sims) {
10   x1 <- rnorm(n)
11   x2 <- rnorm(n)
12   u   <- rnorm(n)
13   y   <- beta_true[1] + beta_true[2] * x1 + beta_true[3] * x2 + u
14
15   fit <- lm(y ~ x1 + x2)
16   beta_hat_store[s, ] <- coef(fit)
17 }
18
19 # Are OLS estimates centred on true values?
20 colMeans(beta_hat_store) # should be close to (2, 3, -1)
21 apply(beta_hat_store, 2, sd) # sampling standard deviations
22
23 # Plot sampling distribution of beta_1
24 ggplot(data.frame(beta1 = beta_hat_store[, 2]), aes(x = beta1)) +
25   geom_histogram(aes(y = ..density..), bins = 40, fill =
26     "steelblue", alpha = 0.7) +
27   geom_vline(xintercept = beta_true[2], color = "red", linewidth =
28     1) +
29   stat_function(fun = dnorm,
30     args = list(mean = beta_true[2], sd =
31       sd(beta_hat_store[, 2])),
32     color = "darkred", linewidth = 1) +
33   labs(title = "Sampling Distribution of OLS Estimator",
34     subtitle = paste0("True beta_1 = ", beta_true[2], " (red
35       line); n_sims = ", n_sims),
36     x = "Estimated beta_1", y = "Density") +
37   theme_minimal()

```

4.12.3 Exercise 4.2: Inference in R

```

1 library(AER)
2 data("CPS1988")
3 CPS1988 <- CPS1988 %>% mutate(log_wage = log(wage))
4
5 model <- lm(log_wage ~ education + experience + I(experience^2),
6   data = CPS1988)
7 summary(model)
8
9 # Manual confidence interval
10 alpha <- 0.05
11 n <- nrow(CPS1988)
12 k <- length(coef(model))
13 t_crit <- qt(1 - alpha/2, df = n - k)
14
15 coef_table <- data.frame(
16   Estimate = coef(model),
17   SE = sqrt(diag(vcov(model)))

```

```

17 ) %>%
18   mutate(
19     t_stat = Estimate / SE,
20     CI_lower = Estimate - t_crit * SE,
21     CI_upper = Estimate + t_crit * SE,
22     p_value = 2 * pt(-abs(t_stat), df = n - k)
23   )
24 print(round(coef_table, 4))

```

4.12.4 Exercise 4.3: F-Test for Joint Significance

```

1 # Test H0: beta_experience = beta_experience^2 = 0
2 # Method 1: Built-in F-test via anova()
3 model_restricted <- lm(log_wage ~ education, data = CPS1988)
4 model_unrestricted <- model
5 anova(model_restricted, model_unrestricted)
6
7 # Method 2: Manual F-statistic
8 SSR_R <- sum(residuals(model_restricted)^2)
9 SSR_U <- sum(residuals(model_unrestricted)^2)
10 q <- 2 # number of restrictions
11 F_stat <- ((SSR_R - SSR_U) / q) / (SSR_U / (n - k))
12 p_val <- pf(F_stat, df1 = q, df2 = n - k, lower.tail = FALSE)
13 cat("F-statistic:", round(F_stat, 3), "\n")
14 cat("p-value:", round(p_val, 4), "\n")

```

4.12.5 Exercise 4.4: Effect of Multicollinearity

```

1 # Simulate data with varying levels of correlation between x1 and x2
2 set.seed(42)
3 n <- 200
4 beta_true <- c(2, 3, -1)
5
6 results <- map_dfr(c(0, 0.5, 0.8, 0.95, 0.99), function(rho) {
7   sims <- replicate(1000, {
8     SIGMA <- matrix(c(1, rho, rho, 1), 2, 2)
9     X_sim <- MASS::mvrnorm(n, mu = c(0, 0), Sigma = SIGMA)
10    u_sim <- rnorm(n)
11    y_sim <- beta_true[1] + beta_true[2] * X_sim[,1] + beta_true[3]
12      * X_sim[,2] + u_sim
13    coef(lm(y_sim ~ X_sim))[2] # coefficient on x1
14  })
15   data.frame(rho = rho, se_beta1 = sd(sims))
16 })
17 print(results)
18
19 ggplot(results, aes(x = rho, y = se_beta1)) +

```

```
20 geom_line(color = "steelblue", linewidth = 1.2) +
21 geom_point(size = 3, color = "steelblue") +
22 labs(
23   title = "Effect of Multicollinearity on Standard Errors",
24   subtitle = "As correlation between x1 and x2 increases, SE of
      beta_1 explodes",
25   x = "Correlation between x1 and x2",
26   y = "Standard Deviation of OLS Estimator (Monte Carlo)"
27 ) +
28 theme_minimal()
```

End of Chapter 4

References

- Davidson, R., and MacKinnon, J. G. (2004). *Econometric Theory and Methods*. Oxford University Press. Chapter 3.
- Wooldridge, J. M. (2019). *Introductory Econometrics* (7th ed.). Cengage. Chapters 4–5.
- Angrist, J. D., and Pischke, J. (2009). *Mostly Harmless Econometrics*. Princeton University Press. Chapter 3.

Chapter 5

Chapter 5: Asymptotic Theory and Robust Inference

5.1 Chapter Overview

Chapter 4 derived exact finite-sample results under the strong assumption that errors are normally distributed. In practice, economic data are rarely normally distributed: wages are right-skewed, binary outcomes are Bernoulli, error variances vary across observations. This chapter develops **asymptotic (large-sample) theory** — statistical results that hold approximately when n is large, without requiring normality or homoskedasticity.

We derive the **asymptotic distribution of OLS**, show how to estimate the asymptotic variance under **heteroskedasticity** (non-constant variance) and **within-cluster correlation**, and provide practical guidance on when and how to use robust standard errors. These are the standard errors used in virtually all applied economic research today.

5.2 Learning Objectives

After completing this chapter, students will be able to:

1. Explain why asymptotic theory is needed in applied econometrics.
2. State and apply the **Law of Large Numbers (LLN)** and **Central Limit Theorem (CLT)** to OLS.
3. Prove **consistency** of OLS under weaker conditions than Gauss-Markov.
4. Derive the **asymptotic distribution** of the OLS estimator.
5. Explain what **heteroskedasticity** is and why it invalidates classical standard errors.
6. Compute and interpret **heteroskedasticity-robust (HC) standard errors**.
7. Explain **cluster-robust standard errors** and when clustering is necessary.

8. Provide practical guidance on standard error choice in applied work.

5.3 5.1 Why Asymptotic Theory?

The Gauss-Markov assumptions, particularly normality (Assumption 4.5), are often implausible:

- **Discrete outcomes:** Earnings rounded to the nearest dollar, binary indicators of employment, counts of accidents. Error distributions are clearly non-normal.
- **Non-spherical errors:** Wages vary more for high earners (heteroskedasticity); students in the same class share correlated outcomes (clustering); annual firm data have autocorrelated shocks.
- **Misspecified functional form:** When the CEF is non-linear, OLS residuals systematically deviate from normality.

When the classical assumptions fail, we need **asymptotic theory** — results that hold as $n \rightarrow \infty$ under weaker conditions. The key tools are the **Law of Large Numbers** (sample averages converge to population means) and the **Central Limit Theorem** (standardized sample averages are approximately normal for large n).

5.4 5.2 Consistency of OLS

Definition 5.1 (Consistency): An estimator $\hat{\beta}_n$ is **consistent** for β if $\hat{\beta}_n \xrightarrow{p} \beta$ as $n \rightarrow \infty$. That is, for any $\epsilon > 0$:

$$\lim_{n \rightarrow \infty} P(\|\hat{\beta}_n - \beta\| > \epsilon) = 0$$

Theorem 5.1 (Consistency of OLS): Suppose the following conditions hold:

1. The model is linear: $y_i = \mathbf{x}_i^T \beta + u_i$.
2. The data $\{(y_i, \mathbf{x}_i)\}$ are i.i.d.
3. **Predeterminedness:** $E[\mathbf{x}_i u_i] = \mathbf{0}$ (orthogonality condition).
4. $E[\mathbf{x}_i \mathbf{x}_i^T]$ is finite and non-singular.

Then $\hat{\beta}^{OLS} \xrightarrow{p} \beta$.

Proof:

$$\hat{\beta} = \beta + \left(\frac{1}{n} \mathbf{X}^T \mathbf{X} \right)^{-1} \left(\frac{1}{n} \mathbf{X}^T \mathbf{u} \right)$$

By the LLN:

$$\frac{1}{n} \mathbf{X}^T \mathbf{X} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^T \xrightarrow{p} E[\mathbf{x}_i \mathbf{x}_i^T] \equiv \mathbf{S}_{XX}$$

$$\frac{1}{n} \mathbf{X}^T \mathbf{u} = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i u_i \xrightarrow{p} E[\mathbf{x}_i u_i] = \mathbf{0}$$

By Slutsky's theorem and the continuous mapping theorem:

$$\hat{\beta} \xrightarrow{p} \beta + \mathbf{S}_{XX}^{-1} \mathbf{0} = \beta \quad \square$$

5.4.1 5.2.1 Consistency vs. Unbiasedness

Key distinction: Consistency only requires the weaker condition $E[\mathbf{x}_i u_i] = \mathbf{0}$ (predeterminedness / zero covariance), not strict exogeneity $E[u_i | \mathbf{X}] = \mathbf{0}$.

- Strict exogeneity $\Rightarrow$ predeterminedness, but NOT vice versa.
- Predeterminedness is enough for consistency; strict exogeneity is needed for unbiasedness.

This means OLS can be **consistent but biased in small samples** — the bias vanishes as $n \rightarrow \infty$.

Practical implication: In many panel data models (with lagged dependent variables), OLS is biased in small samples but consistent in large samples. For small panels, this bias can be substantial.

5.5 5.3 Asymptotic Normality and the Sandwich Variance

Theorem 5.2 (Asymptotic Normality of OLS): Under the i.i.d. assumptions of Theorem 5.1, plus $E[\mathbf{x}_i \mathbf{x}_i^T u_i^2]$ finite:

$$\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(\mathbf{0}, \mathbf{S}_{XX}^{-1} \mathbf{\Omega} \mathbf{S}_{XX}^{-1})$$

where $\mathbf{S}_{XX} = E[\mathbf{x}_i \mathbf{x}_i^T]$ and $\mathbf{\Omega} = E[\mathbf{x}_i \mathbf{x}_i^T u_i^2] = E[u_i^2 \mathbf{x}_i \mathbf{x}_i^T]$.

Proof: The key step is applying the **Multivariate Central Limit Theorem (MCLT)** to $\frac{1}{\sqrt{n}} \mathbf{X}^T \mathbf{u}$:

$$\frac{1}{\sqrt{n}} \mathbf{X}^T \mathbf{u} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{x}_i u_i \xrightarrow{d} N(\mathbf{0}, \mathbf{\Omega})$$

By the LLN, $\frac{1}{n} \mathbf{X}^T \mathbf{X} \xrightarrow{p} \mathbf{S}_{XX}$. By Slutsky's theorem:

$$\sqrt{n}(\hat{\beta} - \beta) = \left(\frac{1}{n} \mathbf{X}^T \mathbf{X} \right)^{-1} \frac{1}{\sqrt{n}} \mathbf{X}^T \mathbf{u} \xrightarrow{d} \mathbf{S}_{XX}^{-1} \cdot N(\mathbf{0}, \mathbf{\Omega}) = N(\mathbf{0}, \mathbf{S}_{XX}^{-1} \mathbf{\Omega} \mathbf{S}_{XX}^{-1}) \quad \square$$

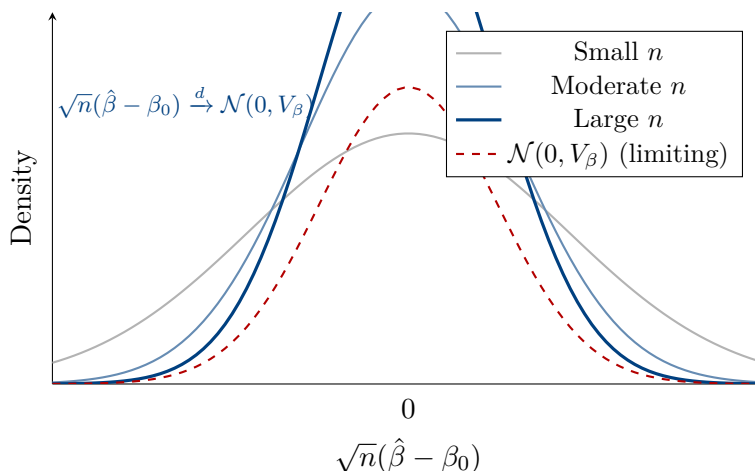


Figure 5.1: Asymptotic normality of the OLS estimator. As the sample size n increases, the scaled and centered OLS estimator $\sqrt{n}(\hat{\beta} - \beta_0)$ converges in distribution to a normal random variable. The limiting distribution $\mathcal{N}(0, V_\beta)$ (dashed red) provides the basis for inference in large samples.

5.5.1 5.3.1 The Sandwich Variance Estimator

The matrix $\mathbf{S}_{XX}^{-1} \mathbf{\Omega} \mathbf{S}_{XX}^{-1}$ is called the **sandwich variance** (or **Huber-White variance**). In finite samples:

$$\text{Var}(\hat{\beta}) \approx \frac{1}{n} \mathbf{S}_{XX}^{-1} \mathbf{\Omega} \mathbf{S}_{XX}^{-1} = (\mathbf{X}^T \mathbf{X})^{-1} \left[\mathbf{X}^T \mathbf{\Omega} \mathbf{X} \right] (\mathbf{X}^T \mathbf{X})^{-1}$$

Under homoskedasticity: $E[u_i^2 \mid \mathbf{x}_i] = \sigma^2$, so $\mathbf{\Omega} = \sigma^2 \mathbf{S}_{XX}$, and the sandwich variance simplifies to:

$$\frac{\sigma^2}{n} \mathbf{S}_{XX}^{-1} \approx \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

This is the classical variance formula — consistent with the Gauss-Markov result.

Under heteroskedasticity: $E[u_i^2 \mid \mathbf{x}_i] = \sigma_i^2 \neq \sigma^2$, the classical formula gives the wrong variance. We need to estimate $\mathbf{\Omega}$ consistently.

5.6 5.4 Heteroskedasticity-Robust Standard Errors

Definition 5.2 (Heteroskedasticity): Errors are heteroskedastic if $V(u_i \mid \mathbf{x}_i)$ varies across observations.

Example 5.1: In a wage regression, it is common for high earners to have greater dispersion in wages than low earners — the variance of u_i may be an increasing function of predicted wages.

5.6.1 5.4.1 Consequences of Ignoring Heteroskedasticity

If we use the classical variance formula $\hat{\sigma}^2(\mathbf{X}^T \mathbf{X})^{-1}$ under heteroskedasticity:

- OLS is still **consistent** (consistency does not require spherical errors).
- But standard errors are **wrong**: they may be too large or too small.
- t -tests and F -tests are **invalid**: rejection rates under the null are not α .

5.6.2 5.4.2 The Heteroskedasticity-Robust (HC) Estimator

White (1980) proposed a consistent estimator of the sandwich variance that does not require spherical errors:

$$\widehat{\text{Var}}_{HC}(\hat{\beta}) = (\mathbf{X}^T \mathbf{X})^{-1} \left(\sum_{i=1}^n \hat{u}_i^2 \mathbf{x}_i \mathbf{x}_i^T \right) (\mathbf{X}^T \mathbf{X})^{-1}$$

This is a sample analogue of the sandwich formula, replacing $E[u_i^2 \mathbf{x}_i \mathbf{x}_i^T]$ with $\frac{1}{n} \sum_{i=1}^n \hat{u}_i^2 \mathbf{x}_i \mathbf{x}_i^T$.

Why does this work? By the consistency of OLS, $\hat{u}_i \approx u_i$ for large n . So $\hat{u}_i^2 \mathbf{x}_i \mathbf{x}_i^T$ is a consistent estimator of $E[u_i^2 \mathbf{x}_i \mathbf{x}_i^T]$.

5.6.3 5.4.3 Small-Sample Corrections (HC1, HC2, HC3)

In small samples, $\hat{u}_i^2$ is a biased downward estimate of u_i^2 (because OLS fits the residuals). Several corrections are used in practice:

Version	Formula	Motivation
HC0	$\hat{u}_i^2$	Original White (1980) estimator
HC1	$\frac{n}{n-k} \hat{u}_i^2$	Degrees-of-freedom correction
HC2	$\frac{\hat{u}_i^2}{1-h_{ii}}$	Corrects for leverage
HC3	$\frac{\hat{u}_i^2}{(1-h_{ii})^2}$	Better in very small samples

where $h_{ii} = [\mathbf{P}_\mathbf{X}]_{ii}$ is the i -th diagonal of the hat matrix (leverage of observation i).

In practice: Use HC3 for small samples ($n < 250$), HC1 for larger samples. The `sandwich` and `lmtest` packages in R provide all variants.

The Rule of Thumb: In modern applied econometrics, always report heteroskedasticity-robust standard errors unless you have a strong theoretical reason to believe errors are homoskedastic.**

5.7 5.5 Cluster-Robust Standard Errors

5.7.1 5.5.1 When Clustering Matters

Clustering arises when observations are not independent. Consider:

$$y_{ig} = \mathbf{x}_{ig}^T \boldsymbol{\beta} + u_{ig}$$

where i indexes individuals within group (cluster) g . If individuals within the same group share common unobservables (e.g., students in the same school share a teacher quality shock), then $\text{Cov}(u_{ig}, u_{jg}) \neq 0$ for $i \neq j$.

The variance-covariance structure is block-diagonal:

$$E[\mathbf{u}\mathbf{u}^T] = \begin{pmatrix} \boldsymbol{\Sigma}_1 & \mathbf{0} & \cdots & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Sigma}_2 & \cdots & \mathbf{0} \\ \vdots & & \ddots & \vdots \\ \mathbf{0} & \mathbf{0} & \cdots & \boldsymbol{\Sigma}_G \end{pmatrix}$$

where $\boldsymbol{\Sigma}_g = E[\mathbf{u}_g \mathbf{u}_g^T]$ allows arbitrary within-cluster correlation.

5.7.2 5.5.2 The Cluster-Robust Sandwich Estimator

Definition 5.3 (Cluster-Robust SE):

$$\widehat{\text{Var}}_{CR}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^T \mathbf{X})^{-1} \left(\sum_{g=1}^G \mathbf{X}_g^T \hat{\mathbf{u}}_g \hat{\mathbf{u}}_g^T \mathbf{X}_g \right) (\mathbf{X}^T \mathbf{X})^{-1}$$

where $\mathbf{X}_g$ and $\hat{\mathbf{u}}_g$ are the design matrix and residuals for cluster g .

Key difference from HC: HC sums over individual observations; CR sums over clusters. Within each cluster, we allow arbitrary covariance in residuals.

5.7.3 5.5.3 Practical Guidance on When to Cluster

Clustering is needed when:

1. **Treatment is at a higher level than outcomes.** E.g., a minimum wage law applies to a state, but outcomes are individual workers. Cluster at the state level.
1. **Panel data with repeated observations.** Each individual appears multiple times; errors are autocorrelated over time. Cluster at the individual level.
1. **Assignment of treatment is clustered.** E.g., a program assigns villages to treatment; outcomes are individual household members. Cluster at the village level.

Rule of thumb: Cluster at the level at which treatment varies (or at which observations are correlated).

Warning — few clusters: Cluster-robust SEs rely on $G \rightarrow \infty$. With few clusters ($G < 20$), they are unreliable. In such cases, use the **wild cluster bootstrap** (see Cameron and Miller, 2015).

Example 5.2 (Card and Krueger, 1994): In their study of the effect of a New Jersey minimum wage increase on fast-food employment, Card and Krueger cluster at the restaurant level — each restaurant is surveyed before and after the policy change, so two observations per restaurant are correlated.

5.8 5.6 When to Cluster? Practical Guidance

A common question from students: "Should I cluster?" The answer depends on the data structure, not the size of the coefficient:

Situation	Cluster?	Level
Random sample, i.i.d. observations	No	—
Survey with multi-stage sampling	Yes	Primary sampling unit
Difference-in-differences	Yes	State/region/treatment unit
Panel data	Yes	Individual/firm
Cross-section with geographic variation	Often	Geographic unit
RCT with cluster randomization	Yes	Randomization unit

Never choose whether to cluster based on whether it changes your results. Cluster based on the structure of the data.

5.9 5.7 Empirical Illustration: Wages and Education Revisited

Consider the CPS wage regression:

$$\log(\text{wage}_i) = \beta_1 + \beta_2 \cdot \text{educ}_i + \beta_3 \cdot \text{exp}_i + \beta_4 \cdot \text{exp}_i^2 + u_i$$

Table 5.1: OLS Estimates with Different Standard Errors

	Classical SE	HC1 SE	HC3 SE
Education	0.0709 (0.0010)	0.0709 (0.0011)	0.0709 (0.0011)
Experience	0.0343 (0.0010)	0.0343 (0.0013)	0.0343 (0.0013)
Exp ²	−0.0005 (0.000)	−0.0005 (0.000)	−0.0005 (0.000)

Standard errors in parentheses.

In this large sample, classical and robust SEs are similar because n is large. The key point: in datasets with clear heteroskedasticity patterns, or with clustering, the differences can be substantial.

5.10 Chapter Summary

- Asymptotic theory delivers results about $\hat{\beta}$ under weaker conditions than finite-sample theory.
 - OLS is **consistent** under predeterminedness ($E[\mathbf{x}_i u_i] = \mathbf{0}$) — weaker than strict exogeneity.
 - The **asymptotic distribution** of $\sqrt{n}(\hat{\beta} - \beta)$ is normal with sandwich variance $\mathbf{S}_{XX}^{-1} \mathbf{\Omega} \mathbf{S}_{XX}^{-1}$.
 - **Heteroskedasticity** means $V(u_i | \mathbf{x}_i)$ is not constant. It does not bias OLS but invalidates classical SEs.
 - **White (HC) standard errors** provide consistent inference under heteroskedasticity of unknown form.
 - **Cluster-robust standard errors** are needed when observations within groups are correlated; they cluster the "meat" of the sandwich at the group level.
 - In modern applied work, **always report robust standard errors** unless there is a strong reason to believe errors are spherical.
-

5.11 Key Terms

Consistency — $\hat{\beta} \xrightarrow{p} \beta$ as $n \rightarrow \infty$.

Predeterminedness — $E[\mathbf{x}_i u_i] = \mathbf{0}$; weaker than strict exogeneity.

Asymptotic normality — $\sqrt{n}(\hat{\beta} - \beta) \xrightarrow{d} N(\mathbf{0}, \mathbf{V})$ for some variance matrix $\mathbf{V}$.

Sandwich variance — $\mathbf{S}_{XX}^{-1} \boldsymbol{\Omega} \mathbf{S}_{XX}^{-1}$; the asymptotic variance of OLS.

Heteroskedasticity — non-constant error variance: $V(u_i | \mathbf{x}_i) \neq \sigma^2$.

Heteroskedasticity-robust (HC) SE — consistent standard errors that allow for heteroskedasticity.

Cluster-robust SE — consistent standard errors that allow for within-cluster correlation.

Wild cluster bootstrap — a resampling method for inference with few clusters.

5.12 Exercises

5.12.1 Conceptual Questions

5.1 Explain the difference between **unbiasedness** and **consistency**. Give an example of an estimator that is consistent but biased in small samples.

5.2 Why does heteroskedasticity not cause OLS to be biased or inconsistent? What does it affect?

5.3 A researcher runs a difference-in-differences regression using state-level panel data (50 states, 10 years). A colleague says "You have 500 observations, so clustering doesn't matter." Evaluate this claim.

5.4 What is the "sandwich" structure of the heteroskedasticity-robust variance estimator? Why is it called a sandwich?

5.5 You are analyzing school data where 30 students are observed in each of 50 schools. Should you cluster by student, school, or not at all? Justify your answer.

5.12.2 Analytical Questions

5.6 Prove that under predeterminedness and i.i.d. sampling, OLS is consistent.

5.7 Suppose $V(u_i | x_i) = \sigma^2 x_i^2$. Derive the true variance of OLS in the bivariate regression of y_i on x_i . Show it differs from the classical formula σ^2 / SST_x .

5.8 Show that when $V(u_i | \mathbf{x}_i) = \sigma^2$ (homoskedasticity), the HC0 sandwich estimator is NOT equal to the classical estimator, but converges to the same limit as $n \rightarrow \infty$.

5.9 Consider panel data $y_{it} = \mathbf{x}_{it}^T \boldsymbol{\beta} + \alpha_i + \epsilon_{it}$ where α_i is an individual-specific shock

and ϵ_{it} is i.i.d. If you run OLS pooling all observations without fixed effects, $\hat{\beta}$ is generally biased. But even if you control for fixed effects (making $\hat{\beta}$ consistent), why might you still need to cluster standard errors at the individual level?

5.12.3 Applied Questions

5.10 Using the CPS data, estimate the wage regression with (a) classical SEs, (b) HC1 SEs, and (c) HC3 SEs. Report all three and compare. When do they differ?

5.11 In R, demonstrate via Monte Carlo simulation that the classical t -test has the wrong size (rejection rate $\neq 5\%$ under H_0) when errors are heteroskedastic, but the HC-robust t -test maintains the correct size.

5.13 R Lab 5: Robust Standard Errors in Practice

5.13.1 Setup

```
1 library(tidyverse)
2 library(sandwich) # for vcovHC
3 library(lmtest)   # for coeftest
4 library(AER)
5
6 data("CPS1988")
7 CPS1988 <- CPS1988 %>% mutate(log_wage = log(wage))
8
9 model <- lm(log_wage ~ education + experience + I(experience^2),
10            data = CPS1988)
```

5.13.2 Exercise 5.1: Compare Standard Errors

```
1 # Classical SEs
2 coeftest(model)
3
4 # HCO (White 1980)
5 coeftest(model, vcov = vcovHC(model, type = "HCO"))
6
7 # HC1 (degrees-of-freedom correction)
8 coeftest(model, vcov = vcovHC(model, type = "HC1"))
9
10 # HC3 (best for small samples)
11 coeftest(model, vcov = vcovHC(model, type = "HC3"))
```

5.13.3 Exercise 5.2: Detecting Heteroskedasticity

```

1 # Residuals vs fitted values plot
2 df_resid <- data.frame(
3   fitted    = fitted(model),
4   residuals = residuals(model)
5 )
6
7 ggplot(df_resid, aes(x = fitted, y = residuals)) +
8   geom_point(alpha = 0.2, color = "steelblue") +
9   geom_hline(yintercept = 0, color = "red") +
10  geom_smooth(method = "loess", se = FALSE, color = "darkred") +
11  labs(
12    title = "Residuals vs. Fitted Values",
13    subtitle = "Pattern indicates heteroskedasticity",
14    x = "Fitted Values", y = "Residuals"
15  ) +
16  theme_minimal()
17
18 # Breusch-Pagan test for heteroskedasticity
19 library(lmtest)
20 bptest(model)

```

5.13.4 Exercise 5.3: Cluster-Robust Standard Errors

```

1 library(sandwich)
2
3 # Simulate panel-like clustering by state (using ethnicity as a
4   # proxy cluster)
5 # In real applications, cluster = geographic unit or treatment unit
6
7 # Cluster by 'ethnicity' (as illustrative proxy)
8 vcov_cluster <- vcovCL(model, cluster = ~ CPS1988$ethnicity)
9 coeftest(model, vcov = vcov_cluster)

```

5.13.5 Exercise 5.4: Monte Carlo — Size of Classical vs. Robust t-tests under Heteroskedasticity

```

1 set.seed(42)
2 n_sims <- 5000
3 n      <- 200
4 reject_classical <- numeric(n_sims)
5 reject_robust    <- numeric(n_sims)
6
7 for (s in 1:n_sims) {
8   x <- rnorm(n)
9   u <- rnorm(n, sd = abs(x)) # heteroskedastic: V(u/x) = x^2
10  y <- 1 + 0 * x + u          # H0: beta = 0 is TRUE
11

```

```
12  fit <- lm(y ~ x)
13
14  # Classical t-stat
15  t_classic <- summary(fit)$coefficients["x", "t value"]
16  reject_classical[s] <- abs(t_classic) > 1.96
17
18  # Robust t-stat
19  t_robust <- coeftest(fit, vcov = vcovHC(fit, "HC1"))["x", "t
20    value"]
21  reject_robust[s] <- abs(t_robust) > 1.96
22 }
23 cat("Classical test rejection rate:", round(mean(reject_classical),
24    3),
25    "(should be ~0.05)\n")
26 cat("Robust test rejection rate:    ", round(mean(reject_robust), 3),
27    "(should be ~0.05)\n")
```

Expected finding: The classical test rejects too often (size > 5%) under heteroskedasticity; the robust test maintains approximately the correct size.

End of Chapter 5

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Part II

The Regression Toolkit

Chapter 6

Chapter 6: The Frisch-Waugh-Lovell Theorem and Omitted Variable Bias

6.1 Chapter Overview

Chapter 3 introduced the Frisch-Waugh-Lovell (FWL) Theorem as a result about the algebra of OLS. This chapter makes the theorem earn its keep: it is the key tool for understanding **omitted variable bias (OVB)** — the most important source of endogeneity in applied work. We derive the OVB formula, discuss its direction and magnitude, and explain the difference between bad and good controls. We also cover proxy variables and the concept of "bad controls" — variables that, when included, can actually increase bias rather than reduce it.

6.2 Learning Objectives

After completing this chapter, students will be able to:

1. Formally state and prove the **Frisch-Waugh-Lovell (FWL) Theorem**.
 2. Derive the **omitted variable bias (OVB) formula** and use it to sign the direction of bias.
 3. Explain when a control variable reduces bias vs. when it introduces bias ("bad controls").
 4. Distinguish between "bad controls" and "good controls."
 5. Explain the concept of a **proxy variable** and when it is useful.
 6. Apply these concepts to empirical examples in wages, health, and education.
-

6.3 6.1 Partitioned Regression and the FWL Theorem

We proved the FWL Theorem in Chapter 3. Here we develop its implications more carefully.

Setup: Partition the regression:

$$\mathbf{y} = \mathbf{X}_1\beta_1 + \mathbf{X}_2\beta_2 + \mathbf{u}$$

where $\mathbf{X}_1$ is $n \times k_1$ ("control variables") and $\mathbf{X}_2$ is $n \times k_2$ ("variables of interest").

Theorem 6.1 (Frisch-Waugh-Lovell): The OLS estimate $\hat{\beta}_2$ from the full regression equals the OLS estimate from regressing $\tilde{\mathbf{y}} = \mathbf{M}_{\mathbf{X}_1}\mathbf{y}$ on $\tilde{\mathbf{X}}_2 = \mathbf{M}_{\mathbf{X}_1}\mathbf{X}_2$:

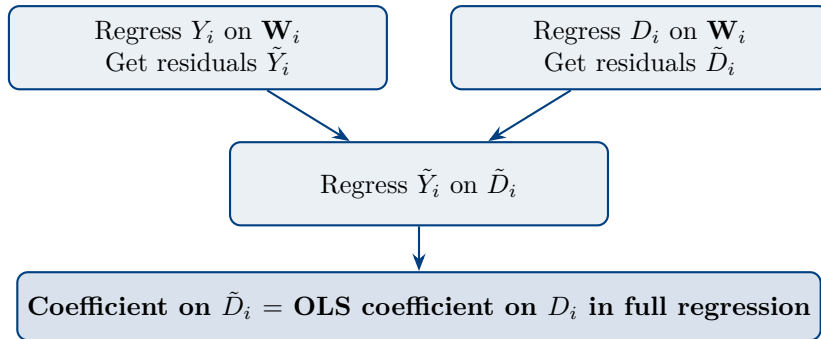
$$\hat{\beta}_2 = (\tilde{\mathbf{X}}_2^T \tilde{\mathbf{X}}_2)^{-1} \tilde{\mathbf{X}}_2^T \tilde{\mathbf{y}}$$

Economic interpretation of "residualizing":

- $\tilde{\mathbf{y}} = \mathbf{M}_{\mathbf{X}_1}\mathbf{y}$: the part of y that is NOT linearly predictable from $\mathbf{X}_1$.
- $\tilde{\mathbf{X}}_2 = \mathbf{M}_{\mathbf{X}_1}\mathbf{X}_2$: the part of $\mathbf{X}_2$ that is NOT linearly predictable from $\mathbf{X}_1$.

The FWL regression asks: *among the remaining variation in $\mathbf{X}_2$ after removing what $\mathbf{X}_1$ can predict, how much of the remaining variation in y does it explain?*

This is exactly what we mean by "controlling for $\mathbf{X}_1$ " — we remove the linear influence of $\mathbf{X}_1$ on both y and $\mathbf{X}_2$ before asking about the relationship between $\mathbf{X}_2$ and y .



$$\text{FWL: } \hat{\beta}_D^{\text{full}} = (\tilde{\mathbf{D}}^T \tilde{\mathbf{D}})^{-1} \tilde{\mathbf{D}}^T \tilde{\mathbf{Y}}$$

where tildes denote residuals after projecting out $\mathbf{W}$

Figure 6.1: The Frisch-Waugh-Lovell (FWL) Theorem as a two-step procedure. The coefficient on D_i in the full regression (controlling for $\mathbf{W}_i$) equals the coefficient from regressing the “residualized” outcome $\tilde{Y}_i$ on the “residualized” treatment $\tilde{D}_i$. This demonstrates that OLS “partials out” the controls.

6.4 6.2 Omitted Variable Bias

The most important implication of the FWL Theorem is for **omitted variable bias (OVB)**: what happens to estimated coefficients when we leave out a variable that belongs in the model?

6.4.1 6.2.1 The Setup

Suppose the true model is:

$$y_i = \beta_1 + \beta_2 x_i + \beta_3 z_i + u_i \quad (\text{Long})$$

but we instead estimate the **short regression**:

$$y_i = \tilde{\beta}_1 + \tilde{\beta}_2 x_i + v_i \quad (\text{Short})$$

What is the relationship between $\hat{\tilde{\beta}}_2$ (from the short regression) and $\hat{\beta}_2$ (from the long regression)?

6.4.2 6.2.2 The OVB Formula

Theorem 6.2 (Omitted Variable Bias): The short-regression coefficient satisfies:

$$\hat{\tilde{\beta}}_2 = \hat{\beta}_2 + \hat{\beta}_3 \cdot \hat{\delta}_{zx}$$

where $\hat{\delta}_{zx}$ is the **auxiliary regression coefficient** from regressing z_i on x_i (the slope from OLS of z on x).

In population (probability limits):

$$\text{plim}(\hat{\tilde{\beta}}_2) = \beta_2 + \beta_3 \cdot \delta_{zx}$$

where $\delta_{zx} = \text{Cov}(x_i, z_i)/V(x_i)$.

The **omitted variable bias** is:

$$\text{OVB} = \underbrace{\beta_3}_{\text{effect of } z \text{ on } y} \cdot \underbrace{\delta_{zx}}_{\text{relationship between } z \text{ and } x}$$

Proof: By OLS algebra and the FWL Theorem. The short regression estimates:

$$\hat{\tilde{\beta}}_2 = \frac{\text{Cov}_s(x_i, y_i)}{V_s(x_i)}$$

Substituting $y_i = \beta_1 + \beta_2 x_i + \beta_3 z_i + u_i$ and using properties of covariance:

$$= \beta_2 + \beta_3 \frac{\text{Cov}(x_i, z_i)}{V(x_i)} + \frac{\text{Cov}(x_i, u_i)}{V(x_i)} = \beta_2 + \beta_3 \delta_{zx} \quad \square$$

(where $\text{Cov}(x_i, u_i) = 0$ by the long regression assumption.)

6.4.3 6.2.3 Signing the Bias

The OVB formula has two factors. To determine the sign of the bias:

$$\text{sign(OVB)} = \text{sign}(\beta_3) \times \text{sign}(\delta_{zx})$$

Effect of z on y (β_3)	Correlation of x and z (δ_{zx})	OVB	Short regression
Positive	Positive	Positive (upward)	Overestimates β_2
Positive	Negative	Negative (downward)	Underestimates β_2
Negative	Positive	Negative (downward)	Underestimates β_2
Negative	Negative	Positive (upward)	Overestimates β_2

Correlation of omitted variable with D

		$\rho_{D\tilde{W}} > 0$	$\rho_{D\tilde{W}} < 0$
Effect of omitted variable	$\delta_W > 0$	$\hat{\delta} > \delta$ (upward bias)	$\hat{\delta} < \delta$ (downward bias)
	$\delta_W < 0$	$\hat{\delta} < \delta$ (downward bias)	$\hat{\delta} > \delta$ (upward bias)

$\text{OVB} = \delta_W \cdot \rho_{D\tilde{W}}$

Figure 6.2: Omitted Variable Bias (OVB) direction table. The sign of the bias in $\hat{\delta}$ depends on (1) whether the omitted variable W has a positive or negative effect on Y (holding D fixed), and (2) whether W is positively or negatively correlated with the treatment D . Bias is upward (red) when these signs agree, downward (blue) when they disagree.

Example 6.1 (Returns to Education): True model: $\log \text{wages} = \beta_1 + \beta_2 \cdot \text{educ} + \beta_3 \cdot \text{ability} + u$.

If we omit ability:

- $\beta_3 > 0$: ability increases wages.
- $\delta_{zx} > 0$: ability is positively correlated with education.
- $\text{OV} > 0$: the short regression **overestimates** the return to education.

This is the classical "ability bias" problem in the returns to schooling literature. Simply regressing wages on years of schooling produces an upward-biased estimate because more able people both get more education and earn more.

Example 6.2 (Class Size and Test Scores): True model: test scores = $\beta_1 + \beta_2 \cdot \text{classsize} + \beta_3 \cdot \text{income} + u$.

If we omit family income:

- $\beta_3 > 0$: higher income families have children with higher scores.
- $\delta_{zx} < 0$: higher income districts have smaller classes.
- $\text{OV} < 0$: the short regression **overestimates** the negative effect of class size (makes classes seem more harmful than they are).

6.4.4 6.2.4 The Multivariate OVB Formula

With multiple omitted variables $\mathbf{z}_i = (z_{1i}, \dots, z_{mi})^T$:

$$\text{plim}(\hat{\beta}_2) = \beta_2 + \beta_z^T \delta_{zx}$$

where β_z is the vector of coefficients on $\mathbf{z}$ in the long regression and δ_{zx} is the vector of auxiliary regression coefficients. OVB can be positive or negative depending on the signs of all $\beta_{z_j} \delta_{z_j x}$ terms.

6.5 6.3 Bad Controls

A natural response to OVB is to "include more controls." But including too many controls — or the wrong controls — can introduce new biases. Variables that seem like good controls can actually be **bad controls**.

6.5.1 6.3.1 What Is a Bad Control?

Definition 6.1 (Bad Control): A variable z_i is a **bad control** if it is itself affected by the treatment variable x_i (i.e., z_i is a "mediator" on the causal pathway from x_i to y_i , or if it shares unobserved determinants with y_i beyond those that caused z_i to be correlated with x_i).

Canonical example (Angrist and Pischke, 2009): Suppose we want to estimate the effect of college education (x) on wages (y). Occupation (z) is a potential control

variable. But:

- College education affects the occupation a worker enters.
- Occupation affects wages.
- Occupation is thus a **mediator** on the causal path: Education $\rightarrow$ Occupation $\rightarrow$ Wages.

If we control for occupation, we are "blocking" part of the causal pathway through which education affects wages. The coefficient on education will **understate** the total effect of education on wages.

6.5.2 6.3.2 The Bias from Controlling for a Mediator

When z_i is a mediator:

$$y_i = \beta_1 + \beta_2^{direct} x_i + \beta_3 z_i + u_i$$

where β_2^{direct} is the *direct* effect of x on y (not through z). But:

$$z_i = \gamma_1 + \gamma_2 x_i + \epsilon_i$$

The **total** effect of x on y is $\beta_2^{direct} + \beta_3 \gamma_2$ (direct + indirect through z). Controlling for z recovers only the direct effect.

Is this wrong? It depends on the question. If you want the **total** causal effect of education on wages, you should NOT control for occupation. If you want the **direct** effect (not mediated by occupation), you should control for it.

6.5.3 6.3.3 Selection into the Control Variable

A more subtle problem arises when the control variable z_i has its own unobserved determinants w_i that also affect y_i :

- $x \rightarrow y$ (the causal path we want)
- $x \rightarrow z$ (treatment affects the control)
- $w \rightarrow z$ (unobserved factor affects control)
- $w \rightarrow y$ (same unobserved factor affects outcome)

Controlling for z can open up a "backdoor" correlation between x and y through w . Including z introduces bias from w that wasn't there before.

Example: Suppose we want the effect of gender (x) on wages (y). A naive control might be job title (z). But:

- Gender affects job title (discrimination in promotion).

- w = unobserved productivity affects both job title and wages.
- Conditioning on job title opens a spurious path: Gender $\rightarrow$ Job Title $\leftarrow$ Productivity $\rightarrow$ Wages.
- This introduces bias rather than removing it.

6.5.4 6.3.4 Rules of Thumb for Control Variables

Good controls: Pre-treatment variables that affect both x and y but are not affected by x . These reduce bias from confounding.

Bad controls: Variables that are:

1. Caused by the treatment x (mediators)
2. Share unobserved determinants with y beyond their relationship with x
3. Selected into by the same process that creates the endogeneity problem

Practical Advice: Think carefully about the causal structure before choosing controls. A useful framework is the Directed Acyclic Graph (DAG)**, which visually represents causal relationships and helps identify good and bad controls.

6.6 6.4 Proxy Variables

When the true omitted variable z_i is unobserved, we may have access to a **proxy** — a variable p_i that is correlated with z_i and satisfies certain conditions.

Definition 6.2 (Proxy Variable): p_i is a valid proxy for z_i if:

1. p_i is correlated with z_i : $\text{Cov}(p_i, z_i) \neq 0$.
2. p_i is uncorrelated with the structural error u_i conditional on z_i : $E[u_i \mid x_i, z_i, p_i] = E[u_i \mid x_i, z_i]$.

Condition (2) says p_i affects y_i only through z_i — once we control for the underlying unobservable z_i , the proxy p_i adds no additional information.

Example: In the wage-education model, IQ score (p_i) may serve as a proxy for cognitive ability (z_i). If IQ only affects wages through its relationship with ability, and doesn't directly affect wages, it satisfies the proxy conditions.

Limitation: The OLS estimate using a proxy will generally not recover the true structural coefficient on x_i , but it reduces OVB compared to omitting the proxy entirely.

6.7 Chapter Summary

- The **FWL Theorem** shows that OLS slope coefficients measure partial effects after "residualizing" on all other controls.
 - **Omitted variable bias** arises when an important determinant of y is both correlated with included regressors and omitted from the model.
 - The **OVB formula** is $OVB = \beta_3 \cdot \delta_{zx}$, the product of the omitted variable's effect on y and its correlation with the included regressor.
 - "More controls are better" is NOT always true. **Bad controls** — mediators or variables that share unobserved determinants with y — can introduce new biases.
 - **Proxy variables** can reduce OVB when the true omitted variable is unobservable but an imperfect proxy is available.
-

6.8 Key Terms

Omitted Variable Bias (OVB) — bias in an estimated coefficient when a variable that belongs in the model is omitted.

Long regression — the correctly specified model including all relevant variables.

Short regression — the misspecified model omitting one or more relevant variables.

Auxiliary regression — the regression of the omitted variable on the included regressors.

OVB formula — $\text{plim}(\hat{\beta}_2) = \beta_2 + \beta_3 \delta_{zx}$.

Bad control — a variable that, when included, introduces bias rather than reducing it.

Mediator — a variable on the causal pathway from the treatment to the outcome.

Proxy variable — an observable variable correlated with an unobservable omitted variable.

Directed Acyclic Graph (DAG) — a visual representation of assumed causal relationships.

6.9 Exercises

6.9.1 Conceptual Questions

6.1 State the OVB formula. What are its two components? Explain the direction of OVB for the ability bias in the returns-to-education literature.

6.2 Explain why "including more controls" does not always reduce bias. Give an example of a bad control.

6.3 You are studying the effect of a job training program (x) on earnings (y). A colleague suggests controlling for whether the participant found a new job after the program (z). Is this a good or bad control? Explain.

6.4 Define a proxy variable. What conditions must it satisfy? Give an example from economics.

6.5 In a regression of wages on education and experience, a researcher finds that adding "number of previous employers" reduces the education coefficient from 0.10 to 0.06. What does this suggest about the omitted variable bias in the original regression?

6.9.2 Analytical Questions

6.6 True model: $y_i = 2 + 3x_i + (-1)z_i + u_i$, where $\text{Cov}(x, z)/V(x) = 0.5$. (a) What is the short-regression coefficient on x ? (b) Is the estimate biased upward or downward relative to the true $\beta_2 = 3$?

6.7 Suppose we have the true model $y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + u_i$, but we estimate $y_i = \gamma_0 + \gamma_1 x_{1i} + \gamma_2 x_{2i} + v_i$ (omitting x_{3i}). Derive the formula for $\text{plim}(\hat{\gamma}_1)$ and $\text{plim}(\hat{\gamma}_2)$.

6.8 Consider a regression of health status on income and region. Suppose region is correlated with both income and health (richer regions have better health care). Describe what happens to the income coefficient if you (a) include region as a control; (b) omit region.

6.9.3 Applied Questions

6.9 Using the CPS data: (a) Estimate the return to education with no controls. (b) Add experience as a control. Does the education coefficient change? In which direction? Is this consistent with OVB? (c) Add ethnicity as a control. Does the education coefficient change? What does this imply?

6.10 Conduct the following experiment in R:

- Generate data: $y_i = 1 + 2x_i + 3z_i + u_i$, $z_i = 0.5x_i + \epsilon_i$.
 - Estimate the short regression (omitting z) and the long regression (including z).
 - Verify the OVB formula numerically.
-

6.10 R Lab 6: Omitted Variable Bias and the FWL Theorem

6.10.1 Exercise 6.1: Demonstrating OVB

```
1 library(tidyverse)
2 set.seed(2024)
3 n <- 1000
4
5 # True parameters
6 beta_0 <- 1; beta_x <- 2; beta_z <- 3
7
8 # Simulate omitted variable correlated with x
9 x <- rnorm(n)
10 z <- 0.5 * x + rnorm(n) # delta_zx ~ 0.5
11 u <- rnorm(n)
12 y <- beta_0 + beta_x * x + beta_z * z + u
13
14 # Short regression (omitting z)
15 short_reg <- lm(y ~ x)
16 cat("Short regression coefficient on x:",
17     round(coef(short_reg)["x"], 3), "\n")
18
19 # Long regression (including z)
20 long_reg <- lm(y ~ x + z)
21 cat("Long regression coefficient on x:", round(coef(long_reg)["x"],
22     3), "\n")
23
24 # Auxiliary regression
25 aux_reg <- lm(z ~ x)
26 delta_zx <- coef(aux_reg)["x"]
27 cat("delta_zx (z on x):", round(delta_zx, 3), "\n")
28
29 # OVB formula
30 ovb_formula <- beta_z * delta_zx
31 cat("OVB formula prediction:", round(ovb_formula, 3), "\n")
32 cat("Actual OVB:", round(coef(short_reg)["x"] - coef(long_reg)["x"],
33     3), "\n")
```

6.10.2 Exercise 6.2: Visualizing Bad Controls

```
1 # Simulate: education -> occupation -> wages; ability -> wages
2 set.seed(42)
3 n <- 2000
4
5 ability <- rnorm(n)
6 education <- 0.5 * ability + rnorm(n)
7 # occupation is a mediator: caused by education AND ability
8 occupation_index <- 0.4 * education + 0.3 * ability + rnorm(n)
9 wages <- 0.5 * education + 0.3 * ability + 0.2 * occupation_index +
10     rnorm(n)
```

```

11 # True total effect of education on wages (should be approx 0.5 +
    0.2*0.4 = 0.58)
12 cat("True total effect:", 0.5 + 0.2 * 0.4, "\n")
13
14 # Without controlling for occupation
15 reg1 <- lm(wages ~ education)
16 cat("OLS without occupation:", round(coef(reg1)["education"], 3),
    "\n")
17
18 # Controlling for occupation (BAD: occupation is a mediator)
19 reg2 <- lm(wages ~ education + occupation_index)
20 cat("OLS with occupation (bad control):",
    round(coef(reg2)["education"], 3), "\n")
21
22 # Controlling for ability (GOOD: pre-treatment confounder)
23 reg3 <- lm(wages ~ education + ability)
24 cat("OLS with ability (good control):",
    round(coef(reg3)["education"], 3), "\n")

```

6.10.3 Exercise 6.3: Verifying FWL Numerically

```

1 library(AER)
2 data("CPS1988")
3 CPS1988 <- CPS1988 %>% mutate(log_wage = log(wage))
4
5 # Full regression
6 full_model <- lm(log_wage ~ education + experience + I(experience^2)
    + ethnicity,
7                   data = CPS1988)
8
9 # FWL: residualize log_wage and education on other controls
10 M_controls_y <- residuals(lm(log_wage ~ experience +
    I(experience^2) + ethnicity, data = CPS1988))
11 M_controls_educ <- residuals(lm(education ~ experience +
    I(experience^2) + ethnicity, data = CPS1988))
12
13 fwl_model <- lm(M_controls_y ~ M_controls_educ)
14
15 cat("Full OLS education coefficient:",
    round(coef(full_model)["education"], 6), "\n")
16 cat("FWL education coefficient: ",
    round(coef(fwl_model)["M_controls_educ"], 6), "\n")

```

End of Chapter 6

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Chapter 7

Chapter 7: Regression with Special Regressors: Dummies, Interactions, and Non-linearities

7.1 Chapter Overview

Real economic relationships rarely look like straight lines. This chapter extends the linear regression toolkit to handle discrete and categorical variables, interactions between variables, polynomial terms, and logarithmic transformations. We also discuss heterogeneous treatment effects and weighted regression. These tools are essential for applied work and appear in virtually every empirical paper.

7.2 Learning Objectives

After completing this chapter, students will be able to:

1. Use **binary dummy variables** to capture discrete categories.
 2. Avoid the **dummy variable trap** (perfect multicollinearity with the intercept).
 3. Include **interaction terms** to allow effects to vary across groups.
 4. Interpret **log-linear**, **log-log**, and **linear-log** regression specifications.
 5. Use **polynomial regression** to capture non-linear relationships.
 6. Interpret **regression with saturated specifications** as estimating the CEF exactly.
 7. Understand how heterogeneous treatment effects are reflected in regression.
 8. Apply **weighted least squares (WLS)** when error variances are known.
-

7.3 7.1 Binary and Categorical Variables

7.3.1 7.1.1 The Binary Dummy Variable

A **binary (dummy) variable** $D_i \in \{0, 1\}$ captures membership in a category:

$$D_i = \begin{cases} 1 & \text{if condition holds} \\ 0 & \text{otherwise} \end{cases}$$

Examples: employed/unemployed, male/female, urban/rural, treated/not treated.

Regression with a single dummy:

$$y_i = \alpha + \rho D_i + u_i$$

Interpretation:

- $\alpha = E[y_i \mid D_i = 0] = \text{mean outcome for the reference group.}$
- $\rho = E[y_i \mid D_i = 1] - E[y_i \mid D_i = 0] = \text{difference in mean outcomes between the two groups.}$

The OLS estimate $\hat{\rho}$ is simply the difference in sample means: $\bar{y}_1 - \bar{y}_0$.

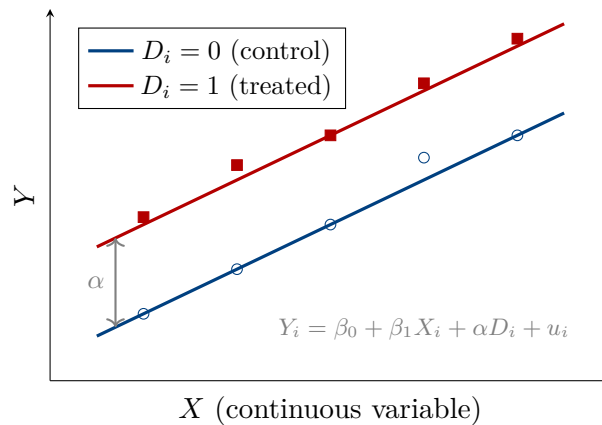


Figure 7.1: A dummy variable shifts the intercept by α while keeping the slope constant. The two regression lines for treated ($D_i = 1$, red) and untreated ($D_i = 0$, blue) are parallel, separated by the treatment effect α .

7.3.2 7.1.2 Categorical Variables with J Categories

For a variable with J categories (e.g., education levels: less than high school, high school, some college, college, graduate), create $J - 1$ dummy variables, omitting one

reference category:

$$y_i = \alpha + \sum_{j=2}^J \rho_j D_{ij} + u_i$$

where $D_{ij} = 1$ if individual i is in category j .

Interpretation: Each ρ_j is the mean difference in y between category j and the reference category (category 1), conditional on other controls.

The Dummy Variable Trap: Including J dummies plus an intercept creates perfect multicollinearity: $\sum_{j=1}^J D_{ij} = 1 = \text{intercept}$. Always omit one category.

Example 7.1 (Education and Wages): Create dummies for education groups:

- $D_{\text{HS},i} = 1$ if high school graduate (reference category: less than HS)
- $D_{\text{SC},i} = 1$ if some college
- $D_{\text{BA},i} = 1$ if bachelor's degree
- $D_{\text{MA},i} = 1$ if graduate degree

Coefficients $\hat{\rho}_{\text{HS}}, \hat{\rho}_{\text{SC}}, \hat{\rho}_{\text{BA}}, \hat{\rho}_{\text{MA}}$ are the wage premia for each group relative to those with less than a high school diploma.

7.4 7.2 Interaction Terms

7.4.1 7.2.1 Why Interactions?

An **interaction term** allows the effect of one variable to depend on the value of another. Without interactions, we impose that the slope coefficient on x is the same for all individuals. This is often too restrictive.

7.4.2 7.2.2 Interaction Between Two Dummy Variables

$$y_i = \alpha + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 (D_{1i} \times D_{2i}) + u_i$$

The expected value of y_i for each combination:

D_1	D_2	$E[y_i]$
0	0	α
1	0	$\alpha + \beta_1$
0	1	$\alpha + \beta_2$
1	1	$\alpha + \beta_1 + \beta_2 + \beta_3$

The interaction coefficient β_3 is the **additional effect** of having both $D_1 = 1$ and $D_2 = 1$ beyond the sum of their individual effects.

Example 7.2: Regress wages on a female dummy (D_1), a college dummy (D_2), and their interaction. The interaction coefficient captures whether the college wage premium is different for women than for men.

7.4.3 7.2.3 Interaction Between a Dummy and a Continuous Variable

$$y_i = \alpha + \beta_1 D_i + \beta_2 x_i + \beta_3 (D_i \times x_i) + u_i$$

Interpretation:

- When $D_i = 0$: $E[y_i | x_i, D_i = 0] = \alpha + \beta_2 x_i$ (slope = β_2).
- When $D_i = 1$: $E[y_i | x_i, D_i = 1] = (\alpha + \beta_1) + (\beta_2 + \beta_3)x_i$ (slope = $\beta_2 + \beta_3$).

The interaction allows for different slopes in the two groups. β_3 is the **differential slope** for the group with $D_i = 1$.

Example 7.3: Allow the return to education to differ by gender. Regress log wages on education, a female dummy, and the education $\times$ female interaction. The interaction coefficient gives the female-male difference in returns to education.

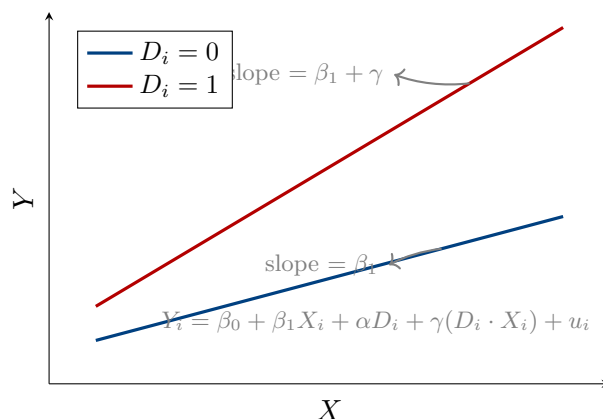


Figure 7.2: An interaction term $D_i \cdot X_i$ allows the slope of X to differ between groups. When $\gamma \neq 0$, the two regression lines are *not* parallel: the effect of X on Y depends on treatment status.

7.4.4 7.2.4 The Saturated Regression Model

A **saturated** model includes a separate indicator for every possible combination of the categorical variables. For two binary variables D_1 and D_2 :

$$y_i = \alpha + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 (D_{1i} \times D_{2i}) + u_i$$

This is saturated for two binary variables. The OLS estimates of $E[y \mid D_1, D_2]$ for each cell are exactly the sample cell means. **The saturated model estimates the CEF exactly** (for discrete regressors).

7.5 7.3 Polynomial and Logarithmic Transformations

7.5.1 7.3.1 Polynomial Regression

To capture non-linearities in a continuous variable, include higher-order polynomial terms:

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \cdots + \beta_p x_i^p + u_i$$

This is still a **linear** regression (linear in the parameters), so OLS applies directly.

Application: Experience-Earnings Profile

In the labour economics literature, the relationship between experience and wages is typically hump-shaped — wages rise rapidly at first, then more slowly, eventually declining near retirement. A quadratic specification captures this:

$$\log(\text{wage}_i) = \beta_0 + \beta_1 \text{exp}_i + \beta_2 \text{exp}_i^2 + \dots$$

The peak of the experience-earnings profile occurs at:

$$\text{exp}^* = -\frac{\hat{\beta}_1}{2\hat{\beta}_2}$$

Example 7.4: Typical estimates: $\hat{\beta}_1 \approx 0.04$, $\hat{\beta}_2 \approx -0.0007$. Peak at $\text{exp}^* \approx 0.04/(2 \times 0.0007) \approx 28$ years of experience.

7.5.2 7.3.2 Logarithmic Transformations

Logarithmic transformations are ubiquitous in economics. They:

- Reduce skewness in right-tailed distributions (wages, income, firm size).
- Allow interpretation of coefficients as **percentage changes**.
- Provide a natural way to estimate **elasticities**.

Three standard specifications:

Model	Equation	Interpretation of $\hat{\beta}$
Log-linear	$\log(y_i) = \beta_0 + \beta_1 x_i + u_i$	1-unit $\uparrow$ in $x \rightarrow 100\hat{\beta}_1\%$ change in y
Linear-log	$y_i = \beta_0 + \beta_1 \log(x_i) + u_i$	1% $\uparrow$ in $x \rightarrow \hat{\beta}_1/100$ unit change in y
Log-log	$\log(y_i) = \beta_0 + \beta_1 \log(x_i) + u_i$	1% $\uparrow$ in $x \rightarrow \hat{\beta}_1\%$ change in y (elasticity)

Example 7.5 (Mincer Earnings Equation): The classic Mincer (1974) equation uses a log-linear specification:

$$\log(\text{wage}_i) = \beta_0 + \beta_1 \cdot \text{schooling}_i + \beta_2 \cdot \text{exp}_i + \beta_3 \cdot \text{exp}_i^2 + u_i$$

The coefficient $\hat{\beta}_1 \approx 0.10$ implies each additional year of schooling raises wages by approximately 10%.

Approximation note: $\log(y_1) - \log(y_0) \approx (y_1 - y_0)/y_0$ for small changes. For large changes ($>20\%$), the approximation becomes less accurate and we should use $\exp(\hat{\beta}_1) - 1$ for the exact percentage change.

7.6 7.4 Heterogeneous Treatment Effects and Regression

7.6.1 7.4.1 What Regression Recovers with Heterogeneous Effects

In Chapter 1, we assumed constant treatment effects: $Y_{1i} - Y_{0i} = \rho$ for all i . In reality, treatment effects vary across individuals. What does OLS estimate in this case?

Consider:

$$Y_i = \alpha + \rho_i D_i + \eta_i$$

where $\rho_i = Y_{1i} - Y_{0i}$ is the individual treatment effect and $\eta_i = Y_{0i} - E[Y_{0i}]$.

If D_i is randomly assigned, one can show that OLS estimates:

$$\hat{\rho}^{OLS} \xrightarrow{p} E[\rho_i] + \frac{\text{Cov}(\rho_i, D_i(1 - D_i))}{E[D_i(1 - D_i)]}$$

Under random assignment (D_i independent of ρ_i), this simplifies to:

$$\hat{\rho}^{OLS} \xrightarrow{p} E[\rho_i] = \text{ATE}$$

OLS recovers the ATE under random assignment, even with heterogeneous effects.

7.6.2 7.4.2 Regression Weighted Average of Effects

More generally, Angrist (1998) shows that with heterogeneous effects and a continuous treatment, OLS computes a **variance-weighted average** of causal effects:

$$\hat{\beta}^{OLS} \xrightarrow{p} \frac{E[\omega_i \cdot \beta_i]}{E[\omega_i]}$$

where $\omega_i = (x_i - E[x_i])^2$ and β_i is the individual-specific treatment effect. Observations with more variation in x_i receive higher weight.

Implication: OLS gives greater weight to the treatment effect for individuals whose x_i is far from the mean. This is not necessarily the same as the average effect for the typical individual.

7.7 7.5 Weighting and Weighted Least Squares

7.7.1 7.5.1 When to Weight?

Weighted regression assigns observation i weight $w_i > 0$ and minimizes:

$$\sum_{i=1}^n w_i (y_i - \mathbf{x}_i^T \boldsymbol{\beta})^2$$

The WLS estimator is:

$$\hat{\boldsymbol{\beta}}^{WLS} = (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{y}$$

where $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$.

Two main justifications for weighting:

1. **Known heteroskedasticity:** If $V(u_i) = \sigma^2/w_i$, then WLS with weights w_i is equivalent to OLS on a transformed model with homoskedastic errors, making WLS the efficient estimator (BLUE under heteroskedasticity).
1. **Survey weights:** Complex surveys oversample certain groups. Survey weights adjust for differential sampling probabilities to ensure estimates are representative of the population.

Example 7.6: Suppose $V(u_i) = \sigma^2 x_i$ (variance proportional to x_i). Use WLS with $w_i = 1/x_i$. This makes the transformed errors homoskedastic: $u_i^* = u_i/\sqrt{x_i}$ has constant variance σ^2 .

7.7.2 7.5.2 Debate on Weighting in Applied Work

A subtle issue (Solon, Haider, and Wooldridge, 2015): survey weights adjust for sampling design. But should we use them to estimate conditional means?

- If the model is **correctly specified** (CEF is truly linear with the same coefficients for all groups), weighting is unnecessary — unweighted OLS is more efficient.
- If the model is **misspecified** (the CEF is non-linear or slopes vary across groups), weighted estimates may better describe the population-representative effect.

There is no universal answer. Practical guidance: report both weighted and unweighted estimates; if they differ substantially, investigate why.

7.8 Chapter Summary

- **Dummy variables** capture categorical distinctions; always omit one reference category to avoid the dummy variable trap.
 - **Interaction terms** allow effects to vary across groups or with the level of another variable.
 - The **saturated model** (full set of interactions for discrete regressors) estimates the CEF exactly at each cell.
 - **Polynomial terms** capture non-linearities; logarithmic transformations facilitate elasticity interpretation.
 - With heterogeneous treatment effects, OLS estimates a **variance-weighted average** of effects.
 - **Weighted least squares (WLS)** is appropriate under known heteroskedasticity or with survey weights.
-

7.9 Key Terms

Dummy variable — a binary $\{0, 1\}$ variable indicating group membership.

Reference category — the omitted category in a set of dummy variables.

Dummy variable trap — perfect multicollinearity from including all category dummies with an intercept.

Interaction term — the product of two regressors, allowing the effect of one to vary with the other.

Saturated model — a model that includes all possible combinations of categorical variables, estimating the CEF exactly.

Log-linear model — $\log(y) = \mathbf{x}^T\boldsymbol{\beta} + u$; coefficients are approximate percentage changes.

Elasticity — coefficient from a log-log model; measures percentage change in y per 1% change in x .

Mincer earnings equation — the classic log-linear wage regression with schooling and a quadratic in experience.

Weighted Least Squares (WLS) — minimizes a weighted sum of squared residuals.

7.10 Exercises

7.10.1 Conceptual Questions

7.1 You include a gender dummy (1 = female) in a wage regression. The coefficient is -0.15 . Interpret this coefficient. What does it mean? What doesn't it tell you?

7.2 Explain the dummy variable trap. Why is it a problem? How do you avoid it?

7.3 A researcher runs: $\log(\text{wage}) = \beta_0 + \beta_1\text{educ} + \beta_2\text{female} + \beta_3(\text{educ} \times \text{female}) + u$. Interpret each coefficient.

7.4 In the quadratic experience-earnings model, derive the formula for the experience level at which predicted wages are maximized.

7.5 "A log-log regression coefficient is the elasticity of y with respect to x ." Is this exactly correct? When is it an approximation?

7.10.2 Analytical Questions

7.6 Suppose $y_i = \alpha + \beta_1 D_{1i} + \beta_2 D_{2i} + \beta_3 (D_{1i} \times D_{2i}) + u_i$ where D_1 = male (0=female), D_2 = college (0=no college).

Write the expected value of y for each of the four groups: (female, no college), (female, college), (male, no college), (male, college). Express each in terms of $\alpha, \beta_1, \beta_2, \beta_3$.

7.7 Show that the OLS estimator for WLS with weights w_i solves $\hat{\boldsymbol{\beta}}^{WLS} = (\mathbf{X}^T \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{W} \mathbf{y}$.

7.8 Prove that the saturated model with two binary regressors produces fitted values equal to the sample cell means.

7.10.3 Applied Questions

7.9 Using CPS data, estimate the Mincer earnings equation. Report: (a) the return to education; (b) the return to experience at 5, 15, and 25 years; (c) the peak experience level.

7.10 Add an interaction between education and ethnicity. Does the return to education differ significantly by ethnicity?

7.11 R Lab 7: Dummies, Interactions, and Non-linearities

```
1 library(tidyverse)
2 library(AER)
3 data("CPS1988")
4 CPS1988 <- CPS1988 %>%
5   mutate(log_wage = log(wage),
6          ethnicity_dummy = as.integer(ethnicity == "afam"))
7
8 # 7.1: Mincer earnings equation
9 mincer <- lm(log_wage ~ education + experience + I(experience^2),
10             data = CPS1988)
11 summary(mincer)
12
13 # Peak experience
14 beta_exp <- coef(mincer)["experience"]
15 beta_exp2 <- coef(mincer)["I(experience^2)"]
16 peak_exp <- -beta_exp / (2 * beta_exp2)
17 cat("Peak experience:", round(peak_exp, 1), "years\n")
18
19 # 7.2: Interaction with ethnicity dummy
20 mincer_int <- lm(log_wage ~ education * ethnicity + experience +
21                 I(experience^2),
22                 data = CPS1988)
23 summary(mincer_int)
24
25 # 7.3: Plot experience-earnings profile
26 exp_grid <- seq(0, 45, by = 1)
27 # Predicted log wages at mean education, varying experience
28 newdata <- data.frame(
29   education = mean(CPS1988$education),
30   experience = exp_grid,
31   ethnicity = "cauc"
32 )
33 newdata$pred <- predict(mincer_int, newdata = newdata)
34
35 ggplot(newdata, aes(x = experience, y = pred)) +
36   geom_line(color = "steelblue", linewidth = 1.2) +
37   labs(
38     title = "Experience-Earnings Profile",
39     subtitle = "Quadratic specification at mean education",
40     x = "Years of Experience",
41     y = "Predicted Log Wage"
42   ) +
43   theme_minimal()
44
45 # 7.4: Log-log regression for wage elasticity w.r.t. education
```

```
44 log_log <- lm(log_wage ~ log(education) + experience +  
    I(experience^2), data = CPS1988)  
45 summary(log_log)  
46 cat("Elasticity of wage w.r.t. education:",  
    round(coef(log_log)["log(education)"], 3), "\n")
```

End of Chapter 7

References

- Angrist, J. D. (1998). Estimating the labor market impact of voluntary military service using social security data on military applicants. *Econometrica*, 66(2), 249–288.
- Mincer, J. (1974). *Schooling, Experience, and Earnings*. NBER.
- Solon, G., Haider, S. J., and Wooldridge, J. M. (2015). What are we weighting for? *Journal of Human Resources*, 50(2), 301–316.
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Part III

Identification Strategies

Chapter 8

Chapter 8: Instrumental Variables and Two-Stage Least Squares

8.1 Chapter Overview

Chapters 2–7 developed regression analysis as a tool for estimating conditional means and partial effects. But we repeatedly noted that OLS identifies causal effects only under strict exogeneity — an assumption that fails whenever there is omitted variable bias, reverse causality, or measurement error. This chapter introduces **Instrumental Variables (IV)** — arguably the most important identification strategy in applied economics — as a way to estimate causal effects when OLS is endogenous.

The key insight: if we can find a variable Z_i (the "instrument") that is correlated with the endogenous regressor X_i but affects the outcome Y_i only through X_i , then we can use the exogenous variation in Z_i to "purge" the endogeneity from X_i .

8.2 Learning Objectives

After completing this chapter, students will be able to:

1. Explain the **endogeneity problem** and why OLS fails.
 2. Define the two conditions for a valid instrument: **relevance** and **exclusion**.
 3. Derive the **IV estimator** (Wald estimator) in the bivariate case.
 4. Derive and interpret the **Two-Stage Least Squares (2SLS)** estimator in the general case.
 5. Derive the asymptotic distribution of 2SLS.
 6. Explain the **weak instruments** problem and its consequences.
 7. Use the **first-stage F -statistic** to test for weak instruments.
 8. Apply IV methods to canonical empirical examples (Angrist and Krueger, 1991).
-

8.3 8.1 The Endogeneity Problem

Definition 8.1 (Endogeneity): The regressor X_i is **endogenous** in the regression $Y_i = \alpha + \rho X_i + u_i$ if $\text{Cov}(X_i, u_i) \neq 0$.

When this holds, OLS is biased and inconsistent:

$$\text{plim}(\hat{\rho}^{OLS}) = \rho + \frac{\text{Cov}(X_i, u_i)}{V(X_i)} \neq \rho$$

The three main sources of endogeneity are:

1. **Omitted variables:** u_i contains an omitted variable A_i correlated with X_i (e.g., ability in the wage regression).
 2. **Reverse causality:** Y_i causes X_i as well as vice versa (e.g., income and health).
 3. **Measurement error:** X_i is measured with error correlated with the true X_i^* (attenuation bias).
-

8.4 8.2 The Instrumental Variables (IV) Estimator

8.4.1 8.2.1 What Is an Instrument?

An **instrument** Z_i must satisfy two conditions:

Condition 1 — Relevance (First Stage): $\text{Cov}(Z_i, X_i) \neq 0$.

The instrument must be correlated with the endogenous regressor.

Condition 2 — Exclusion Restriction (Exogeneity): $\text{Cov}(Z_i, u_i) = 0$.

The instrument must be uncorrelated with the structural error — it affects Y_i only through X_i .

Graphically: $Z_i \rightarrow X_i \rightarrow Y_i$ (with Z_i having no direct path to Y_i).

8.4.2 8.2.2 The Wald Estimator (Bivariate Case)

In the bivariate model $Y_i = \alpha + \rho X_i + u_i$ with a single binary instrument $Z_i \in \{0, 1\}$:

$$\hat{\rho}^{IV} = \frac{E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0]}{E[X_i | Z_i = 1] - E[X_i | Z_i = 0]} = \frac{\text{Reduced Form}}{\text{First Stage}}$$

This is the **Wald estimator**: the ratio of the effect of the instrument on the outcome to the effect of the instrument on the endogenous regressor.

Why this works: If Z_i is randomly assigned (or as good as random), it affects Y_i only through X_i . Dividing the total effect of Z_i on Y_i by the effect of Z_i on X_i "scales" the outcome effect to be per unit of X_i — yielding the causal effect of X_i on Y_i .

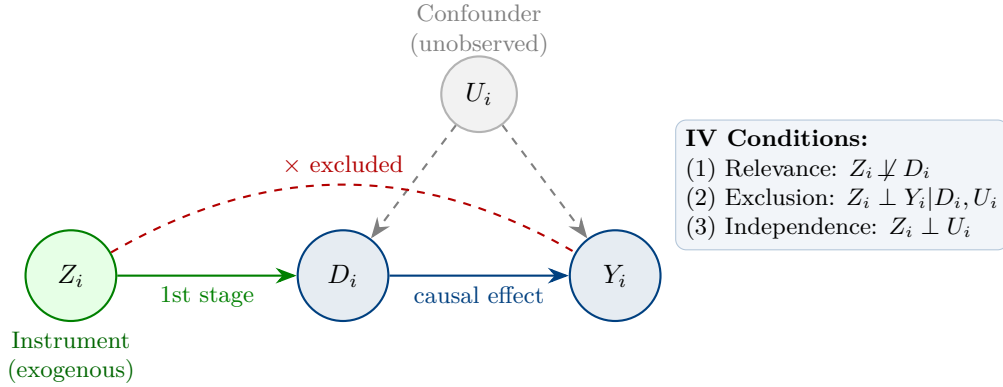


Figure 8.1: DAG for an instrumental variables (IV) strategy. The instrument Z_i (green) affects the outcome Y_i *only* through the treatment D_i (exclusion restriction, red dashed), is correlated with D_i (relevance), and is independent of the unobserved confounder U_i .

8.4.3 8.2.3 Consistency of the IV Estimator

Proof: Using sample moments:

$$\hat{\rho}^{IV} = \frac{\text{Cov}_s(Z_i, Y_i)}{\text{Cov}_s(Z_i, X_i)}$$

Substituting $Y_i = \alpha + \rho X_i + u_i$:

$$\frac{\text{Cov}(Z_i, Y_i)}{\text{Cov}(Z_i, X_i)} = \frac{\text{Cov}(Z_i, \alpha + \rho X_i + u_i)}{\text{Cov}(Z_i, X_i)} = \frac{\rho \text{Cov}(Z_i, X_i) + \text{Cov}(Z_i, u_i)}{\text{Cov}(Z_i, X_i)}$$

Under the exclusion restriction $\text{Cov}(Z_i, u_i) = 0$: $\hat{\rho}^{IV} \xrightarrow{p} \rho$. $\square$

8.5 8.3 Two-Stage Least Squares (2SLS): The General Case

With multiple endogenous variables and multiple instruments, we use **Two-Stage Least Squares (2SLS)**.

8.5.1 8.3.1 The General Setup

The **structural equation** is:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}$$

where $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$: $\mathbf{X}_1$ contains the exogenous regressors (controls) and $\mathbf{X}_2$ contains the k_2 endogenous variables.

The **instrument set** is $\mathbf{Z} = [\mathbf{X}_1, \mathbf{Z}_2]$: the same exogenous controls $\mathbf{X}_1$ plus $\ell \geq k_2$ excluded instruments $\mathbf{Z}_2$.

For **identification**: $\ell \geq k_2$ (order condition — at least as many instruments as endogenous variables).

8.5.2 8.3.2 The Two Stages

Stage 1 (First Stage): Regress each endogenous variable on all instruments:

$$\mathbf{X}_2 = \mathbf{Z}\Pi + \mathbf{V}$$

Obtain fitted values: $\hat{\mathbf{X}}_2 = \mathbf{Z}(\mathbf{Z}^T\mathbf{Z})^{-1}\mathbf{Z}^T\mathbf{X}_2 = \mathbf{P}_Z\mathbf{X}_2$

This decomposes $\mathbf{X}_2$ into a part predicted by the instruments ($\hat{\mathbf{X}}_2$, exogenous variation) and a residual ($\hat{\mathbf{V}}$, endogenous variation).

Stage 2 (Second Stage): Regress $\mathbf{y}$ on $\hat{\mathbf{X}} = [\mathbf{X}_1, \hat{\mathbf{X}}_2]$:

$$\hat{\beta}^{2SLS} = (\hat{\mathbf{X}}^T\hat{\mathbf{X}})^{-1}\hat{\mathbf{X}}^T\mathbf{y}$$

8.5.3 8.3.3 The 2SLS Estimator in Matrix Form

The 2SLS estimator can be written more compactly:

$$\hat{\beta}^{2SLS} = (\mathbf{X}^T\mathbf{P}_Z\mathbf{X})^{-1}\mathbf{X}^T\mathbf{P}_Z\mathbf{y}$$

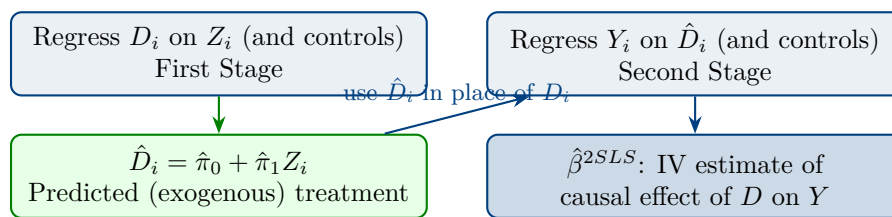
Key insight: 2SLS projects both $\mathbf{X}$ and $\mathbf{y}$ onto the column space of $\mathbf{Z}$ before running the regression. This replaces the endogenous variation in $\mathbf{X}$ with the exogenous variation from the instruments.

Special case: When $\mathbf{Z} = \mathbf{X}$ (all regressors are exogenous), $\mathbf{P}_Z = \mathbf{P}_X$ and 2SLS = OLS.

8.6 8.4 Asymptotic Properties of 2SLS

Theorem 8.1 (Consistency of 2SLS): Under the conditions:

1. Linear model $\mathbf{y} = \mathbf{X}\beta + \mathbf{u}$
2. Instrument relevance: $E[\mathbf{z}_i\mathbf{x}_i^T]$ has rank k
3. Exclusion restriction: $E[\mathbf{z}_i u_i] = \mathbf{0}$



Key: $\hat{D}_i$ captures only the exogenous variation in D_i induced by Z_i , purging the endogeneity caused by U_i .

Figure 8.2: Two-Stage Least Squares (2SLS). In the first stage, we project D_i onto the instrument Z_i to obtain the “clean” predicted values $\hat{D}_i$. In the second stage, we regress Y_i on $\hat{D}_i$, using only the exogenous variation in treatment to identify the causal effect.

Then $\hat{\beta}^{2SLS} \xrightarrow{p} \beta$.

Theorem 8.2 (Asymptotic Distribution of 2SLS):

$$\sqrt{n}(\hat{\beta}^{2SLS} - \beta) \xrightarrow{d} N(\mathbf{0}, \mathbf{V}^{2SLS})$$

where under homoskedasticity:

$$\mathbf{V}^{2SLS} = \sigma^2 \left(E[\mathbf{x}_i \mathbf{z}_i^T] \left(E[\mathbf{z}_i \mathbf{z}_i^T] \right)^{-1} E[\mathbf{z}_i \mathbf{x}_i^T] \right)^{-1}$$

In practice, use the 2SLS residuals $\hat{\mathbf{u}}^{2SLS} = \mathbf{y} - \mathbf{X}\hat{\beta}^{2SLS}$ (note: NOT the residuals from Stage 2) to estimate σ^2 and use robust/cluster SEs.

Important warning: Never use software’s default Stage-2 SEs — they use Stage-2 residuals and are wrong. Always use the correct 2SLS standard errors (or use dedicated IV commands: `ivreg()` in R, `ivregress` in Stata).

8.7 8.5 Instrument Validity: Relevance and Exclusion

8.7.1 8.5.1 Testing Relevance: The First-Stage F -Statistic

The relevance assumption can be **tested**: run the first stage and report the F -statistic for the joint significance of the excluded instruments.

Rule of thumb (Staiger and Stock, 1997): First-stage $F > 10$ is often cited as a threshold for “strong” instruments.

The weak instruments problem (preview): When instruments are only weakly correlated with the endogenous variable (F close to 0), 2SLS has poor finite-sample properties:

- The 2SLS estimator is biased toward OLS.
- Standard errors are misleading; t -tests are severely size-distorted.

8.7.2 8.5.2 The Exclusion Restriction Cannot Be Tested

The exclusion restriction $E[\mathbf{z}_i u_i] = \mathbf{0}$ is **fundamentally untestable** (in the sense that u_i is unobserved). Researchers must argue for it on substantive grounds.

Testing over-identifying restrictions (when $\ell > k_2$): When we have more instruments than needed, we can test whether different instruments give similar estimates — the **Sargan-Hansen test**. Rejection suggests at least one instrument is invalid.

However: The Sargan-Hansen test only checks consistency across instruments, not absolute validity. If all instruments are invalid in the same way, it will not detect the problem.

8.8 8.6 Weak Instruments: Diagnosis and Remedies

8.8.1 8.6.1 The Weak Instruments Problem

Definition 8.2 (Weak Instruments): Instruments are weak if the first-stage relationship is weak — the instruments explain little variation in the endogenous variable beyond controls.

With weak instruments, even modest violations of the exclusion restriction cause large bias. The bias of 2SLS relative to OLS is approximately:

$$\text{Bias}(2SLS) \approx \frac{1}{F} \text{Bias}(OLS)$$

where F is the first-stage F -statistic. With $F = 5$, 2SLS is biased about 20% of the way toward OLS.

8.8.2 8.6.2 Detection

- **First-stage F -statistic:** Always report and examine. $F < 10$ is cause for concern.
- **Partial R^2 of instruments:** The fraction of variation in the endogenous variable explained by the excluded instruments alone.

- **Anderson-Rubin confidence sets:** Inference that is robust to weak instruments.

8.8.3 8.6.3 Remedies

- Find stronger instruments.
 - Use **Limited Information Maximum Likelihood (LIML)**, which is median-unbiased under many-instrument asymptotics.
 - Use **Anderson-Rubin tests** for inference robust to weak instruments.
-

8.9 8.7 Empirical Illustration: Angrist and Krueger (1991)

8.9.1 8.7.1 The Research Question

Angrist and Krueger (1991) study the return to education using compulsory schooling laws as an instrument. Their key insight: US students born early in the year (Q1) start school at an older age and can legally drop out having completed less education than students born later in the year (Q4). Therefore, **quarter of birth** creates quasi-random variation in education levels.

Instrument: Quarter of birth (Q1 vs. Q4).

Endogenous variable: Years of schooling.

Outcome: Log weekly earnings.

8.9.2 8.7.2 Instrument Validity

Relevance: Q1 births have on average 0.1–0.2 fewer years of schooling than Q4 births. This is the "season of birth" effect documented in the data. The first-stage relationship, while small, is statistically significant.

Exclusion restriction: Quarter of birth should affect earnings *only* through its effect on schooling. (This is plausible if Q1 vs. Q4 birth is essentially random and birth quarter has no direct effect on labor market outcomes.)

8.9.3 8.7.3 Results

OLS estimate: Return to schooling $\approx 7\text{--}8\%$ per year.

IV/2SLS estimate (using Q1 birth as instrument): Return to schooling $\approx 8\text{--}10\%$ per year.

The OLS and IV estimates are remarkably similar. This suggests (though doesn't prove) that ability bias in OLS may be smaller than often assumed, or that the instrument is strong enough to identify a similar population parameter.

Box 8.1 — The Angrist-Krueger (1991) Paper

Angrist and Krueger use quarter of birth interacted with year of birth and state of birth as instruments — generating many instruments. Card (1995) later pointed out that many weak instruments can lead to biased 2SLS estimates toward OLS. This prompted important work on weak instruments and LIML. The original paper remains a canonical example of creative instrument design.

8.9.4 8.7.4 Reduced Form and First Stage

A useful way to present IV results is to show:

1. **First Stage:** Quarter of birth $\rightarrow$ Education (must be significant).
2. **Reduced Form:** Quarter of birth $\rightarrow$ Log wages (directly).
3. **2SLS:** The ratio, rescaling reduced form to "per year of education."

If the reduced form is insignificant, the instrument has no explanatory power — the IV estimate has no credibility.

8.10 Chapter Summary

- **Endogeneity** ($\text{Cov}(X_i, u_i) \neq 0$) makes OLS biased and inconsistent.
 - An instrument Z_i must satisfy **relevance** ($\text{Cov}(Z, X) \neq 0$) and **exclusion** ($\text{Cov}(Z, u) = 0$).
 - The **Wald estimator** is the ratio of the reduced form to the first stage.
 - **2SLS** generalizes IV to multiple endogenous variables and instruments by projecting regressors onto the instrument space.
 - **Weak instruments** cause large bias toward OLS; always report the first-stage F -statistic.
 - The exclusion restriction is untestable; defend it on substantive grounds.
 - Angrist and Krueger (1991) is a canonical application using quarter of birth as an instrument for education.
-

8.11 Key Terms

Endogeneity — $\text{Cov}(X_i, u_i) \neq 0$; OLS is biased and inconsistent.

Instrument — variable Z_i correlated with X_i but uncorrelated with u_i .

Relevance — $\text{Cov}(Z_i, X_i) \neq 0$.

Exclusion restriction — $\text{Cov}(Z_i, u_i) = 0$.

Wald estimator — $\hat{\rho}^{IV} = \frac{\text{Cov}(Z, Y)}{\text{Cov}(Z, X)}$.

Two-Stage Least Squares (2SLS) — IV estimator for the general case with multiple instruments and endogenous variables.

First stage — regression of endogenous variable(s) on all instruments.

Reduced form — regression of outcome on instruments directly.

Weak instruments — instruments with small first-stage F -statistic; cause bias and size distortion.

Sargan-Hansen test — test of over-identifying restrictions; checks consistency across instruments.

8.12 Exercises

8.12.1 Conceptual Questions

8.1 Explain why OLS is biased when $\text{Cov}(X_i, u_i) \neq 0$. Give three economic examples of endogeneity.

8.2 State the two conditions for a valid instrument. Can we test both in practice? Explain.

8.3 Describe the two stages of 2SLS in plain language. What is "purged" in the first stage?

8.4 Why is the first-stage F -statistic important? What is the consequence of ignoring weak instruments?

8.5 "The exclusion restriction is satisfied if the instrument is randomly assigned." Evaluate this claim.

8.12.2 Analytical Questions

8.6 In the bivariate model $Y_i = \alpha + \rho X_i + u_i$, with a single instrument Z_i : (a) Derive the Wald estimator. (b) Show it is consistent when $\text{Cov}(Z_i, u_i) = 0$ and $\text{Cov}(Z_i, X_i) \neq 0$. (c) What is the probability limit when $\text{Cov}(Z_i, u_i) \neq 0$?

8.7 Show that the 2SLS estimator equals $\hat{\beta}^{2SLS} = (\mathbf{X}^T \mathbf{P}_Z \mathbf{X})^{-1} \mathbf{X}^T \mathbf{P}_Z \mathbf{y}$.

8.8 In the Angrist-Krueger example with $Z_i = 1(\text{Q1 birth})$ and $X_i = \text{years of schooling}$:

- $E[Y_i | Z_i = 1] = 5.80, E[Y_i | Z_i = 0] = 5.90$
- $E[X_i | Z_i = 1] = 12.68, E[X_i | Z_i = 0] = 12.79$

Compute the Wald estimate of the return to schooling.

8.12.3 Applied Questions

8.9 Using the `card.dta` data (Card, 1995): college proximity as an instrument for education. In R: (a) Run OLS of log wages on education and controls. (b) Report the first-stage F -statistic. (c) Run 2SLS using college proximity as the instrument. (d) Compare OLS and 2SLS estimates. Does the direction of change make sense?

8.13 R Lab 8: IV and 2SLS

8.13.1 Setup

```
1 library(tidyverse)
2 library(AER)      # ivreg() for 2SLS
3 library(lmtest)
4 library(sandwich)
5
6 # Card (1995) data: effect of education on wages using college
  # proximity as IV
7 # Available in the ivreg package or can be read from the PS3 data
8 # install.packages("ivreg")
9 library(ivreg)
10 data("SchoolingReturns", package = "ivreg")
11 df <- SchoolingReturns
```

8.13.2 Exercise 8.1: OLS Baseline

```
1 # OLS regression: log wage on education and controls
2 ols_model <- lm(wage ~ education + experience + I(experience^2) +
3   ethnicity +
4   smsa + south, data = df)
5 summary(ols_model)
6 cat("OLS coefficient on education:",
7   round(coef(ols_model)["education"], 4), "\n")
```

8.13.3 Exercise 8.2: First Stage

```
1 # First stage: education on nearcollege (proximity to 4-year
  # college) + controls
2 first_stage <- lm(education ~ nearcollege + experience +
3   I(experience^2) +
4   ethnicity + smsa + south, data = df)
5 summary(first_stage)
```

```

5 cat("First-stage coefficient on nearcollege:",
    round(coef(first_stage)["nearcollege"], 4), "\n")
6 cat("First-stage F-statistic:",
    round(summary(first_stage)$fstatistic[1], 2), "\n")
7
8 # Partial F-test for just the instrument
9 ols_no_iv <- lm(education ~ experience + I(experience^2) + ethnicity
    + smsa + south, data = df)
10 F_partial <- anova(ols_no_iv, first_stage)[["F"]][2]
11 cat("Partial F-statistic for instrument:", round(F_partial, 2), "\n")

```

8.13.4 Exercise 8.3: Reduced Form

```

1 # Reduced form: log wage on nearcollege directly
2 reduced_form <- lm(wage ~ nearcollege + experience + I(experience^2)
    +
3     ethnicity + smsa + south, data = df)
4 summary(reduced_form)
5 cat("Reduced form coefficient:",
    round(coef(reduced_form)["nearcollege"], 4), "\n")

```

8.13.5 Exercise 8.4: 2SLS Estimation

```

1 # 2SLS using ivreg()
2 iv_model <- ivreg(wage ~ education + experience + I(experience^2) +
    ethnicity + smsa + south |
3     nearcollege + experience + I(experience^2) +
4     ethnicity + smsa + south,
5     data = df)
6 summary(iv_model, diagnostics = TRUE)
7 cat("2SLS coefficient on education:",
    round(coef(iv_model)["education"], 4), "\n")
8
9 # Wald estimator (bivariate comparison)
10 wald_est <- coef(reduced_form)["nearcollege"] /
    coef(first_stage)["nearcollege"]
11 cat("Wald estimate:", round(wald_est, 4), "\n")
12 cat("2SLS estimate:", round(coef(iv_model)["education"], 4), "\n")

```

8.13.6 Exercise 8.5: Visualizing First Stage

```

1 ggplot(df, aes(x = factor(nearcollege), y = education)) +
2   geom_boxplot(fill = "steelblue", alpha = 0.7) +
3   labs(
4     title = "First Stage: College Proximity and Years of Education",

```

```
5 subtitle = "Living near a college is associated with more
      schooling",
6 x = "Near College (0 = No, 1 = Yes)",
7 y = "Years of Education"
8 ) +
9 theme_minimal()
```

End of Chapter 8

References

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- Card, D. (1995). Using geographic variation in college proximity to estimate the return to schooling. In L. Christofides et al. (eds.), *Aspects of Labour Market Behaviour*. University of Toronto Press.
- Staiger, D., and Stock, J. H. (1997). Instrumental variables regression with weak instruments. *Econometrica*, 65(3), 557–586.
- Davidson, R., and MacKinnon, J. G. (2004). *Econometric Theory and Methods*. Chapter 8.
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Chapter 9

Chapter 9: Heterogeneous Treatment Effects and the Local Average Treatment Effect

9.1 Chapter Overview

Chapter 8 presented IV as a solution to endogeneity when the exclusion restriction holds. But what does IV actually estimate when treatment effects are heterogeneous — when the effect of the treatment varies across individuals? This chapter answers that question using the **potential outcomes framework** developed in Chapter 1. The central result — the **LATE Theorem** of Imbens and Angrist (1994) — shows that IV with a binary instrument estimates the **Local Average Treatment Effect**: the average causal effect for a specific subpopulation called **compliers** — individuals whose treatment status is affected by the instrument. This has profound implications for interpreting IV estimates and designing empirical research.

9.2 Learning Objectives

After completing this chapter, students will be able to:

1. Explain the limitations of the **constant-effects model** and why heterogeneous effects matter.
2. Define **potential outcomes** for both the outcome and the treatment.
3. Define the four **compliance types** (always-takers, never-takers, compliers, defiers).
4. State the four **LATE assumptions** (independence, exclusion, first stage, monotonicity).
5. State and prove the **LATE Theorem** (Imbens and Angrist, 1994).
6. Interpret IV estimates as LATE, and understand when LATE differs from ATE.

7. Explain the relationship between the **Wald estimator** and LATE.
 8. Apply the LATE framework to canonical examples (Vietnam draft lottery, compulsory schooling).
-

9.3 9.1 The Constant-Effects Model and Its Limitations

In Chapter 8 (and Chapter 1), we wrote the structural equation as:

$$Y_i = \alpha + \rho D_i + u_i$$

where $\rho = Y_{1i} - Y_{0i}$ is the same for everyone — the **constant-effects model**.

This assumption is convenient but unrealistic. In labour economics, the return to education varies by individual ability, occupation, and timing. In health economics, the effect of a treatment varies by disease severity. In program evaluation, training program effects vary by background and motivation.

The question this chapter answers: When effects are heterogeneous and we use an IV estimator, what causal parameter does it recover?

9.4 9.2 The Potential Outcomes Framework Revisited

We now have potential outcomes for **both** the treatment and the outcome.

Potential treatments: For each individual i and each possible value of the instrument $Z_i \in \{0, 1\}$:

D_{0i} = treatment status individual i would take if $Z_i = 0$

D_{1i} = treatment status individual i would take if $Z_i = 1$

Potential outcomes: Given treatment $d \in \{0, 1\}$:

Y_{di} = outcome individual i would experience if $D_i = d$

Observed outcomes (via the switching equations):

$$D_i = D_{0i} + (D_{1i} - D_{0i})Z_i$$

$$Y_i = Y_{0i} + (Y_{1i} - Y_{0i})D_i$$

9.5 9.3 SUTVA

A key background assumption is **Stable Unit Treatment Value Assumption (SUTVA)**:

Assumption 9.1 (SUTVA):

1. **No interference:** One unit's treatment does not affect another's outcomes.
2. **No hidden versions of treatment:** There is only one version of each treatment level.

SUTVA allows us to write individual potential outcomes without conditioning on others' treatments. It fails in settings with spillovers (e.g., vaccination programs that create herd immunity) or when there are multiple treatment variants (e.g., different types of job training).

9.6 9.4 Compliance Types: Always-Takers, Never-Takers, Compliers, Defiers

Given the instrument Z_i and potential treatments D_{0i}, D_{1i} , we can classify all individuals into four compliance types:

Type	D_{0i}	D_{1i}	Behavior
Always-takers	1	1	Take treatment regardless of Z
Never-takers	0	0	Never take treatment regardless of Z
Compliers	0	1	Take treatment when $Z = 1$, not when $Z = 0$
Defiers	1	0	Take treatment when $Z = 0$, not when $Z = 1$

Key insight:

- We can **never directly observe** which type an individual is, because we observe only one of D_{0i} or D_{1i} .
- Always-takers and never-takers are "uninformative" about the causal effect — their treatment status doesn't change with the instrument.
- The instrument only affects the treatment of **compliers** (and defiers).

Example 9.1 (Vietnam Draft Lottery): Consider a lottery that drafts men into military service.

- Always-takers: men who volunteer regardless of whether they're drafted.
- Never-takers: men who avoid service regardless (flee the country, obtain exemptions).

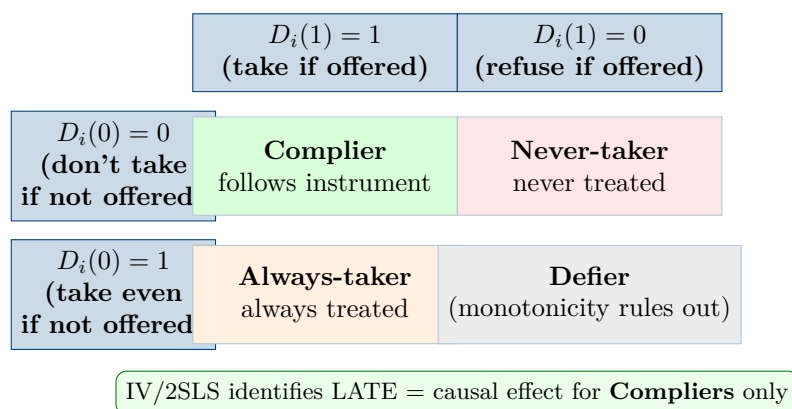


Figure 9.1: The four compliance types, defined by potential treatment status under each instrument value. IV estimates the Local Average Treatment Effect (LATE): the average causal effect for *compliers* — units whose treatment status is switched by the instrument. Under monotonicity (no defiers), LATE is point identified.

- Compliers: men who serve if drafted and don't serve if not drafted.
- Defiers: men who serve only when NOT drafted (unlikely to exist).

9.7 9.5 The Four LATE Assumptions

Assumption 9.2 (Independence / Random Assignment):

$$(Y_{0i}, Y_{1i}, D_{0i}, D_{1i}) \perp\!\!\!\perp Z_i$$

The instrument Z_i is independent of all potential outcomes and potential treatments. This is satisfied by random assignment.

Assumption 9.3 (Exclusion Restriction):

$$Y_{di} = Y_{di}(Z_i) \text{ for } d \in \{0, 1\}$$

Or more precisely: the instrument affects outcomes only through the treatment channel. Z_i has no direct effect on Y_i given D_i .

Assumption 9.4 (First Stage / Relevance):

$$E[D_{1i} - D_{0i}] \neq 0$$

The instrument has a non-zero (on average) effect on the treatment.

Assumption 9.5 (Monotonicity):

$$D_{1i} \geq D_{0i} \text{ for all } i \quad (\text{or } D_{1i} \leq D_{0i} \text{ for all } i)$$

The instrument shifts treatment in the same direction for everyone. This rules out defiers: if $Z = 1$ increases treatment for some, it does not decrease treatment for others.

The combination of the first-stage and monotonicity assumptions implies that the complier fraction is positive: $P(D_{1i} > D_{0i}) > 0$.

9.8 9.6 The LATE Theorem

Theorem 9.1 (Imbens and Angrist, 1994): Under Assumptions 9.2–9.5, the IV/Wald estimator identifies the **Local Average Treatment Effect**:

$$\text{LATE} \equiv E[Y_{1i} - Y_{0i} \mid D_{1i} > D_{0i}] = \frac{E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0]}{E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0]}$$

The LATE is the average treatment effect for the subpopulation of **compliers** — those individuals for whom $D_{1i} > D_{0i}$ (who are induced into treatment by the instrument).

Proof:

Denominator (First Stage):

$$E[D_i \mid Z_i = 1] - E[D_i \mid Z_i = 0]$$

By independence (Assumption 9.2): $E[D_i \mid Z_i = z] = E[D_{zi}]$.

$$= E[D_{1i}] - E[D_{0i}] = E[D_{1i} - D_{0i}]$$

By monotonicity: $D_{1i} - D_{0i} \in \{0, 1\}$ (no defiers means the difference is either 0 or 1). So:

$$= P(D_{1i} > D_{0i}) = P(\text{Complier})$$

Numerator (Reduced Form):

$$E[Y_i \mid Z_i = 1] - E[Y_i \mid Z_i = 0]$$

By independence and the switching equation:

$$\begin{aligned}
 &= E[Y_{D_{1i},i}] - E[Y_{D_{0i},i}] \\
 &= E[Y_{1i}D_{1i} + Y_{0i}(1 - D_{1i})] - E[Y_{1i}D_{0i} + Y_{0i}(1 - D_{0i})] \\
 &= E[(Y_{1i} - Y_{0i})(D_{1i} - D_{0i})]
 \end{aligned}$$

By monotonicity, $D_{1i} - D_{0i} \geq 0$, so:

$$\begin{aligned}
 &= E[(Y_{1i} - Y_{0i})(D_{1i} - D_{0i})] \\
 &= E[Y_{1i} - Y_{0i} \mid D_{1i} > D_{0i}] \cdot P(D_{1i} > D_{0i})
 \end{aligned}$$

(the always-takers and never-takers have $D_{1i} = D_{0i}$, so they contribute zero to the sum).

Combining:

$$\frac{\text{Numerator}}{\text{Denominator}} = \frac{E[Y_{1i} - Y_{0i} \mid D_{1i} > D_{0i}] \cdot P(\text{Complier})}{P(\text{Complier})} = E[Y_{1i} - Y_{0i} \mid D_{1i} > D_{0i}] = \text{LATE} \quad \square$$

9.9 9.7 Interpreting LATE

9.9.1 9.7.1 Who Are Compliers?

Compliers are the people "on the margin" between treatment and control who are moved by the instrument. Their identity depends on what the instrument is:

- **Vietnam draft lottery (Angrist, 1990):** Compliers are men who served in the military because they were drafted, but would not have served voluntarily.
- **Compulsory schooling (Angrist-Krueger, 1991):** Compliers are individuals who stayed in school because they weren't old enough to drop out by the end of the term, and would have dropped out otherwise.
- **College proximity (Card, 1995):** Compliers are individuals who attended college because a college was nearby, and would not have attended otherwise.

9.9.2 9.7.2 LATE vs. ATE vs. ATT

In general, **LATE** $\neq$ **ATE** $\neq$ **ATT** when:

1. Treatment effects are heterogeneous (vary across individuals).
2. The complier group differs systematically from the full population.

Parameter	Population	Method of identification
ATE	Full population	Random assignment to all
ATT	Treated units	Selection on observables (CIA)
LATE	Compliers	Instrumental variable

When does LATE $\approx$ ATE?

- When treatment effects are homogeneous.
- When compliers are representative of the full population.

When does LATE $\neq$ ATE?

- When the instrument selects a non-representative subset. The Vietnam draft lottery compliers are men who didn't volunteer — possibly those with lower "returns" to military service.

9.9.3 9.7.3 LATE and External Validity

The LATE from one instrument is not necessarily the same as the LATE from another instrument — even when both instruments are valid. Different instruments define different complier populations.

Example: Two valid instruments for education:

1. Quarter of birth (Angrist-Krueger): compliers are people who wanted to drop out as soon as legally allowed.
2. College proximity (Card): compliers are people who would have attended college if it were nearby.

These two groups likely have very different returns to education. This means the two IV estimates can legitimately differ — not because one is wrong, but because they estimate different parameters for different populations.

This has profound implications for **external validity**: an IV estimate tells you the causal effect for the compliers defined by that specific instrument, not necessarily the effect for the average person or for a new policy context.

9.10 9.8 2SLS as LATE with Multiple Instruments

When there are multiple instruments $Z_1, Z_2, \dots, Z_\ell$ for a single endogenous variable, 2SLS combines them. Under the LATE framework:

$$\hat{\rho}^{2SLS} \xrightarrow{p} \sum_{j=1}^{\ell} \omega_j \cdot \text{LATE}_j$$

where LATE_j is the LATE defined by instrument Z_j (the effect for compliers of instrument j) and ω_j are weights that sum to one. The weights are proportional to the variance of the first-stage predictions from each instrument.

This means 2SLS with multiple instruments estimates a **weighted average of LATEs** — a complex estimand whose interpretation requires care.

9.11 9.9 Empirical Applications

9.11.1 9.9.1 Application 1: Vietnam Draft Lottery (Angrist, 1990)

Research question: What is the effect of military service on lifetime earnings?

Identification problem: Veterans and non-veterans differ systematically in many ways (selection into service). OLS estimates of the effect of veteran status are confounded by these differences.

Instrument: Vietnam-era draft lottery. Men born in years 1950–1953 were assigned draft lottery numbers 1–365 based on birthday. Low lottery numbers were called first; men with numbers below a threshold were drafted.

Validity:

- **Independence:** Lottery numbers were randomly assigned (approximately).
- **Exclusion:** Lottery number affects earnings only through military service (no direct effect of knowing your lottery number on future earnings).
- **First stage:** Being assigned a low lottery number significantly increases the probability of military service.
- **Monotonicity:** Low numbers increase service (no defiers expected).

Results: IV (Wald) estimates suggest military service reduced civilian earnings by approximately 15% for compliers (white veterans), with effects persisting for at least 15 years. The OLS estimate was biased upward because veterans tend to come from lower-income backgrounds (attenuation by selection).

Who are the compliers? Men who were called up by the draft lottery and served. These are NOT representative of all veterans (who also include volunteers) and NOT

representative of all men (since many draft-eligible men found other ways to avoid service).

9.11.2 9.9.2 Application 2: Compulsory Schooling (Angrist and Krueger, 1991)

As discussed in Chapter 8:

Compliers: Men who would have dropped out of school as soon as legally permitted (when they turned 16 or after completing the grade they were enrolled in at age 16), but who in fact had to stay in school longer because they started school at a younger age (and were thus younger than the dropout cutoff when they reached certain grades).

LATE interpretation: The Angrist-Krueger estimates measure the return to education for this specific group of marginal students — those constrained by compulsory schooling laws. The returns for this group may be higher or lower than for the average student.

9.12 Chapter Summary

- With heterogeneous treatment effects, the question "what does IV estimate?" is non-trivial.
 - The **LATE Theorem** shows that IV identifies the average treatment effect for **compliers** — those whose treatment is affected by the instrument.
 - The four LATE assumptions are: independence, exclusion restriction, first stage, and monotonicity.
 - **Always-takers** and **never-takers** contribute no information to the IV estimate; their effects are "washed out."
 - **LATE** $\neq$ **ATE** when compliers are not representative of the full population.
 - Different instruments define different complier populations and may give different (but all valid) estimates.
 - The LATE framework forces us to be explicit about *which population* our IV estimate describes — crucial for policy relevance.
-

9.13 Key Terms

SUTVA — Stable Unit Treatment Value Assumption; no interference between units, no hidden treatment versions.

Potential treatments — D_{0i} and D_{1i} : treatment under $Z_i = 0$ and $Z_i = 1$.

Compliance types — always-takers, never-takers, compliers, defiers.

Compliers — individuals with $D_{1i} > D_{0i}$; treatment is affected by the instrument.

Monotonicity — no defiers: $D_{1i} \geq D_{0i}$ for all i .

LATE (Local Average Treatment Effect) — $E[Y_{1i} - Y_{0i} \mid D_{1i} > D_{0i}]$; average effect for compliers.

ATE — $E[Y_{1i} - Y_{0i}]$; average effect for the full population.

External validity — whether an estimate from one context generalizes to another.

9.14 Exercises

9.14.1 Conceptual Questions

9.1 Explain in your own words why IV estimates LATE rather than ATE.

9.2 A researcher uses random assignment of a health insurance subsidy as an instrument for health insurance take-up, to estimate the effect of insurance on health outcomes. Who are the compliers in this setting? Who are the always-takers and never-takers? Does LATE have a meaningful policy interpretation?

9.3 Why is monotonicity an important assumption for the LATE theorem? What would happen if defiers existed?

9.4 Two researchers use different instruments to estimate the return to education. Researcher A uses college proximity; Researcher B uses compulsory schooling laws. Both produce valid IV estimates. Explain why the two estimates might differ even though both are valid.

9.5 Evaluate this statement: "Since IV estimates LATE for compliers, and compliers are a small fraction of the population, IV estimates are not useful for policy."

9.14.2 Analytical Questions

9.6 Prove the LATE Theorem carefully. Specifically, show that: (a) The denominator of the Wald estimator equals $P(\text{Complier})$. (b) The numerator equals $E[Y_{1i} - Y_{0i} \mid \text{Complier}] \times P(\text{Complier})$.

9.7 Suppose the population is 20% always-takers ($\text{TE} = \tau_{AT}$), 50% never-takers ($\text{TE} = \tau_{NT}$), and 30% compliers ($\text{TE} = \tau_C$). Express the ATE in terms of τ_{AT} , τ_{NT} , τ_C . Under what conditions does $\text{LATE} = \text{ATE}$?

9.8 In the draft lottery example, suppose:

- $P(\text{Complier}) = 0.16$ (16% of eligible men were induced to serve by the lottery).

- $E[Y_i | Z_i = 1] - E[Y_i | Z_i = 0] = -0.024$ (2.4% reduction in earnings from being in a low-lottery cohort).
- $E[D_i | Z_i = 1] - E[D_i | Z_i = 0] = 0.16$.

Compute the LATE. Interpret it.

9.14.3 Applied Questions

9.9 In R, simulate the LATE framework:

- Create a population of 10,000 with 4 compliance types.
 - Assign a binary instrument Z_i randomly.
 - Compute the Wald estimator and show it equals the average treatment effect for compliers.
-

9.15 R Lab 9: Simulating the LATE Framework

9.15.1 Exercise 9.1: Simulating Compliance Types

```

1 library(tidyverse)
2 set.seed(2024)
3 n <- 10000
4
5 # Define compliance types (known only to God!)
6 type <- sample(c("always_taker", "never_taker", "complier",
7   "defier"),
8   n, replace = TRUE,
9   prob = c(0.20, 0.30, 0.45, 0.05))
10
11 # Potential treatment under Z=0 and Z=1
12 D0 <- ifelse(type %in% c("always_taker", "defier"), 1, 0)
13 D1 <- ifelse(type %in% c("always_taker", "complier"), 1, 0)
14
15 # True heterogeneous treatment effects by type
16 te <- case_when(
17   type == "always_taker" ~ rnorm(n, 5, 2),
18   type == "never_taker" ~ rnorm(n, 3, 2),
19   type == "complier" ~ rnorm(n, 8, 2), # compliers have higher
20   returns
21   type == "defier" ~ rnorm(n, 1, 2)
22 )
23
24 # Potential outcomes
25 Y0 <- rnorm(n, 10, 3) # baseline untreated outcome
26 Y1 <- Y0 + te # treated outcome

```

```
26 # Randomly assign instrument
27 Z <- rbinom(n, 1, 0.5)
28
29 # Observed treatment and outcome
30 D <- ifelse(Z == 1, D1, D0)
31 Y <- ifelse(D == 1, Y1, Y0)
32
33 df_late <- data.frame(Z, D, Y, type, te, D0, D1)
```

9.15.2 Exercise 9.2: Computing the LATE

```
1 # True LATE (average treatment effect for compliers)
2 true_LATE <- mean(te[type == "complier"])
3 cat("True LATE (compliers only):", round(true_LATE, 4), "\n")
4
5 # True ATE
6 true_ATE <- mean(te)
7 cat("True ATE:", round(true_ATE, 4), "\n")
8
9 # Wald estimator (IV)
10 Y_Z1 <- mean(Y[Z == 1])
11 Y_Z0 <- mean(Y[Z == 0])
12 D_Z1 <- mean(D[Z == 1])
13 D_Z0 <- mean(D[Z == 0])
14
15 wald_est <- (Y_Z1 - Y_Z0) / (D_Z1 - D_Z0)
16 cat("Wald (IV) estimate:", round(wald_est, 4), "\n")
17 cat("Complier share:", round(D_Z1 - D_Z0, 4), "\n")
```

9.15.3 Exercise 9.3: IV Regression via ivreg

```
1 library(ivreg)
2 iv_model <- ivreg(Y ~ D | Z, data = df_late)
3 summary(iv_model)
4 cat("ivreg IV estimate:", round(coef(iv_model)["D"], 4), "\n")
```

9.15.4 Exercise 9.4: Distribution of Treatment Effects by Type

```
1 df_late %>%
2   ggplot(aes(x = te, fill = type)) +
3   geom_density(alpha = 0.5) +
4   geom_vline(xintercept = true_LATE, linetype = "dashed", color =
5     "red") +
6   geom_vline(xintercept = true_ATE, linetype = "dotted", color =
7     "blue") +
8   labs(
```

```
7   title = "Heterogeneous Treatment Effects by Compliance Type",
8   subtitle = "Red: LATE (complier mean); Blue: ATE (population
9             mean)",
10  x = "Individual Treatment Effect",
11  y = "Density",
12  fill = "Compliance Type"
13 ) +
    theme_minimal()
```

End of Chapter 9

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Chapter 10

Chapter 10: Panel Data and Fixed Effects

10.1 Chapter Overview

Panel data — repeated observations on the same units over time — are among the most powerful data structures in applied economics. They allow researchers to control for **unobserved individual heterogeneity**: time-invariant characteristics of individuals, firms, or regions that are correlated with both the treatment variable and the outcome, but that we can never directly observe.

This chapter develops the **fixed effects (FE)** estimator, shows how it is related to OLS via the Frisch-Waugh-Lovell theorem, and discusses its assumptions, limitations, and extensions.

10.2 Learning Objectives

After completing this chapter, students will be able to:

1. Explain the **panel data structure** and define key terminology (balanced/unbalanced panels, N , T).
 2. Explain the **unobserved heterogeneity problem** and why pooled OLS fails when individual effects are correlated with regressors.
 3. Derive the **within (fixed effects) estimator** via demeaning.
 4. Show that FE is equivalent to **LSDV (Least Squares Dummy Variables)**.
 5. State the identifying assumption of FE and explain what it rules out.
 6. Contrast the **first-differences estimator** with the within estimator.
 7. Discuss the **random effects estimator** and the **Hausman test**.
 8. Apply fixed effects to the Card (1995) earnings study.
-

10.3 10.1 Why Panel Data? The Unobserved Heterogeneity Problem

10.3.1 10.1.1 The Setup

Suppose we observe individual i in periods $t = 1, 2, \dots, T$. The outcome Y_{it} depends on an observed regressor X_{it} and an **unobserved individual effect** α_i :

$$Y_{it} = \alpha_i + \beta X_{it} + u_{it} \quad (10.1)$$

where:

- α_i is a **time-invariant** unobserved characteristic of individual i (e.g., innate ability, firm quality, state-specific resources).
- u_{it} is an idiosyncratic error term.

10.3.2 10.1.2 The Pooled OLS Problem

If we ignore α_i and run OLS on the stacked dataset:

$$Y_{it} = \alpha + \beta X_{it} + v_{it}$$

where $v_{it} = \alpha_i - \alpha + u_{it}$, then OLS is biased if $\text{Cov}(X_{it}, \alpha_i) \neq 0$.

Example 10.1: Estimating the return to union membership (X_{it}) on wages (Y_{it}). α_i might be unobserved worker productivity — high-productivity workers are both more likely to be unionized and have higher wages independently of union membership. So $\text{Cov}(X_{it}, \alpha_i) > 0$, and pooled OLS overstates the union wage premium.

10.4 10.2 The Panel Data Model

Definition 10.1 (One-Way Fixed Effects Model):

$$Y_{it} = \alpha_i + \beta X_{it} + u_{it}$$

where $i = 1, \dots, N$ and $t = 1, \dots, T$.

Key assumption (strict exogeneity within the panel):

$$E[u_{it} \mid X_{i1}, X_{i2}, \dots, X_{iT}, \alpha_i] = 0 \quad \text{for all } t$$

This says that, conditional on the individual effect α_i , the idiosyncratic error is uncorrelated with regressors in all periods. This is stronger than requiring $E[u_{it} \mid X_{it}, \alpha_i] = 0$.

	$t = 1$	$t = 2$	$t = 3$	$t = T$
$i = 1$	(Y_{it}, X_{it})	(Y_{it}, X_{it})	(Y_{it}, X_{it})	(Y_{it}, X_{it})
$i = 2$	(Y_{it}, X_{it})	(Y_{it}, X_{it})	(Y_{it}, X_{it})	(Y_{it}, X_{it})
$i = N$	(Y_{it}, X_{it})	(Y_{it}, X_{it})	(Y_{it}, X_{it})	(Y_{it}, X_{it})

Balanced panel: N units observed at T time periods $\Rightarrow NT$ observations total

Figure 10.1: Structure of a balanced panel dataset. Each cell (i, t) contains observations on unit i at time t . With N units and T periods, there are NT total observations. Panel data allow researchers to control for time-invariant unobserved heterogeneity by exploiting within-unit variation over time.

No assumption on $\text{Cov}(X_{it}, \alpha_i)$: Unlike pooled OLS, we allow the individual effect to be correlated with the regressors.

10.5 10.3 The Within (Fixed Effects) Estimator

10.5.1 10.3.1 Demeaning

The key insight: **subtract the individual time-mean** to remove α_i .

For individual i , the time-mean of equation (10.1) is:

$$\bar{Y}_i = \alpha_i + \beta \bar{X}_i + \bar{u}_i$$

where $\bar{Y}_i = \frac{1}{T} \sum_{t=1}^T Y_{it}$, etc.

Subtracting from equation (10.1):

$$\underbrace{(Y_{it} - \bar{Y}_i)}_{\tilde{Y}_{it}} = \beta \underbrace{(X_{it} - \bar{X}_i)}_{\tilde{X}_{it}} + \underbrace{(u_{it} - \bar{u}_i)}_{\tilde{u}_{it}}$$

The individual effect α_i is eliminated. This is called the **within transformation** or **demeaning**.

10.5.2 10.3.2 The Within (FE) Estimator

Definition 10.2 (Within Estimator):

$$\hat{\beta}^{FE} = \left(\sum_{i=1}^N \sum_{t=1}^T \tilde{X}_{it}^2 \right)^{-1} \sum_{i=1}^N \sum_{t=1}^T \tilde{X}_{it} \tilde{Y}_{it}$$

In matrix notation: $\hat{\beta}^{FE} = (\tilde{\mathbf{X}}^T \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^T \tilde{\mathbf{Y}}$, where $\tilde{\mathbf{Y}}$ and $\tilde{\mathbf{X}}$ are the demeaned outcome and regressor matrices.

This is **OLS on the demeaned (within-individual) data**.

10.5.3 10.3.3 Connection to the FWL Theorem

The within estimator is a special case of the FWL Theorem. Let $\mathbf{X}_1$ be the matrix of individual dummies (one per individual) and $\mathbf{X}_2 = X_{it}$. Then:

$$\hat{\beta}^{FE} = \hat{\beta}_2^{FWL} = (\tilde{\mathbf{X}}^T \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^T \tilde{\mathbf{Y}}$$

where $\tilde{\mathbf{X}} = \mathbf{M}_{X_1} \mathbf{X}$ and $\tilde{\mathbf{Y}} = \mathbf{M}_{X_1} \mathbf{Y}$ are the residuals from projecting onto the individual dummy space — which is exactly demeaning.

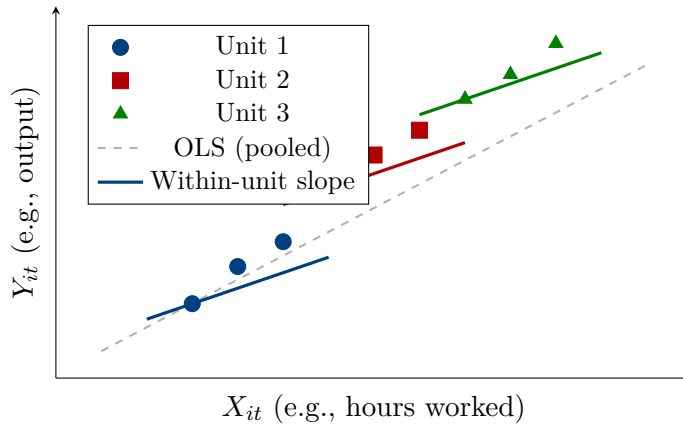


Figure 10.2: Within vs. between variation in panel data. OLS on pooled data (dashed gray) uses both between-unit and within-unit variation, and is biased when unobserved heterogeneity (unit fixed effects) correlates with X . Fixed effects estimation uses only within-unit variation (solid colored lines, common slope) to identify the causal effect.

10.5.4 10.3.4 Identifying Assumption of FE

Assumption 10.1 (No Time-Varying Omitted Variables):

The FE estimator is consistent if, conditional on the individual effect α_i , there are no time-varying confounders correlated with X_{it} in u_{it} .

What FE controls for: Any individual-specific, time-invariant unobservable.

What FE cannot control for: Time-varying confounders (variables that vary both across individuals and over time, and are correlated with X_{it}).

Example of what FE handles: Studying the effect of union membership on wages. FE removes the effect of permanent individual ability (time-invariant). What remains is the association between changes in union status and changes in wages, within the same individual over time.

Example of what FE cannot handle: Studying the effect of a new employment policy on wages, when individuals self-select into regions that adopt the policy at different times based on their expectations of local economic conditions (time-varying, endogenous selection).

10.6 10.4 The LSDV Representation

Proposition 10.1 (LSDV Equivalence): The within estimator is algebraically identical to the OLS estimator from a regression of Y_{it} on X_{it} and N individual dummy variables (LSDV = Least Squares Dummy Variables):

$$Y_{it} = \sum_{j=1}^N \alpha_j D_{ij} + \beta X_{it} + u_{it}$$

where $D_{ij} = 1$ if $j = i$ (individual i 's dummy variable).

Proof: By the FWL theorem, the coefficient on X_{it} from the LSDV regression equals the coefficient from regressing $\mathbf{M}_{X_1} \mathbf{Y}$ on $\mathbf{M}_{X_1} \mathbf{X}$, where $\mathbf{X}_1$ is the matrix of individual dummies. Projecting onto individual dummies means subtracting individual means — which is demeaning. $\square$

Practical note: With large N , LSDV is computationally expensive (requires inverting a large matrix). The within estimator is computationally equivalent but much faster.

10.7 10.5 The First-Differences Estimator

An alternative to demeaning is **first differencing**: subtract period $t - 1$ from period t .

$$\Delta Y_{it} = \beta \Delta X_{it} + \Delta u_{it}$$

where $\Delta Y_{it} = Y_{it} - Y_{i,t-1}$, etc. Individual effects are again eliminated.

FD estimator: OLS on the first-differenced data.

When FD $\equiv$ FE: When $T = 2$.

When FD $\neq$ FE ($T > 2$):

- Under strict exogeneity, both are consistent.
 - If u_{it} is serially uncorrelated, FE is more efficient.
 - If u_{it} follows a random walk ($u_{it} = u_{i,t-1} + \epsilon_{it}$), FD is more efficient.
-

10.8 10.6 Two-Way Fixed Effects

In many panel settings, we want to control for both individual effects AND time effects — for example, macro shocks that affect all individuals simultaneously.

Two-way FE model:

$$Y_{it} = \alpha_i + \gamma_t + \beta X_{it} + u_{it}$$

where γ_t is a **time fixed effect** (common to all individuals in period t).

Estimation: Include individual and time dummies (LSDV) or demean by subtracting both individual means and time means (within transformation in two dimensions).

Application: The two-way FE model is the backbone of the **Differences-in-Differences** estimator studied in Chapter 11.

10.9 10.7 Random Effects and the Hausman Test

10.9.1 10.7.1 The Random Effects (RE) Model

Definition 10.3 (Random Effects): The RE model treats α_i as a random variable drawn from a distribution, uncorrelated with X_{it} :

$$\alpha_i \perp X_{it}$$

Under this assumption, OLS on the pooled data is consistent but inefficient (errors within individuals are correlated). The RE estimator uses a GLS transformation to achieve efficiency.

10.9.2 10.7.2 FE vs. RE: The Hausman Test

Hausman (1978) Test:

- Under H_0 : $\text{Cov}(X_{it}, \alpha_i) = 0$ (RE is consistent and efficient; FE is consistent but inefficient).

- Under H_1 : $\text{Cov}(X_{it}, \alpha_i) \neq 0$ (RE is inconsistent; FE is consistent).

The test statistic is:

$$H = (\hat{\beta}^{FE} - \hat{\beta}^{RE})^T [\text{Var}(\hat{\beta}^{FE}) - \text{Var}(\hat{\beta}^{RE})]^{-1} (\hat{\beta}^{FE} - \hat{\beta}^{RE}) \sim \chi_k^2$$

If H is large (reject H_0), use FE. If not, RE may be preferred.

Practical advice: In applied economics, FE is used far more commonly than RE. The assumption that individual effects are uncorrelated with regressors is strong and often implausible. Use RE only when you have a strong theoretical reason to believe the assumption holds.

10.10 10.8 Empirical Illustration: Returns to Education (Card, 1995)

Card (1995) uses panel data to study whether the returns to education are different from OLS estimates once unobserved individual ability is controlled for.

Data: National Longitudinal Survey (NLS) following the same workers over time.

Key finding: FE estimates of the return to education are *higher* than OLS estimates — not lower, as ability bias would predict. This suggests that:

1. Men who complete more education may have had *lower* ability (conditional on measured characteristics) — a negative $\text{Cov}(X_{it}, \alpha_i)$.
2. Or: OLS ability bias is smaller than typically assumed, while measurement error in education attenuates OLS estimates downward.

This is a sobering finding: the "obvious" direction of ability bias may not be correct once we look within individuals over time.

10.11 Chapter Summary

- **Panel data** allow us to control for time-invariant unobserved heterogeneity.
- **Pooled OLS** is biased when individual effects are correlated with regressors.
- The **within (FE) estimator** eliminates individual effects by demeaning; it is equivalent to OLS with individual dummies (LSDV).
- FE's identifying assumption is that there are no time-varying confounders; it controls for everything time-invariant but nothing that varies over time.
- The **first-differences estimator** is an alternative to FE; they are equivalent when $T = 2$.

- **Two-way FE** controls for both individual and time effects.
 - The **Hausman test** helps choose between FE and RE, but FE is generally preferred in applied economics.
 - Card (1995) shows FE can produce higher (not lower) estimates of returns to education — reversing the naive prediction about ability bias.
-

10.12 Key Terms

Panel data — data with multiple observations per unit over time.

Individual fixed effect (α_i) — time-invariant unobserved characteristic of unit i .

Within transformation (demeaning) — subtracting individual time-means.

Within (FE) estimator — OLS on demeaned data; eliminates individual effects.

LSDV — Least Squares Dummy Variables; OLS with individual dummies; algebraically equivalent to FE.

First-differences (FD) estimator — OLS on first-differenced data; also eliminates individual effects.

Two-way FE — controls for both individual and time effects.

Random effects (RE) — treats individual effects as random draws uncorrelated with regressors; more efficient but requires stronger assumptions.

Hausman test — tests whether $\text{Cov}(X_{it}, \alpha_i) = 0$; guides FE vs. RE choice.

10.13 Exercises

10.13.1 Conceptual Questions

10.1 Explain why pooled OLS is biased in the presence of unobserved individual heterogeneity. What is the sign of the bias in the returns-to-union-membership example?

10.2 What does FE "control for"? What does it NOT control for? Give an example of a confounder that FE cannot address.

10.3 Why is FE equivalent to LSDV? Explain using the FWL theorem.

10.4 Compare the FE and FD estimators. When are they equivalent? When do they differ?

10.5 Explain the Hausman test. What is its null hypothesis? What should you do if the test rejects?

10.13.2 Analytical Questions

10.6 Show formally that the within estimator eliminates the individual fixed effect α_i .

10.7 In the two-period panel ($T = 2$), show that FD and FE give identical estimates.

10.8 Consider the model $Y_{it} = \alpha_i + \beta X_{it} + u_{it}$. Suppose $X_{it} = \alpha_i + \epsilon_{it}$ (the regressor depends on the individual effect). Show that pooled OLS is biased and derive the direction of bias.

10.9 Show that including individual dummy variables in a regression and applying OLS gives the same coefficient as the within estimator (LSDV equivalence).

10.13.3 Applied Questions

10.10 Using a wages panel dataset in R, estimate: (a) pooled OLS; (b) FE (within estimator); (c) first differences. Compare and interpret the results.

10.11 Apply the Hausman test to your estimates. What do you conclude about the appropriate model?

10.14 R Lab 10: Fixed Effects Estimation

10.14.1 Setup

```
1 library(tidyverse)
2 library(plm)      # panel data estimation
3 library(lmtest)
4 library(sandwich)
5
6 # Install and load data
7 # install.packages("plm")
8 data("Wages", package = "plm")
9 head(Wages)
10 # Variables: id, time, lwage (log wage), ed (education), exp
    (experience),
11 #             union (union status), black, etc.
```

10.14.2 Exercise 10.1: Pooled OLS

```
1 # Convert to panel data format
2 wages_panel <- pdata.frame(Wages, index = c("id", "time"))
3
4 # Pooled OLS
5 pool_model <- plm(lwage ~ ed + exp + I(exp^2) + union + black,
6                   data = wages_panel, model = "pooling")
7 summary(pool_model)
```

10.14.3 Exercise 10.2: Fixed Effects (Within Estimator)

```
1 # Fixed Effects (Within) Model
2 fe_model <- plm(lwage ~ ed + exp + I(exp^2) + union + black,
3                 data = wages_panel, model = "within")
4 summary(fe_model)
5 cat("FE coefficient on union:", round(coef(fe_model)["union"], 4),
6     "\n")
7 cat("Pool OLS coefficient on union:",
8     round(coef(pool_model)["union"], 4), "\n")
```

10.14.4 Exercise 10.3: First Differences

```
1 # First Differences
2 fd_model <- plm(lwage ~ ed + exp + I(exp^2) + union + black,
3                 data = wages_panel, model = "fd")
4 summary(fd_model)
```

10.14.5 Exercise 10.4: LSDV (verify FE equivalence)

```
1 # LSDV using lm() -- add factor(id) as dummy variables
2 # WARNING: this is slow for large N!
3 # Use a small subset for illustration
4 wages_small <- Wages[Wages$id <= 50, ]
5
6 lsdv_model <- lm(lwage ~ ed + exp + I(exp^2) + union + factor(id),
7                 data = wages_small)
8 fe_small <- plm(lwage ~ ed + exp + I(exp^2) + union,
9                 data = pdata.frame(wages_small, index = c("id",
10                  "time")),
11                 model = "within")
12
13 cat("LSDV union coefficient:", round(coef(lsdv_model)["union"], 6),
14     "\n")
15 cat("FE union coefficient: ", round(coef(fe_small)["union"], 6),
16     "\n")
```

10.14.6 Exercise 10.5: Random Effects and Hausman Test

```
1 # Random Effects
2 re_model <- plm(lwage ~ ed + exp + I(exp^2) + union + black,
3                 data = wages_panel, model = "random")
4 summary(re_model)
5
6 # Hausman Test
7 phtest(fe_model, re_model)
```

10.14.7 Exercise 10.6: Two-Way Fixed Effects

```
1 # Two-Way FE (individual + time effects)
2 twofe_model <- plm(lwage ~ ed + exp + I(exp^2) + union + black,
3                   data = wages_panel, model = "within", effect =
4                   "twoways")
5 summary(twofe_model)
```

End of Chapter 10

References

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Chapter 11

Chapter 11: Differences-in-Differences

11.1 Chapter Overview

Differences-in-Differences (DiD) is the workhorse identification strategy of modern empirical economics. It exploits variation in the timing and geography of policy changes to estimate causal effects: we compare the change in outcomes over time for units that were exposed to a treatment (the "treatment group") to the change in outcomes over time for units that were not (the "control group"). The difference of these two differences is the DiD estimator.

The key identifying assumption is **parallel trends**: in the absence of treatment, the treatment and control groups would have followed the same trend over time. When this assumption holds, the DiD estimator is unbiased. Testing it — and understanding when it is plausible — is the central challenge of DiD research.

11.2 Learning Objectives

After completing this chapter, students will be able to:

1. Set up the **2×2 DiD framework** (two groups, two periods).
2. State the **parallel trends assumption** and explain its meaning.
3. Derive the **DiD estimator** from simple mean comparisons.
4. Formulate DiD as a **regression with interaction terms**.
5. Explain the role of **clustering** standard errors in DiD.
6. Use **event study plots** to assess parallel pre-trends.
7. Identify key threats to DiD validity: anticipation, policy endogeneity, compositional changes.
8. Describe the complications of **staggered DiD** and recent methodological advances.
9. Apply DiD to the Card and Krueger (1994) minimum wage study.

11.3 11.1 The 2×2 Differences-in-Differences Setup

11.3.1 11.1.1 Notation

Consider two groups and two time periods:

- **Groups:** $g \in \{0, 1\}$ where $g = 1$ is the treatment group, $g = 0$ is the control group.
- **Periods:** $t \in \{0, 1\}$ where $t = 0$ is the pre-treatment period, $t = 1$ is the post-treatment period.
- **Treatment:** Group 1 receives the treatment in period 1; group 0 never receives treatment.

We define potential outcomes $Y_{gt}(d)$: the mean outcome in group g , period t , if treatment status were d .

11.3.2 11.1.2 The Estimand

The object of interest is the **Average Treatment Effect on the Treated (ATT)** in the post-period:

$$\text{ATT} = E[Y_{i1}(1) - Y_{i1}(0) \mid g_i = 1]$$

This is the average effect of treatment on the treated units in the post-treatment period. The key challenge: we do not observe $Y_{i1}(0)$ for treated units — we need a counterfactual for what the treatment group's outcome would have been had they not been treated.

11.3.3 11.1.3 The Four Observed Cell Means

Period	Control Group ($g = 0$)	Treatment Group ($g = 1$)
Pre ($t = 0$)	$\bar{Y}_{00}$	$\bar{Y}_{10}$
Post ($t = 1$)	$\bar{Y}_{01}$	$\bar{Y}_{11}$

11.4 11.2 The Parallel Trends Assumption

Assumption 11.1 (Parallel Trends):

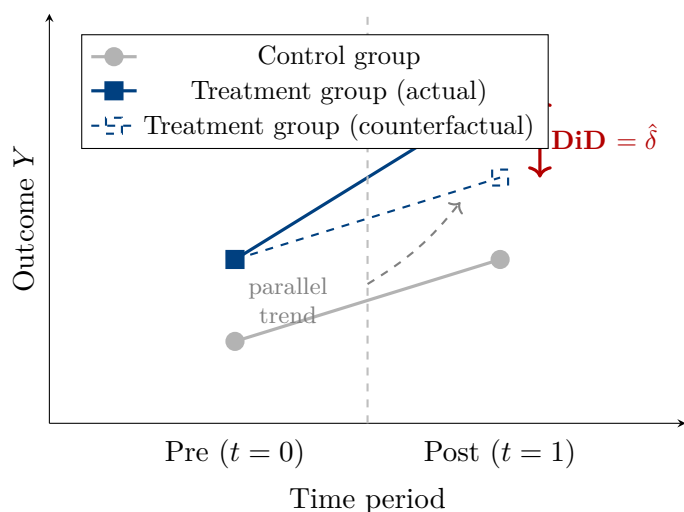


Figure 11.1: The 2×2 Differences-in-Differences design. Both groups trend upward by 1 unit from pre to post under parallel trends. The treatment group (blue) actually rises by 2 units, so the DiD estimate $\hat{\delta} = (4.5 - 2.5) - (2.5 - 1.5) = 1$ captures the treatment effect. The dashed blue line shows the counterfactual trend the treatment group would have followed absent treatment.

$$E[Y_t(0) \mid g = 1, t = 1] - E[Y_t(0) \mid g = 1, t = 0] = E[Y_t(0) \mid g = 0, t = 1] - E[Y_t(0) \mid g = 0, t = 0]$$

In words: **in the absence of treatment, the treatment group would have experienced the same change in outcomes as the control group.**

This does NOT require the two groups to start at the same level. It only requires them to share the same time trend in untreated potential outcomes.

Visual test: Plot outcome trends for both groups in the pre-treatment periods. If the two groups have similar pre-trends (parallel), the assumption is more credible.

11.5 11.3 The DiD Estimator

11.5.1 11.3.1 Deriving the Estimator

The observed outcome for a treated unit in the post-period is:

$$E[Y_{i1} \mid g_i = 1, t = 1] = E[Y_{i1}(1) \mid g_i = 1]$$

We need $E[Y_{i1}(0) \mid g_i = 1]$ — the counterfactual. Under parallel trends:

$$\begin{aligned}
 E[Y_{i1}(0) \mid g_i = 1] &= E[Y_{i0}(0) \mid g_i = 1] + \underbrace{(E[Y_t(0) \mid g = 0, t = 1] - E[Y_t(0) \mid g = 0, t = 0])}_{\text{control group trend}} \\
 &= \bar{Y}_{10} + (\bar{Y}_{01} - \bar{Y}_{00})
 \end{aligned}$$

Therefore:

$$\begin{aligned}
 \text{ATT} &= E[Y_{i1}(1) \mid g_i = 1] - E[Y_{i1}(0) \mid g_i = 1] \\
 &= \bar{Y}_{11} - [\bar{Y}_{10} + (\bar{Y}_{01} - \bar{Y}_{00})] \\
 &= \boxed{(\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00})}
 \end{aligned}$$

This is the DiD estimator: the difference in before-after changes between the treatment and control groups.

11.5.2 11.3.2 Decomposing the DiD

$$\hat{\delta}^{DiD} = \underbrace{(\bar{Y}_{11} - \bar{Y}_{10})}_{\text{Change in treatment group}} - \underbrace{(\bar{Y}_{01} - \bar{Y}_{00})}_{\text{Change in control group}}$$

Comparison	What it measures
$(\bar{Y}_{11} - \bar{Y}_{10})$	Pre-post change for treated group = Treatment effect + Time trend
$(\bar{Y}_{01} - \bar{Y}_{00})$	Pre-post change for control group = Time trend
DiD	Treatment effect (time trend differenced out)

The control group provides the counterfactual trend. Subtracting it removes the common time trend, leaving only the causal effect.

11.5.3 11.3.3 Identification Under Parallel Trends

Proposition 11.1: Under Assumption 11.1 (Parallel Trends) and the overlap condition ($P(g = 1) > 0$, $P(g = 0) > 0$):

$$\hat{\delta}^{DiD} \xrightarrow{P} \text{ATT}$$

11.6 11.4 DiD as a Regression

The DiD estimator has a natural regression formulation.

Regression DiD:

$$Y_{igt} = \alpha + \beta_g G_i + \beta_t T_t + \delta(G_i \times T_t) + u_{igt}$$

where:

- $G_i = 1$ if unit i is in the treatment group
- $T_t = 1$ if period t is the post-treatment period
- $G_i \times T_t$ is the interaction term

Interpreting coefficients:

Group	Period	$E[Y]$
Control ($G = 0$)	Pre ($T = 0$)	α
Control ($G = 0$)	Post ($T = 1$)	$\alpha + \beta_t$
Treatment ($G = 1$)	Pre ($T = 0$)	$\alpha + \beta_g$
Treatment ($G = 1$)	Post ($T = 1$)	$\alpha + \beta_g + \beta_t + \delta$

Therefore:

- β_g : pre-existing level difference between treatment and control.
- β_t : common time trend (post-pre change for the control group).
- δ : the **DiD estimator** — the additional change for the treatment group in the post period = **ATT under parallel trends**.

Verification: $\delta = (\alpha + \beta_g + \beta_t + \delta) - (\alpha + \beta_t) - (\alpha + \beta_g) + \alpha = \hat{\delta}^{DiD}$. ✓

11.6.1 11.4.1 With Additional Controls

To increase precision and relax the parallel trends assumption:

$$Y_{igt} = \alpha_g + \gamma_t + \delta(G_i \times T_t) + \mathbf{X}_{igt}^T \boldsymbol{\theta} + u_{igt}$$

where α_g are group fixed effects, γ_t are time fixed effects, and $\mathbf{X}_{igt}$ are time-varying controls.

Note: With multiple time periods and groups, this becomes the **two-way fixed effects DiD** model of Chapter 10.

11.7 11.5 Standard Errors in DiD: Clustering

Standard errors in DiD settings need careful treatment. The treatment is typically assigned at the group level (e.g., a state adopts a policy), but outcomes are measured at the individual level. This creates **within-group correlation** in residuals that, if ignored, leads to severely underestimated standard errors.

Bertrand, Duflo, and Mullainathan (2004) showed that ignoring clustering can lead to rejection rates of 45% at the nominal 5% level when the true effect is zero.

Solution: Always cluster standard errors at the level of treatment assignment (e.g., state, school, firm).

Rule: Cluster at the group $\times$ time level that the policy variation comes from. If treatment varies at the state level, cluster at the state level even if outcomes are at the individual level.

11.8 11.6 Testing Parallel Trends: Event Study Plots

The parallel trends assumption is the key identifying assumption of DiD. While it cannot be directly tested (we never observe the counterfactual trend for the treatment group), we can assess its plausibility using **pre-treatment trends**.

11.8.1 11.6.1 The Event Study Specification

Replace the single post-period interaction with a set of interactions for each time period:

$$Y_{igt} = \alpha_g + \gamma_t + \sum_{k \neq -1} \delta_k (G_i \times \mathbb{I}[t = k]) + \mathbf{X}_{igt}^T \boldsymbol{\theta} + u_{igt}$$

where k indexes periods relative to the treatment date, and $k = -1$ is the omitted (reference) period just before treatment.

The δ_k coefficients:

- For $k < 0$ (pre-treatment): should be zero if parallel trends holds.
- For $k \geq 0$ (post-treatment): these are the dynamic treatment effects.

Event study plot: Plot $\hat{\delta}_k$ with confidence intervals against time k . If pre-treatment coefficients are near zero (statistically and practically), parallel trends is more plausible.

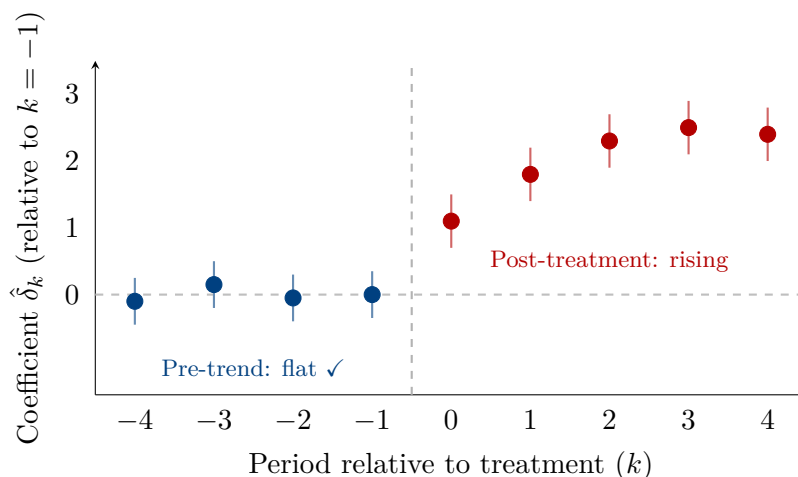


Figure 11.2: An event study (dynamic DiD) plot. Coefficients $\hat{\delta}_k$ measure the treatment group's outcome relative to the control group, normalized to zero at $k = -1$ (one period before treatment). Flat pre-treatment coefficients (blue) support the parallel trends assumption. Rising post-treatment coefficients (red) reveal the treatment effect growing over time.

11.8.2 11.6.2 Caution: Pre-trends Tests Are Noisy

Even if the parallel trends assumption holds, pre-trends tests have low power with few pre-treatment periods. Conversely, a non-significant pre-trend test does not guarantee that parallel trends holds. The test is necessary but not sufficient evidence.

11.9 11.7 Threats to DiD Validity

11.9.1 11.7.1 Violation of Parallel Trends

The most common threat. If treatment group would have experienced a different trend even without treatment, the DiD estimate is biased.

Example: If firms that adopt a new technology (treatment) were on a faster growth trajectory before adoption, the DiD estimator confounds the treatment effect with the pre-existing growth differential.

Mitigation: Use multiple pre-treatment periods to document parallel pre-trends; use control groups that are more similar to the treatment group.

11.9.2 11.7.2 Anticipation Effects

If treated units change behavior in anticipation of the treatment before it officially begins, the pre-treatment outcomes are already affected by the treatment. This biases the DiD downward (understates the total effect).

Example: A minimum wage increase announced 6 months before it takes effect. Firms may start adjusting employment before the law takes effect.

Mitigation: Extend the pre-treatment window to see if anticipation effects appear.

11.9.3 11.7.3 Policy Endogeneity

The treatment is often adopted in response to current or expected outcomes. If states with declining fast-food employment adopt a minimum wage increase, the control group (states with stable employment) will show different trends.

Mitigation: Find settings where treatment assignment is exogenous — natural experiments, sharp rules, randomized rollouts.

11.9.4 11.7.4 Compositional Changes

If the composition of the treatment and control groups changes over time (e.g., selective migration in response to a policy), the pre-post comparison may reflect composition changes rather than treatment effects.

11.10 11.8 Staggered DiD and Recent Advances

11.10.1 11.8.1 Staggered Treatment Adoption

In many real settings, different units adopt treatment at different times (staggered rollout). For example, US states adopt a minimum wage increase in different years. The two-way FE estimator in this setting can be written as:

$$Y_{it} = \alpha_i + \gamma_t + \delta D_{it} + u_{it}$$

where $D_{it} = 1$ if unit i is treated by period t .

Problem: Recent work (Callaway and Sant’Anna, 2021; Goodman-Bacon, 2021; de Chaisemartin and D’Haultfoeuille, 2020) has shown that in staggered DiD settings, the two-way FE δ is a weighted average of 2×2 DiD estimates, but with **potentially negative weights**. If treatment effects are heterogeneous (vary over time or across groups), the two-way FE estimate can be a misleading average or even have the wrong sign.

11.10.2 11.8.2 Solutions

Modern approaches decompose the staggered DiD into a series of 2×2 comparisons:

- **Callaway and Sant’Anna (2021):** Compute group-time average treatment effects $ATT(g, t)$ for each adoption cohort g and time period t , then aggregate.
- **Sun and Abraham (2021):** Introduce a fully saturated interaction model.
- **de Chaisemartin and D’Haultfoeuille (2020):** Propose the "did_multiplegt" estimator robust to heterogeneous effects.

11.11 11.9 Empirical Illustration: Card and Krueger (1994)

11.11.1 11.9.1 The Research Question

In April 1992, New Jersey raised its minimum wage from \$4.25 to \$5.05 per hour. Pennsylvania did not change its minimum wage. Card and Krueger (1994) use this natural experiment to study the effect of the minimum wage increase on fast-food employment.

Treatment group: Fast-food restaurants in New Jersey. **Control group:** Fast-food restaurants in eastern Pennsylvania. **Pre-treatment period:** February 1992. **Post-treatment period:** November 1992.

11.11.2 11.9.2 Identification

The parallel trends assumption requires that, absent the New Jersey minimum wage increase, employment trends in New Jersey and Pennsylvania fast-food restaurants would have been similar.

Plausibility: New Jersey and Pennsylvania are geographically contiguous, compete in similar labor markets, and have similar industry mix. This makes parallel trends more plausible than comparing New Jersey to, say, California.

11.11.3 11.9.3 Results

	New Jersey (treatment)	Pennsylvania (control)	DiD
Feb (pre)	20.44 employees	23.33 employees	—
Nov (post)	21.03 employees	21.17 employees	—
Change	+0.59	−2.16	+2.76

DiD estimate: $0.59 - (-2.16) = 2.76$ additional full-time equivalent employees in New Jersey restaurants.

This suggests the minimum wage increase actually *increased* employment in New Jersey fast-food restaurants — directly contrary to the standard competitive model prediction.

Why? Card and Krueger argue: (1) fast-food labor markets are monopsonistic, so minimum wages can increase employment; (2) the positive effect might reflect the positive income effect on low-wage households increasing demand for fast food.

11.11.4 11.9.4 Criticism and Response

Neumark and Wascher (1994) challenged the results using payroll data (vs. Card-Krueger's phone survey data). Their estimates showed negative employment effects.

Methodological lesson: The choice of data source, control group, and period significantly affects DiD estimates. Always subject your design to sensitivity checks.

11.12 Chapter Summary

- **DiD** compares the change in outcomes before and after treatment for treated units versus control units.
 - The **parallel trends assumption** is the key identifying assumption: control group trend serves as the counterfactual for the treatment group.
 - **DiD regression:** $Y = \alpha + \beta_g G + \beta_t T + \delta(G \times T) + u$; the coefficient δ is the ATT.
 - **Event study plots** assess pre-treatment parallel trends — necessary but not sufficient evidence.
 - **Cluster standard errors** at the level of treatment assignment; ignoring clustering severely understates uncertainty.
 - Key threats: violations of parallel trends, anticipation effects, policy endogeneity, compositional changes.
 - **Staggered DiD** settings require caution: two-way FE can have wrong-sign weights under heterogeneous effects.
 - **Card and Krueger (1994)** is the canonical DiD application, showing minimum wages may not reduce fast-food employment.
-

11.13 Key Terms

Differences-in-Differences (DiD) — $(\bar{Y}_{11} - \bar{Y}_{10}) - (\bar{Y}_{01} - \bar{Y}_{00})$; the change in outcomes for the treated group minus the change for the control group.

Parallel trends assumption — in the absence of treatment, treated and control groups would have experienced the same trend.

ATT (DiD context) — $E[Y_{i1}(1) - Y_{i1}(0) \mid g_i = 1]$; effect of treatment on treated units in the post-period.

Event study — regression estimating dynamic treatment effects before and after treatment.

Pre-trends test — testing whether pre-treatment DiD coefficients are zero; assesses parallel trends plausibility.

Staggered DiD — DiD with variation in timing of treatment adoption across units.

11.14 Exercises

11.14.1 Conceptual Questions

11.1 State the parallel trends assumption in your own words. Why is it called "parallel"?

11.2 Explain why the DiD estimator is not simply the before-after change for the treatment group.

11.3 A researcher uses DiD to study the effect of a state-level policy on individual outcomes. She has 50 states and 10,000 individual observations. Should she cluster at the individual or state level? Why?

11.4 Describe an event study plot. What would you conclude from (a) flat pre-trends with a jump at treatment; (b) rising pre-trends in the treatment group before treatment?

11.5 Explain the "negative weights" problem in staggered DiD. Why is the two-way FE estimator potentially misleading?

11.14.2 Analytical Questions

11.6 Using the Card-Krueger data in the table above, compute the DiD estimate by hand.

11.7 Show that the OLS coefficient on $G_i \times T_t$ in the regression $Y = \alpha + \beta_g G + \beta_t T + \delta(G \times T) + u$ equals the DiD estimator.

11.8 Suppose parallel trends fails: the treatment group would have grown by 3% in

the post-period even without treatment, while the control group grows by 1%. The observed DiD is 5%. What is the true ATT?

11.14.3 Applied Questions

11.9 Using publicly available Card-Krueger data, replicate their DiD estimate in R.

11.10 Generate event study plots for a dataset with multiple pre-treatment periods.

11.15 R Lab 11: Differences-in-Differences

11.15.1 Setup

```
1 library(tidyverse)
2 library(fixest) # fast fixed effects with robust SE
3 library(ggplot2)
4
5 # Simulate DiD data
6 set.seed(42)
7 n_units <- 200 # 100 treatment, 100 control
8 n_periods <- 6 # periods -3, -2, -1 (pre) and 0, 1, 2 (post)
9
10 df <- expand_grid(
11   unit = 1:n_units,
12   period = -3:2
13 ) %>%
14   as_tibble() %>%
15   mutate(
16     treated = unit > 100, # 101-200 are
17     # treated
18     post = period >= 0, # periods 0,1,2 are
19     # post
20     time_trend = 0.5 * period, # common linear
21     # time trend
22     unit_fe = rnorm(n_units)[unit], # unit-specific
23     # fixed effect
24     te = ifelse(treated & post, 3, 0), # true ATT = 3
25     noise = rnorm(n_units * n_periods, 0, 1.5),
26     Y = 10 + unit_fe + time_trend + te + noise
27   )
```

11.15.2 Exercise 11.1: Visualizing Parallel Trends

```
1 df %>%
2   group_by(treated, period) %>%
3   summarise(mean_Y = mean(Y), .groups = "drop") %>%
```

```

4  ggplot(aes(x = period, y = mean_Y, color = treated, group =
    treated)) +
5  geom_line(linewidth = 1.2) +
6  geom_point(size = 3) +
7  geom_vline(xintercept = -0.5, linetype = "dashed", color =
    "gray50") +
8  labs(
9    title      = "Outcome Trends: Treatment vs. Control Group",
10   subtitle   = "Dashed line = treatment date. Pre-trends look
    parallel.",
11   x = "Period", y = "Mean Outcome",
12   color = "Treated"
13 ) +
14 theme_minimal()

```

11.15.3 Exercise 11.2: The 2×2 DiD

```

1  # Use only pre (-1) and post (0) periods
2  df_2x2 <- df %>% filter(period %in% c(-1, 0))
3
4  # Cell means
5  cell_means <- df_2x2 %>%
6    group_by(treated, post) %>%
7    summarise(mean_Y = mean(Y), .groups = "drop")
8  print(cell_means)
9
10 # DiD by hand
11 did_manual <- with(cell_means,
12   (mean_Y[treated == TRUE & post == TRUE] -
13    mean_Y[treated == TRUE & post == FALSE]) -
14   (mean_Y[treated == FALSE & post == TRUE] -
15    mean_Y[treated == FALSE & post == FALSE])
16 )
17 cat("DiD estimate (manual):", round(did_manual, 3), "\n")
18
19 # DiD via regression
20 did_reg <- lm(Y ~ treated * post, data = df_2x2)
21 summary(did_reg)

```

11.15.4 Exercise 11.3: Regression DiD with Fixed Effects

```

1  # Full panel DiD with unit and time fixed effects
2  fe_did <- feols(Y ~ i(post, treated) | unit + period,
3    data = df, cluster = ~unit)
4  summary(fe_did)
5  cat("FE DiD estimate:",
    round(coef(fe_did)["post::TRUE:treatedTRUE"], 3), "\n")

```

11.15.5 Exercise 11.4: Event Study Plot

```
1 # Event study: dynamic treatment effects by period
2 event_study <- feols(Y ~ i(period, treated, ref = -1) | unit +
3   period,
4   data = df, cluster = ~unit)
5
6 iplot(event_study,
7   xlab = "Periods Relative to Treatment",
8   main = "Event Study: Dynamic Treatment Effects",
9   sub = "Reference period = -1. Treatment begins at period 0.")
```

11.15.6 Exercise 11.5: Card and Krueger Replication

```
1 # Read Card-Krueger data (available at various sources)
2 # Here we use the 'wooldridge' package
3 # install.packages("wooldridge")
4 library(wooldridge)
5 data("fastfood", package = "wooldridge")
6
7 # DiD regression
8 ck_model <- lm(fte ~ nj + d + nj:d, data = fastfood)
9 summary(ck_model)
10 cat("DiD estimate (Card-Krueger):", round(coef(ck_model)["nj:d"],
11   3), "\n")
```

End of Chapter 11

References

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Chapter 12

Chapter 12: Matching and Selection on Observables

12.1 Chapter Overview

Matching estimators provide an alternative to regression for estimating average treatment effects when treatment is not randomly assigned. The key assumption is **selection on observables (CIA — Conditional Independence Assumption)**: after conditioning on observed covariates, treatment is independent of potential outcomes. This chapter develops the theory and practice of matching, connects it to regression, and introduces propensity score methods.

12.2 Learning Objectives

After completing this chapter, students will be able to:

1. State the **Conditional Independence Assumption (CIA)** and explain what it requires.
 2. Explain the **propensity score theorem** and why it simplifies matching.
 3. Describe and implement **nearest-neighbor matching**.
 4. Explain the connection between regression and matching.
 5. Apply the **propensity score** for matching, weighting, and regression.
 6. Check the **overlap (common support)** condition and deal with violations.
 7. Evaluate the quality of matching using **balance checks**.
 8. Discuss when matching is more or less appropriate than other methods.
-

12.3 12.1 The Conditional Independence Assumption (CIA)

Definition 12.1 (Conditional Independence Assumption / Ignorability):

$$\{Y_{0i}, Y_{1i}\} \perp\!\!\!\perp D_i \mid \mathbf{X}_i$$

Given observed covariates $\mathbf{X}_i$, the treatment D_i is independent of potential outcomes. Conditioning on $\mathbf{X}_i$ "explains away" all selection into treatment.

This implies:

$$E[Y_{0i} \mid D_i = 1, \mathbf{X}_i = \mathbf{x}] = E[Y_{0i} \mid D_i = 0, \mathbf{X}_i = \mathbf{x}]$$

Within cells defined by $\mathbf{X}_i = \mathbf{x}$, treatment is effectively random.

When is CIA plausible?

The CIA is plausible when:

- Treatment assignment is driven by observable characteristics (e.g., eligibility rules based on measurable criteria).
- The researcher has rich data capturing all important determinants of selection.
- There are no hidden selection mechanisms (no selection on unobservables).

When is CIA implausible?

- Treatment is self-selected based on unobservable motivation or ability.
- Assignment rules are discretionary and based on unrecorded factors.
- There is reverse causality.

Example 12.1: A job training program recruits participants based on age, education, and prior employment status — all observable. If the data contains these variables, CIA may be plausible: conditional on observables, participants and non-participants are comparable.

12.4 12.2 The Overlap Condition

Assumption 12.1 (Overlap / Common Support):

$$0 < P(D_i = 1 \mid \mathbf{X}_i = \mathbf{x}) < 1 \quad \text{for all } \mathbf{x} \text{ in the support of } \mathbf{X}$$

Every individual, regardless of their covariate values, has a positive probability of being in both the treatment and control groups.

Overlap fails when:

- Some covariate values are found only among treated units (no valid control group members to compare to).

- Perfect predictors of treatment exist.

Dealing with overlap violations:

- Trim the sample to the common support region.
 - Use only observations where propensity scores overlap between treatment and control.
 - Report results with and without trimming; check sensitivity.
-

12.5 12.3 Sub-classification and Direct Matching

12.5.1 12.3.1 Sub-classification

The simplest matching approach: divide the sample into cells defined by discrete values of $\mathbf{X}_i$, estimate the treatment effect within each cell, then average across cells.

$$\hat{\tau}^{sub} = \sum_{\mathbf{x}} \hat{w}(\mathbf{x}) [\bar{Y}_1(\mathbf{x}) - \bar{Y}_0(\mathbf{x})]$$

where $\bar{Y}_d(\mathbf{x})$ is the mean outcome for group d in cell $\mathbf{x}$, and $\hat{w}(\mathbf{x})$ is the fraction of the treatment group in cell $\mathbf{x}$.

Problem: With many continuous covariates, there may be no exact matches (the "curse of dimensionality"). Propensity score matching is the solution.

12.5.2 12.3.2 Nearest-Neighbor Matching

Match each treated unit to the control unit(s) with the most similar covariates (measured by distance in $\mathbf{X}$ -space):

$$\hat{\tau}^{NNM} = \frac{1}{N_1} \sum_{i:D_i=1} \left[Y_i - \frac{1}{|J(i)|} \sum_{j \in J(i)} Y_j \right]$$

where $J(i)$ is the set of matched control units for treated unit i .

Key choices in nearest-neighbor matching:

- **Distance metric:** Euclidean? Mahalanobis?
- **Number of matches M :** 1-to-1 or 1-to- M ? More matches reduce variance but increase bias.
- **With or without replacement:** Matching with replacement reduces bias; without replacement is simpler.
- **Caliper:** Restrict matches to be within a maximum distance.

12.6 12.4 The Propensity Score Theorem

With many covariates, exact matching is impossible. Rosenbaum and Rubin (1983) showed that matching on a single scalar — the **propensity score** — suffices.

Definition 12.2 (Propensity Score):

$$p(\mathbf{x}) = P(D_i = 1 \mid \mathbf{X}_i = \mathbf{x})$$

The propensity score is the conditional probability of receiving treatment given observed covariates.

Theorem 12.1 (Propensity Score Theorem, Rosenbaum and Rubin 1983):

If the CIA holds with $\mathbf{X}_i$, then it also holds conditional on $p(\mathbf{X}_i)$:

$$\{Y_{0i}, Y_{1i}\} \perp\!\!\!\perp D_i \mid p(\mathbf{X}_i)$$

Why this is powerful: Instead of matching on a k -dimensional covariate vector, we only need to match on a single number — the propensity score.

12.6.1 12.4.1 Estimating the Propensity Score

In practice, $p(\mathbf{x})$ is unknown and must be estimated. Common methods:

- **Logistic regression:** $\hat{p}(\mathbf{x}) = \frac{\exp(\mathbf{x}^T \hat{\gamma})}{1 + \exp(\mathbf{x}^T \hat{\gamma})}$
 - **Probit regression**
 - **Machine learning:** Random forests, LASSO (for high-dimensional $\mathbf{X}$)
-

12.7 12.5 Propensity Score Matching Estimators

12.7.1 12.5.1 Propensity Score Nearest-Neighbor Matching

Match each treated unit to the control unit with the closest propensity score:

$$\hat{\tau}^{PSM} = \frac{1}{N_1} \sum_{i:D_i=1} [Y_i - Y_{j(i)}]$$

where $j(i) = \arg \min_{j:D_j=0} |p(\hat{\mathbf{X}}_i) - p(\hat{\mathbf{X}}_j)|$.

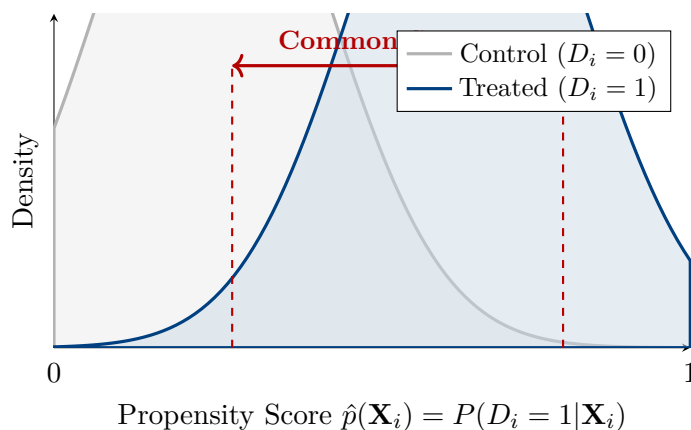


Figure 12.1: Propensity score distributions for treated and control groups. The *common support* (overlap) region (between the red dashed lines) is where both treated and control units exist at similar propensity scores. Matching and IPW estimators are only valid in the common support region; observations outside it must be trimmed or reweighted.

12.7.2 12.5.2 Inverse Probability Weighting (IPW)

An elegant alternative: weight observations by the inverse of the propensity score so that the reweighted control group "looks like" the treatment group.

$$\hat{\tau}_{ATT}^{IPW} = \frac{1}{N_1} \sum_{i=1}^n \left[D_i Y_i - \frac{(1 - D_i) \hat{p}(\mathbf{X}_i) Y_i}{1 - \hat{p}(\mathbf{X}_i)} \right]$$

Intuition: Control units with high propensity scores (similar to treated units) receive higher weight; those with very different characteristics receive low weight.

12.7.3 12.5.3 Doubly Robust Estimation

Modern practice combines propensity score weighting with regression — the **doubly robust** (DR) estimator:

$$\hat{\tau}_{ATT}^{DR} = \frac{1}{N_1} \sum_{i:D_i=1} [(Y_i - \hat{\mu}_0(\mathbf{X}_i))] + \frac{1}{N_1} \sum_{i:D_i=0} \frac{\hat{p}(\mathbf{X}_i)}{1 - \hat{p}(\mathbf{X}_i)} [(Y_i - \hat{\mu}_0(\mathbf{X}_i))]$$

where $\hat{\mu}_0(\mathbf{X}_i)$ is the predicted untreated outcome from a regression model.

Double robustness: The DR estimator is consistent if **either** the propensity score model **or** the outcome regression model is correctly specified — but not necessarily both.

12.8 12.6 Regression as a Form of Matching

Regression and matching are deeply related. Under the CIA, both identify the ATT (or ATE). The difference lies in how they handle the comparison:

- **Regression** fits a global linear model and uses extrapolation.
- **Matching** compares treated units only to "nearby" controls (local comparison).

Angrist and Pischke (2009) show that OLS is a form of matching where each control observation receives a weight proportional to how much its $\mathbf{X}$ differs from the treatment group distribution.

When does regression fail relative to matching?

When there is poor overlap — the treatment and control groups have very different covariate distributions — regression extrapolates from the control group to predict untreated potential outcomes for treated units. This extrapolation can be badly wrong if the CEF is non-linear.

Matching avoids this by restricting comparisons to the common support.

12.9 12.7 Balance Checks and Assessment

The CIA requires that treatment and control groups are comparable after matching. **Balance checks** verify that the matched groups have similar covariate distributions.

Formal balance check: Test whether the mean (and sometimes distribution) of each covariate differs between treated and matched controls.

Standardized Difference:

$$\text{SDiff}_j = \frac{\bar{X}_{1j} - \bar{X}_{0j}}{\sqrt{(\hat{\sigma}_{1j}^2 + \hat{\sigma}_{0j}^2)/2}}$$

Values below 0.1 are generally considered good balance.

Love plot: A visual display of standardized differences before and after matching.

12.10 12.8 Practical Guidance: When to Match?

Situation	Recommended approach
Rich observable data, CIA plausible	Propensity score matching or IPW
Concern about model extrapolation	Match first, then regress on matched sample
Many covariates, CIA uncertain	Add controls in regression; test sensitivity
CIA implausible (unobservables matter)	Use IV, DiD, or RDD instead
Extreme overlap problems	Match with caliper; report trimmed results

12.11 Chapter Summary

- **Matching** identifies causal effects by comparing treated and control units with similar observed covariates.
- The **CIA** (selection on observables) is the key identifying assumption: conditional on $\mathbf{X}$, treatment is independent of potential outcomes.
- The **overlap condition** requires that for every covariate value, both treated and control units exist.
- The **propensity score theorem** allows matching on a single scalar (the probability of treatment) instead of the full covariate vector.
- **Doubly robust estimators** are consistent if either the propensity score model or the outcome model is correct.
- **Regression is a form of matching** but extrapolates; matching restricts comparisons to the common support.
- **Balance checks** verify that matching has achieved covariate balance.
- CIA is a strong assumption. When unobservables matter, use IV, DiD, or RDD.

12.12 Key Terms

CIA (Conditional Independence Assumption) — treatment is independent of potential outcomes given observed covariates.

Overlap — every unit has positive probability of receiving each treatment level.

Propensity score — $p(\mathbf{x}) = P(D = 1 \mid \mathbf{X} = \mathbf{x})$.

Propensity score theorem — matching on the propensity score achieves CIA when matching on $\mathbf{X}$ does.

Nearest-neighbor matching — each treated unit is matched to the control unit(s) with the most similar propensity score.

Inverse Probability Weighting (IPW) — reweight control units by $p(\mathbf{X})/(1 - p(\mathbf{X}))$.

Doubly robust estimator — consistent if either the propensity score model or the outcome model is correct.

Balance check — assessment of whether treatment and control groups have similar covariate distributions after matching.

Standardized difference — normalized measure of covariate balance.

12.13 Exercises

12.13.1 Conceptual Questions

12.1 State the CIA. What does it require about the selection process? Give an example where it is plausible and one where it is not.

12.2 Explain the propensity score theorem. Why is matching on the propensity score sufficient when CIA holds for the full vector $\mathbf{X}$?

12.3 What is the overlap condition? What goes wrong when it fails?

12.4 Explain double robustness. Why is it a desirable property?

12.5 A researcher argues that regression and matching give the same estimates under CIA. Is this true? When might they differ?

12.13.2 Applied Questions

12.6 In R, estimate the average treatment effect of union membership on wages using propensity score matching. Steps: (a) Estimate a logistic regression for treatment (union = 1) on observable controls. (b) Compute the propensity score. (c) Check overlap. (d) Implement nearest-neighbor matching. (e) Assess balance. (f) Compute the ATT.

12.14 R Lab 12: Matching and Propensity Scores

12.14.1 Setup

```
1 library(tidyverse)
2 library(MatchIt)      # matching
3 library(cobalt)       # balance assessment
4 library(WeightIt)     # IPW
5
```

```

6 data("lalonge", package = "MatchIt") # LaLonde (1986) job training
  data

```

12.14.2 Exercise 12.1: Descriptive Analysis

```

1 # Summary statistics by treatment status
2 lalonge %>%
3   group_by(treat) %>%
4   summarise(
5     n      = n(),
6     age    = mean(age),
7     educ   = mean(educ),
8     re78   = mean(re78),
9     .groups = "drop"
10  )

```

12.14.3 Exercise 12.2: Propensity Score Estimation

```

1 # Estimate propensity score via logistic regression
2 ps_model <- glm(treat ~ age + educ + race + married + nodegree +
3   re74 + re75,
4               data = lalonge, family = binomial)
5 lalonge$pscore <- predict(ps_model, type = "response")
6
7 # Check overlap
8 ggplot(lalonge, aes(x = pscore, fill = factor(treat))) +
9   geom_density(alpha = 0.5) +
10  labs(title = "Propensity Score Overlap", x = "Propensity Score",
11        fill = "Treated") +
12  theme_minimal()

```

12.14.4 Exercise 12.3: Nearest-Neighbor Matching

```

1 m_out <- matchit(treat ~ age + educ + race + married + nodegree +
2   re74 + re75,
3               data = lalonge, method = "nearest", ratio = 1)
4 summary(m_out)
5
6 # Balance
7 bal.tab(m_out)
8 love.plot(m_out, threshold = 0.1)

```

12.14.5 Exercise 12.4: ATT from Matched Sample

```
1 # Compute ATT from matched data
2 matched_data <- match.data(m_out)
3 att_matching <- mean(matched_data$re78[matched_data$treat == 1]) -
4                 mean(matched_data$re78[matched_data$treat == 0])
5 cat("ATT (matching):", round(att_matching, 2), "\n")
6
7 # Compare to OLS
8 ols_att <- lm(re78 ~ treat + age + educ + race + married + nodegree
9              + re74 + re75,
10             data = lalonde)
11 cat("OLS estimate of treatment:", round(coef(ols_att)["treat"], 2),
12     "\n")
```

12.14.6 Exercise 12.5: IPW Estimator

```
1 w_out <- weightit(treat ~ age + educ + race + married + nodegree +
2                 re74 + re75,
3                 data = lalonde, method = "ps", estimand = "ATT")
4 summary(w_out)
5 # ATT via IPW regression
6 library(jtools)
7 lm_ipw <- lm(re78 ~ treat, data = lalonde, weights = w_out$weights)
8 cat("IPW ATT:", round(coef(lm_ipw)["treat"], 2), "\n")
```

End of Chapter 12

References

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Chapter 13

Chapter 13: Regression Discontinuity Design

13.1 Chapter Overview

Regression Discontinuity Design (RDD) exploits sharp thresholds in treatment assignment rules to identify causal effects. When treatment is determined by whether a continuous "running variable" crosses a threshold — a test score above 70% grants a scholarship; income below a cutoff qualifies for a social program — we can compare outcomes just above and just below the threshold to estimate a causal effect. The key insight is that individuals just above and just below the threshold are nearly identical in all characteristics, except their treatment status.

RDD has become one of the most widely used and credible identification strategies in applied economics, partly because its identifying assumptions are relatively transparent and can be subjected to rigorous validity tests.

13.2 Learning Objectives

After completing this chapter, students will be able to:

1. Explain the **sharp RDD** setup and the identifying assumption of **continuity**.
2. Identify the **treatment effect estimated by RDD** and explain why it is a local effect.
3. Derive the **RDD estimator** as a difference in regression limits at the cutoff.
4. Describe and implement **local linear regression** for RDD estimation.
5. Explain **bandwidth selection** and the bias-variance tradeoff.
6. Describe the **fuzzy RDD** and its relationship to IV.
7. Conduct **validity tests**: density test, covariate balance, placebo cutoffs.

8. Apply RDD to canonical examples: Thistlethwaite-Campbell (1960), regression in test-score-based policy contexts.
-

13.3 13.1 The Regression Discontinuity Idea

13.3.1 13.1.1 Setup

Let R_i be the **running variable** (also called the "forcing variable" or "assignment variable"). Treatment D_i is assigned based on whether R_i crosses a threshold c :

$$D_i = \mathbb{I}[R_i \geq c] \quad (\text{Sharp RDD})$$

Examples:

- Test scores (R) determine scholarship eligibility (D): threshold at 70%.
- Age (R) determines pension eligibility (D): threshold at age 65.
- Birth weight (R) determines neonatal intensive care admission (D): threshold at 1500g.
- Vote share (R) determines election winner (D): threshold at 50%.

13.3.2 13.1.2 The Local Nature of RDD

RDD identifies a **local** causal effect — the treatment effect for individuals near the threshold, not for the full population. This is both a limitation and a strength:

- **Limitation:** May not generalize to individuals far from the cutoff.
 - **Strength:** Near the cutoff, individuals just above and below are nearly identical — making the identifying assumption very credible.
-

13.4 13.2 The Sharp RDD

13.4.1 13.2.1 The CEF Near the Cutoff

For the sharp RDD, the average treatment effect at the threshold c is:

$$\tau_{RDD} = \lim_{r \downarrow c} E[Y_i \mid R_i = r] - \lim_{r \uparrow c} E[Y_i \mid R_i = r]$$

This is the **jump** in the conditional mean of Y at the threshold c .

Definition 13.1 (Sharp RDD Estimand):

$$\tau_{RDD} = E[Y_{1i} - Y_{0i} \mid R_i = c]$$

This is the ATT for individuals at the threshold — those right on the margin of receiving treatment.

13.4.2 13.2.2 The Identifying Assumption: Continuity

Assumption 13.1 (Continuity): The conditional expectations $E[Y_{0i} \mid R_i = r]$ and $E[Y_{1i} \mid R_i = r]$ are continuous in r at $r = c$.

Interpretation: In the absence of the threshold, the outcome would change smoothly as the running variable changes. There is no other discontinuity in outcomes at c .

This rules out:

- Other policies that change at exactly $r = c$.
- Individuals precisely manipulating R_i to sort above or below c .
- Discontinuous changes in any confounder at c .

13.4.3 13.2.3 Identification

Under Assumption 13.1:

$$\begin{aligned} & \lim_{r \downarrow c} E[Y_i \mid R_i = r] - \lim_{r \uparrow c} E[Y_i \mid R_i = r] \\ &= E[Y_{1i} \mid R_i = c] - E[Y_{0i} \mid R_i = c] = \tau_{RDD} \end{aligned}$$

The key insight: by continuity, the average untreated potential outcome $E[Y_{0i} \mid R_i = r]$ is approximately the same for individuals just above and just below the threshold. So the observed jump in Y at the cutoff isolates the treatment effect.

13.5 13.3 Estimation: Local Linear Regression

13.5.1 13.3.1 Fitting Regression on Each Side

The naive approach: fit a separate polynomial regression on each side of the cutoff, then compare the intercepts at c :

Left side ($R_i < c$): $Y_i = f_L(R_i) + u_i$

Right side ($R_i \geq c$): $Y_i = f_R(R_i) + u_i$

$$\hat{\tau}_{RDD} = \hat{f}_R(c) - \hat{f}_L(c)$$

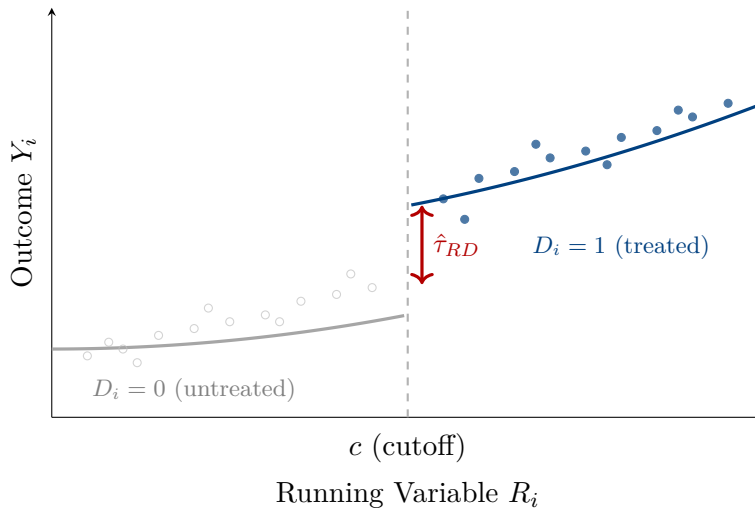


Figure 13.1: A sharp Regression Discontinuity Design. Treatment $D_i = \mathbf{1}[R_i \geq c]$ switches on at the cutoff c . Units just left of c are nearly identical to units just right, except for treatment status. The RDD estimate $\hat{\tau}_{RD}$ equals the jump in the outcome at the cutoff, computed by comparing the right- and left-hand limits of the regression function.

13.5.2 13.3.2 Local Linear Regression

The most common approach: use **local linear regression** (LLR), which fits a linear regression within a bandwidth h on each side of the cutoff.

Definition 13.2 (Local Linear Regression Estimator):

For the right limit: $\hat{\mu}_R = \hat{\beta}_{R0}$ from OLS of Y_i on $(1, R_i - c)$ using only observations with $c \leq R_i \leq c + h$.

For the left limit: $\hat{\mu}_L = \hat{\beta}_{L0}$ from OLS of Y_i on $(1, R_i - c)$ using only observations with $c - h \leq R_i < c$.

$$\hat{\tau}_{RDD}^{LLR} = \hat{\mu}_R - \hat{\mu}_L$$

In one regression (common implementation):

$$Y_i = \alpha + \tau D_i + \beta_1(R_i - c) + \beta_2 D_i(R_i - c) + u_i$$

using only observations with $|R_i - c| \leq h$. The coefficient $\hat{\tau}$ is the RDD estimate.

This specification allows the slope of Y on R to differ on either side of the cutoff, which is important for flexibility.

13.5.3 13.3.3 Kernels and Weighting

To give more weight to observations near the threshold, use **kernel regression**. A common choice is the **triangular kernel**: $K(r) = (1 - |r|)\mathbb{I}[|r| \leq 1]$, which gives zero weight to observations at the bandwidth edge and maximum weight to observations at the cutoff.

Implementing triangular kernel in practice: compute weights $w_i = 1 - |R_i - c|/h$ and use WLS.

13.6 13.4 Bandwidth Selection

13.6.1 13.4.1 The Bias-Variance Tradeoff

The choice of bandwidth h involves a fundamental tradeoff:

- **Larger h :** More observations, lower variance, but more bias (farther observations are less comparable).
- **Smaller h :** Fewer observations, higher variance, but less bias (observations are more comparable).

13.6.2 13.4.2 Optimal Bandwidth Selection

The **Mean Squared Error (MSE)**-optimal bandwidth balances bias and variance:

$$h^* = \arg \min_h [\text{Bias}(\hat{\tau})^2 + \text{Var}(\hat{\tau})]$$

Calonico, Cattaneo, and Titiunik (2014) (CCT) propose a data-driven bandwidth selector based on minimizing the asymptotic MSE of the RDD estimator. The `rdrobust` package in R implements this.

Good practice: Report results under multiple bandwidths. If estimates are sensitive to bandwidth choice, this is a concern. The CCT optimal bandwidth provides a principled default.

13.7 13.5 The Fuzzy RDD and Its Relationship to IV

13.7.1 13.5.1 Setup

In the **fuzzy RDD**, crossing the threshold changes the *probability* of treatment but does not perfectly determine it:

$$P(D_i = 1 \mid R_i = r) = \begin{cases} g_1(r) & \text{if } r \geq c \\ g_0(r) & \text{if } r < c \end{cases}$$

where there is a jump $g_1(c) - g_0(c) \neq 0$ but this jump is not a full $0 \rightarrow 1$ transition.

Example: A test score above a threshold makes individuals *eligible* for a program, but they must actively enroll. Some who score above the threshold don't enroll; some below find other ways to enroll.

13.7.2 13.5.2 Fuzzy RDD as IV

The fuzzy RDD estimator is:

$$\hat{\tau}^{fuzzy} = \frac{\lim_{r \downarrow c} E[Y \mid R = r] - \lim_{r \uparrow c} E[Y \mid R = r]}{\lim_{r \downarrow c} P(D = 1 \mid R = r) - \lim_{r \uparrow c} P(D = 1 \mid R = r)}$$

This is exactly the **Wald estimator** with the threshold indicator $\mathbb{I}[R_i \geq c]$ as the instrument:

- Numerator: reduced form (effect of threshold on Y).
- Denominator: first stage (effect of threshold on D).

The fuzzy RDD therefore inherits the LATE interpretation: it estimates the average effect of treatment for compliers — individuals near the threshold who are induced to take (or not take) treatment by the threshold rule.

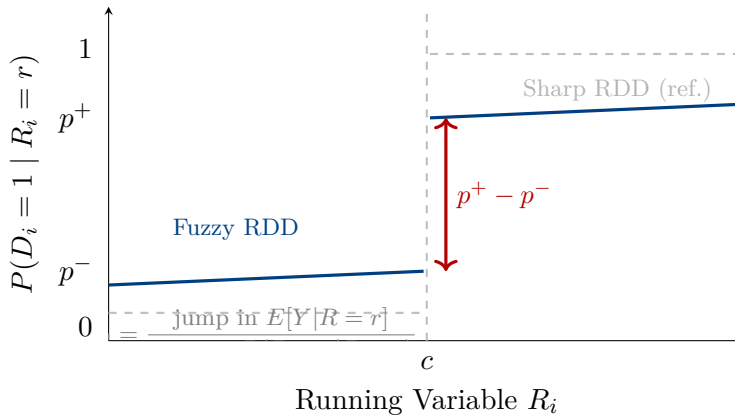


Figure 13.2: Fuzzy RDD: the probability of treatment $P(D_i = 1 \mid R_i = r)$ jumps discontinuously at the cutoff c but does not go from 0 to 1 (as in sharp RDD). The fuzzy RDD estimator is the ratio of the jump in outcomes to the jump in treatment probability, and is equivalent to a local IV estimator at the cutoff.

13.8 13.6 Validity Tests

13.8.1 13.6.1 The Density Test (McCrary, 2008)

If individuals can manipulate the running variable to sort above the threshold (e.g., to get a scholarship), the assumption of quasi-random assignment near the threshold is violated.

Test: If there is manipulation, there will be a discontinuity in the **density of R_i** at c — a "bunching" above the threshold.

The **McCrary (2008) density test** tests for a discontinuity in the density of the running variable at the cutoff. Rejection is evidence of manipulation and undermines the RDD design.

In R: `rddensity` package implements this test.

13.8.2 13.6.2 Covariate Balance

If the RDD is valid, covariates that are predetermined relative to the running variable (age, education, gender, baseline outcomes) should not jump at the threshold. **Test:** Run the same RDD regression using predetermined covariates as the outcome. Non-zero results indicate a problem.

13.8.3 13.6.3 Placebo Cutoffs

If we use an artificial cutoff away from the true cutoff, we should find no discontinuity in outcomes (since there is no treatment at the placebo cutoff). Significant effects at placebo cutoffs suggest confounds.

13.9 13.7 Empirical Illustration: Thistlethwaite and Campbell (1960)

13.9.1 13.7.1 The Original RDD Paper

Thistlethwaite and Campbell (1960) studied whether receiving a National Merit Award affected students' career aspirations. Awards were granted based on test scores exceeding a threshold.

Running variable: Test score. **Treatment:** Receiving a National Merit Award. **Outcome:** Career and educational aspirations.

Students just above and just below the threshold had nearly identical academic preparation (the difference in test score is negligible), but those above received recognition and financial support. Any discontinuity in outcomes at the threshold can be attributed to the award.

This paper introduced RDD to social science. While Thistlethwaite and Campbell did not use modern estimation methods, their design intuition was correct.

13.9.2 13.7.2 Modern Applications of RDD

RDD has been applied to a vast range of questions:

Study	Running Variable	Threshold	Outcome
Lee (2008)	Vote share	50%	Re-election incumbent a
Card et al. (2008)	Age	65 (Medicare eligibility)	Health insurance, health
Angrist and Lavy (1999)	Class size formula	Maimonides rule	Student achievement
Dell (2010)	Poverty index	Eligibility cutoff	Economic development

13.10 Chapter Summary

- **RDD** exploits threshold-based treatment assignment to identify causal effects.
 - The **continuity assumption** (outcomes are smooth in the running variable absent treatment) is the key identifying condition.
 - The **RDD estimand** is the treatment effect at the threshold — a local effect for individuals right at the margin.
 - **Local linear regression** within a bandwidth estimates the jump in conditional means at the cutoff.
 - **Bandwidth selection** involves a bias-variance tradeoff; the CCT method provides a principled choice.
 - The **fuzzy RDD** is an IV estimator using the threshold indicator as an instrument; it estimates the LATE for compliers near the threshold.
 - **Validity tests** (density, covariate balance, placebo cutoffs) assess the plausibility of the continuity assumption.
 - RDD estimates are local and may not generalize to populations far from the threshold.
-

13.11 Key Terms

Running variable — the continuous variable whose value determines treatment.

Cutoff — the threshold value of the running variable at which treatment changes.

Sharp RDD — treatment is perfectly determined by the running variable crossing the threshold.

Fuzzy RDD — crossing the threshold changes the probability but not the certainty of treatment.

Continuity assumption — conditional means are continuous in the running variable at the cutoff.

Local linear regression (LLR) — fitting a linear regression within a bandwidth on each side of the cutoff.

Bandwidth — the window around the cutoff used for estimation.

McCrary density test — test for a discontinuity in the density of the running variable at the cutoff.

Placebo cutoffs — testing for discontinuities at false cutoffs to assess confounds.

13.12 Exercises

13.12.1 Conceptual Questions

13.1 Explain the continuity assumption in plain language. What does it mean about individuals near the threshold?

13.2 Why is the effect estimated by RDD "local"? Does this make RDD less useful than DiD or IV?

13.3 Explain why the fuzzy RDD is equivalent to IV with the threshold indicator as the instrument. What is the LATE interpretation of the fuzzy RDD?

13.4 Describe three validity tests for RDD. What does each test check?

13.5 A researcher studies the effect of receiving a college scholarship (awarded to students scoring above 80 on a national exam) on college enrollment. The density of test scores shows a sharp spike just above 80. What does this suggest? How does it affect the validity of the RDD?

13.12.2 Analytical Questions

13.6 Suppose the true outcome equation is $Y_i = \tau D_i + g(R_i) + u_i$, where $g(r)$ is a smooth function. Derive the RDD estimator and show it identifies τ under the continuity assumption.

13.7 In a fuzzy RDD, the reduced form estimate is 0.08 and the first stage estimate is 0.40. Compute the LATE. What is the LATE interpretation?

13.12.3 Applied Questions

13.8 Using the `rdrobust` package in R, implement a sharp RDD: (a) Plot the running variable and outcomes around the cutoff. (b) Conduct a density test for manipulation. (c) Estimate the RDD effect using local linear regression with the CCT optimal bandwidth. (d) Check covariate balance at the cutoff. (e) Assess sensitivity to bandwidth choice.

13.13 R Lab 13: Regression Discontinuity Design

13.13.1 Setup

```
1 library(tidyverse)
2 library(rdrobust)      # RDD estimation and inference
3 library(rddensity)     # McCrary density test
4 library(ggplot2)
5
6 # Simulate RDD data
7 set.seed(42)
8 n <- 2000
9 c <- 0 # cutoff
10
11 R <- runif(n, -1, 1)    # running variable
12 D <- as.integer(R >= c) # sharp treatment
13
14 # True treatment effect: tau = 0.5
15 tau <- 0.5
16 Y <- 2 + tau * D + 0.8 * R + rnorm(n, sd = 0.5)
17
18 df_rdd <- data.frame(R = R, D = D, Y = Y)
```

13.13.2 Exercise 13.1: Visualizing the Discontinuity

```
1 # Bin means plot
2 df_rdd <- df_rdd %>%
3   mutate(bin = cut(R, breaks = 40))
4
5 bin_means <- df_rdd %>%
6   group_by(bin, D) %>%
7   summarise(Y_mean = mean(Y), R_mean = mean(R), .groups = "drop")
8
9 ggplot(bin_means, aes(x = R_mean, y = Y_mean, color = factor(D))) +
10   geom_point(size = 2) +
11   geom_vline(xintercept = 0, linetype = "dashed") +
12   geom_smooth(method = "lm", se = FALSE) +
13   labs(
14     title = "RDD: Outcome vs. Running Variable",
```

```

15 subtitle = "Clear jump at cutoff (R = 0)",
16 x = "Running Variable (R)", y = "Mean Outcome", color = "Treated"
17 ) +
18 theme_minimal()

```

13.13.3 Exercise 13.2: Density Test

```

1 # McCrary density test
2 density_test <- rddensity(R, c = c)
3 summary(density_test)
4 rdplotdensity(density_test, R)

```

13.13.4 Exercise 13.3: Local Linear RDD Estimation

```

1 # RDD estimate with optimal bandwidth
2 rdd_est <- rdrobust(Y, R, c = c, kernel = "triangular", bwselect =
  "mserd")
3 summary(rdd_est)
4
5 cat("RDD estimate:", round(rdd_est$coef["Conventional"], 3), "\n")
6 cat("95% CI: [", round(rdd_est$ci["Robust", 1], 3), ",",
  round(rdd_est$ci["Robust", 2], 3), "]\n")
7
8 cat("Optimal bandwidth:", round(rdd_est$bws["h", "left"], 3), "\n")

```

13.13.5 Exercise 13.4: Covariate Balance

```

1 # Generate a predetermined covariate
2 X_cov <- 0.3 * R + rnorm(n) # correlated with R but not jumping at
  cutoff
3
4 # RDD of covariate on running variable (should find no discontinuity)
5 rdd_cov <- rdrobust(X_cov, R, c = c)
6 summary(rdd_cov)
7 cat("Covariate RDD estimate (should be ~0):",
  round(rdd_cov$coef["Conventional"], 3), "\n")

```

13.13.6 Exercise 13.5: Sensitivity to Bandwidth

```

1 # Estimate over a range of bandwidths
2 h_grid <- seq(0.1, 0.8, by = 0.05)
3 results <- map_dfr(h_grid, function(h) {
4   est <- rdrobust(Y, R, c = c, h = h)
5   data.frame(h = h, estimate = est$coef["Conventional"],
6             ci_low = est$ci["Robust", 1], ci_high =
7               est$ci["Robust", 2])

```

```
7 })
8
9 ggplot(results, aes(x = h, y = estimate)) +
10   geom_ribbon(aes(ymin = ci_low, ymax = ci_high), alpha = 0.2, fill
11     = "steelblue") +
12   geom_line(color = "steelblue", linewidth = 1) +
13   geom_hline(yintercept = tau, color = "red", linetype = "dashed") +
14   labs(
15     title = "RDD Estimate Across Bandwidths",
16     subtitle = "Red dashed = true effect. Estimate is stable across
17       bandwidths.",
18     x = "Bandwidth (h)", y = "RDD Estimate"
19   ) +
20   theme_minimal()
```

End of Chapter 13

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Notation Index

Symbol	Meaning
Y_i, y_i	Outcome variable for unit i
D_i	Binary treatment indicator
Y_{0i}, Y_{1i}	Potential outcomes (untreated, treated)
β	Parameter vector
$\hat{\beta}^{OLS}$	OLS estimator
$\mathbf{X}$	$n \times k$ design matrix
$\mathbf{P}_\mathbf{X}$	Projection matrix: $\mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$
$\mathbf{M}_\mathbf{X}$	Annihilator: $\mathbf{I} - \mathbf{P}_\mathbf{X}$
$\hat{\mathbf{u}}$	OLS residual vector
σ^2	Error variance
$\hat{\sigma}^2$	Estimated error variance: $SSR/(n - k)$
R^2	Coefficient of determination
Ω	Sandwich “meat”: $E[u_i^2 \mathbf{x}_i \mathbf{x}_i^T]$
$p(\mathbf{x})$	Propensity score: $P(D_i = 1 \mathbf{X}_i = \mathbf{x})$
$\xrightarrow{p}$	Convergence in probability
$\xrightarrow{d}$	Convergence in distribution
ATT	Average Treatment Effect on the Treated
ATE	Average Treatment Effect
LATE	Local Average Treatment Effect
IV	Instrumental Variable
2SLS	Two-Stage Least Squares
FE	Fixed Effects
DiD	Differences-in-Differences
RDD	Regression Discontinuity Design
FWL	Frisch-Waugh-Lovell (Theorem)
OVb	Omitted Variable Bias
CIA	Conditional Independence Assumption